science	professor	title	year	school	subject	total	impact	publications	affiliations	toys	reasoning	web
Mathematics	Alex Kumjian	Professor of Mathematics & Statistics			Operator algebras, C*-algebras, higher rank graphs, groupoids, K-theory	5	9	Higher Rank Graph C*-Algebras (foundational); Graphs, Groupoids, and Cuntz-Krieger Algebras; C*-algebras of extensions of groupoids by group bundles		Computational software		
Mathematics	Chris Rogers	Associate Professor of Mathematics & Statistics			Homotopy theory, homological algebra, algebraic topology, mathematical physics	17	137	Complete L-infinity algebras and their homotopy theory; Which homotopy algebras come from transfer? (2021); Higher symplectic geometry and mathematical physics		Computational algebra software		https://hoalg.net/chrisrogers/
Mathematics	Deena Schmidt	Associate Professor of Mathematics & Statistics		Cornell University (2007)	Mathematical biology, applied probability, networks (neuro/epi), population genetics	4	31	Sleep-wake dynamics in mammalian development (2024); Stochastic models in neuroscience and epidemiology; NSF and Simons Foundation funded network research	Mathematical Biology Group; Integrative Neuroscience Program; TechWise (Google)	Computational biology software, network analysis tools, HPC	TechWise Program director; math+computing+data science bridge; Google partnership; accessible to students	https://www.deenaschmidt.com/
Mathematics	Diana Moss	Associate Professor of Mathematics & Statistics; Teaching	2020	University of Nevada, Reno	Mathematics education, children's mathematical thinking, teacher education	10	14	Tensions between objectivism and constructivism in online STEM education (2022); Pre-service teachers as learners of math and science (2022); Role of representation in understanding fractions as measures (2022)	Preschool to College Research Group	Education research tools, assessment software		
Mathematics	Edward Keppelmann	Associate Professor of Mathematics & Statistics			Topology, periodic points, Nielsen theory, solvmanifolds, nilmanifolds			Wecken property for periodic points on the Klein bottle; Linearizations for maps of nilmanifolds and solvmanifolds (Handbook chapter); Nielsen numbers and fixed point theory		Computational software		
Mathematics	Eric Olson	Associate Professor of Mathematics & Statistics	2008	Indiana University (1999)	Navier-Stokes equations, dynamical systems, fractal dimensions, turbulence, data assimilation	24	1480	Research on Navier-Stokes equations and turbulence; Fractal dimensions and data assimilation; Fluid mechanics and bioremediation modeling	Applied Mathematics Group	HPC clusters, simulation software		
Mathematics	Jeff Mortensen	Professor of Mathematics & Statistics; Teaching & Associate Chair	1993	UC Davis (1993)	Analysis, fractional calculus, differential equations			Fractional vector calculus for fractional advection-dispersion (Physica A 2006); Vector Grunwald formula for fractional derivatives; Fractional conservation of mass equations		Computational software		
Mathematics	Jonathan Beardsley	Assistant Professor of Mathematics & Statistics		Johns Hopkins University (2016)	Algebraic topology, category theory, homotopy theory, chromatic homotopy theory, K-theory	9	12	Skeleta and Categories of Algebras (Advances in Mathematics 2024); Publications in Algebraic & Geometric Topology; Simons Foundation Collaboration Grant ($42K, 2021)		Computational algebra software		
Mathematics	Mark Pinsky	Professor of Mathematics & Statistics			Nonlinear systems, dynamical systems, control theory, stability analysis, bifurcation theory	2	53	Stability Criteria for Linear Systems with Uncertain Periodic Coefficients (SIAM); New approach to the Lure problem in absolute stability (SIAM 2003); Optimal Exponential Feedback Stabilization of Planar Systems (Automatica 2011)		Computational software, simulation tools		
Mathematics	Mehmet Gumus	Associate Professor of Mathematics & Statistics; Teaching	2019	Southern Illinois U Carbondale (2017)	Linear algebra, matrix computations, machine learning, STEM mentoring	6	45	Trace inequalities and matrix computations; ML applications to storage systems; Linear algebra and computational methods		Computational software, ML frameworks		
Mathematics	Mihye Ahn	Associate Professor of Mathematics & Statistics		North Carolina State University (2010)	Statistics, functional MRI analysis, neuroimaging, brain network analysis	9	92	Sparse Reduced Rank Framework for Group Analysis of fMRI Data (Statistica Sinica 2015); LCN: Random Graph Model for Community Detection in Brain Networks (2017); Weighted functional linear Cox regression model (2021)	Integrative Neuroscience Faculty	fMRI data pipelines, statistical computing (R/Python), HPC	Biostatistics bridges math+biology; clinical trial design; collaborative opportunities	https://sites.google.com/site/mihyeunr/
Mathematics	Minggen Lu	Professor of Health Analytics & Biostatistics	2010		Biostatistics, health analytics, nonparametric regression, survival analysis, longitudinal data			Advanced statistical methods for high-dimensional biomedical data Panel count data analysis in epidemiology Machine learning approaches to health analytics and prediction	School of Public Health	Statistical computing, HPC, ML platforms, biostatistical tools	Health analytics+ML; bridges math+computing; high-dimensional data methods	
Mathematics	Nakisa Ghanbarian	Teaching Assistant Professor of Mathematics & Statistics	2024	University of Nevada, Reno (2024)	Pure mathematics, matrix analysis, knot theory	1	4	The non-orientable 4-genus for knots with 10 crossings (2022); The non-orientable 4-genus for knots with 8 or 9 crossings; Band-unknotting numbers research		Computational topology software		
Mathematics	Perry Williams	Associate Professor of Natural Resources & Environmental Science		UNR CABNR	Statistical ecology, quantitative ecology, decision-making in natural resource management	29	192	Statistical methods for wildlife monitoring Decision theory in natural resource management Quantitative ecological modeling	NRES; CABNR	R/Python statistical computing, Bayesian modeling tools		https://perrywilliams.us/
Mathematics	Raul Rojas-Gonzalez	Professor of Mathematics & Statistics; Gibson Endowed	2015	Freie Universität Berlin (1988)	AI, robotics, autonomous vehicles, ML, computational geometry, history of mathematics	12	165	GroundGrid: LiDAR Point Cloud Ground Segmentation (2024); Design and Experimental Analysis of an Adaptive Cruise Control (2022); Path Following with Deep Reinforcement Learning for Autonomous Cars (2021)	Autonomous Systems Research Group	LiDAR sensors, autonomous vehicle platforms, GPU clusters, robotic hardware		
Mathematics	Stanislav Jabuka	Associate Professor of Mathematics & Statistics		Michigan State University (2002)	Low-dimensional topology, knot theory, 3- and 4-manifolds, Heegaard Floer homology	9	8	Knot graphs and Gromov Hyperbolicity (Math. Zeitschrift 2022); On a nonorientable analogue of the Milnor Conjecture (2021); Publications in J. Knot Theory and Its Ramifications		Computational topology software		

Mathematics	Thomas Quint	Foundation Professor of Mathematics & Statistics; 2025	1996	Stanford University (1988)	Game theory, mathematical economics, graph theory, Shapley values	2	20	Measures of Powerlessness in Simple Games; Evaluating Presidential Elections Using Game Theory; ~30 journal articles and 2 books on game theory		Computational software		
Mathematics	Tin-Yau Tam	Professor of Mathematics & Statistics; & Chair; Weeks Endowed Chair	2014	University of Hong Kong (1986)	Matrix theory, Lie groups/ algebras, multilinear algebra, numerical ranges, operator theory	17	39	Color Image Set Recognition Based on Quaternionic Grassmannians (2025); Matrix Inequalities and Their Extensions to Lie Groups (monograph); Extensions of three matrix inequalities to semisimple Lie groups		Computational software (MATLAB, Mathematica)		
Mathematics	Tomasz J. Kozubowski	Professor of Mathematics & Statistics		UC Santa Barbara	Stochastic modeling, distribution theory, Laplace distribution, heavy-tailed distributions, financial math	28	49	The Laplace Distribution and Generalizations (2001, 944 citations); A computational approach to estimation of discrete Pareto parameters (2023); Fractional Laplace motion (Advances in Applied Probability)		Statistical computing (R, SAS)		
Mathematics	Valentin Deaconu	Associate Professor of Mathematics & Statistics	2010	University of Iowa (1995)	Operator algebras, groupoids, C*-algebras, self-similar actions	9	2	Cohomology of ample groupoids (2025); The Koopman representation for self-similar groupoid actions (2022); On groupoids and C*-algebras from self-similar actions		Computational software		
Mathematics	Wei Yang	Foundation Professor of Epidemiology & Biostatistics	2005		Epidemiology, biostatistics, environmental health, population surveillance, statistical methods	200	7359	Statistical methods for clinical and translational research Population-based health surveillance systems and analysis Environmental epidemiology: exposure assessment and disease modeling	School of Public Health; Nevada CSES	Statistical software (R, SAS, Python), epidemiological modeling	Foundation Professor; epidemiology+biostatistics; strong quantitative methods	
Mathematics	Yinghan Chen	Associate Professor of Mathematics & Statistics			Psychometrics, Bayesian statistics, cognitive diagnosis models, latent class models, Monte Carlo methods	16	95	Bayesian Estimation of Attribute Hierarchy for Cognitive Diagnosis Models (2023); Bayesian Inference for Unknown Number of Attributes in Restricted Latent Class Models; DINA model research (NSF $229,951)		Statistical computing (R, MCMC clusters)		
Computing	Alireza Tavakkoli	Associate Professor of Computer Science & Engineering	2023	University of Nevada, Reno (2009)	Computer vision, robot vision, VR/AR/MR, biomedical imaging, remote sensing	200	2048	Guided/Unguided Conditional Diffusion for 3D Point Cloud Generation (2025); Bayesian Graphical Models for Joint Degeneration Analysis (2024); Feature Representation for Retinal Disease Detection	Computer Vision Laboratory (CVL); Integrative Neuroscience Faculty	GPU clusters, VR/AR headsets, 3D scanners, medical imaging systems, cameras	BRAIN Core ML as a service	https://cvl.cse.unr.edu/
Computing	Anikita Shukla	Assistant Professor of Computer Science & Engineering	2024	IIIT-Delhi (2021)	Computer vision, deep learning, geometric methods, AI for science, AI for wildlife conservation	30	533	XPRIZE Rainforest winner ($5M, 1st place); Polynomial Implicit Neural Representations (CVPR 2023); Tiger re-identification; Geometric priors for ICF generative models (NeurIPS 2021 workshop)		GPU clusters, cameras, imaging sensors	AI for Science focus; postdoc at ASU; XPRIZE Rainforest team; open to undergrads; directly relevant to ML research	https://ankita-shukla.github.io/
Computing	Xiaoxue (Eira) Zhang	Assistant Professor of Computer Science & Engineering	2024	UC Santa Cruz (2024)	Computer networks, network security, IoT, quantum networks, blockchains	10	36	Aggregated Payment Channel Networks (IEEE/ACM ToN 2024); network security for IoT		Network testbeds, network analysis tools	Brand new hire (PhD May 2024); USTC undergrad; advisor Chen Qian; relevant to Cybersecurity program	https://xzhan330z.github.io/
Computing	Batyr Charyyev	Assistant Professor of Computer Science & Engineering	2022	Stevens Institute of Technology (2022)	Networking, IoT, traffic fingerprinting, security, privacy, edge computing	10	25	Towards Securing Data Transfers Against Silent Data Corruption; Edge computing for wildfire detection ($500K NSF); Network science and IoT security		IoT sensor networks, edge computing devices, network analysis tools		https://www.cse.unr.edu/~batyr/
Computing	David Feil-Seifer	Associate Professor of Computer Science & Engineering	2013	USC Viterbi (2011)	Socially assistive robotics, human-robot interaction, robotics for ASD intervention	45	167	Defining Socially Assistive Robotics (seminal work); Toward SAR for Augmenting ASD Interventions; Benchmarks for evaluating socially assistive robots	Social Robotics Lab	Social robots (NAO, Pepper-type), sensors, actuators, eye-tracking, ML hardware	Exceptional undergrad mentoring (100+ undergrads); motion capture system; accessible	https://www.cse.unr.edu/~dave/
Computing	Eelke Folmer	Professor of Computer Science & Engineering; & Chair of Computer Science	2006	U of Groningen (2005)	HCI, accessibility technology, VR/AR, wearables, indoor navigation for blind users	15	431	Augmented reality navigation: a survey Universal usability of virtual reality Navigation systems for blind users using smartphone technology	VR/AR Core Facility; Neuroscience Center; NSF, NIH, Google funded	VR/AR systems, wearables, motion capture	CSE Chair; VR/AR facility; NSF+NIH+Google funded; accessible HCI research	https://www.eelke.com/
Computing	Emily Hand	Assistant Professor of Computer Science & Engineering	2018	University of Maryland (2018)	Computer vision, human perception, ML, facial recognition, assistive technology	28	120	Transformation of face recognition technology; Wearable devices for facial recognition; Prior positions at NASA Ames, JPL, Naval Research Lab	Machine Perception Lab; Integrative Neuroscience Faculty	GPU clusters, cameras, wearable prototyping, ML infrastructure	ML/vision with social impact; NASA background; open to undergrads	https://www.machineperceptionlab.com/
Computing	Engin Arslan	Assistant Professor of Computer Science & Engineering	2017	University at Buffalo, SUNY (2016)	HPC, networking, edge/cloud computing, distributed systems, quantum networking	8	45	Demystifying Data Transfer Performance in HPC Networks (IEEE 2023); Falcon: Fair and Efficient Online File Transfer; NSF CAREER ($529K, 2022)		Network testbeds, HPC clusters, cloud infrastructure		https://www.cse.unr.edu/~earslan/
Computing	Feng Yan	Assistant Professor of Computer Science & Engineering	2016	College of William and Mary (2016)	ML systems, deep learning, neural architecture search, federated learning, serverless computing	77	2275	Noctua: Automated Fine-grained Consistency Analysis (EuroSys 2024); Gradient similarity for large-scale training (NeurIPS 2021); Gradient compression for DNN training (SOSP 2021)	Intelligent Data & Systems (IDS) Lab; $20M NSF Wildfire project	GPU/TPU clusters, cloud computing infrastructure, distributed systems	ML systems; leads ML on $20M NSF wildfire project; directly relevant to Computing olympiad	https://www.cse.unr.edu/~fyan/research.html
Computing	Frederick C. Harris Jr.	Foundation Professor of Computer Science & Engineering; Assoc. Dean	1995	Clemson University (1994)	Parallel computation, data science, software engineering, computer graphics, VR	87	344	Multi-scale Transformer Pyramid Networks for Time Series Forecasting (IEEE 2024); 200+ peer-reviewed papers in parallel computing; Computational neuroscience and graphics	HPCVIS Lab; Nevada EPSCoR ($20M NSF)	GPU clusters, HPC systems, VR headsets, visualization hardware		https://www.cse.unr.edu/~fredh/
Computing	George Bebis	Foundation Professor of Computer Science & Engineering		University of Central Florida (1996)	Computer vision, image processing, pattern recognition, machine learning, evolutionary computing	39	200	On-road vehicle detection: a review (934 citations); Vision-based hand pose estimation: A review (682 citations); 3D point cloud classification and mammography analysis	Computer Vision Laboratory (CVL)	GPU clusters, cameras, image sensors, ML infrastructure	Major CV lab; GPU access; open to undergrads; directly relevant to Computing olympiad	https://cvl.cse.unr.edu/

Computing	Hanif Livani	Associate Professor of Electrical & Biomedical Engineering	2014	UNR College of Engineering	Machine learning for power systems, smart grids, microgrids, cyber-security for power systems	81	384	Distributed Cooperative Control System for Smart Microgrids Large-Signal Stability Analysis of Inverter-based Microgrids via Sum of Squares Adaptive neural network-based intelligent secondary control for microgrids	EBME; IEEE	Microgrid testbed, power system simulation		https://www.unr.edu/ebme/research/power-
Computing	Hao Xu	Associate Professor of Electrical & Biomedical Engineering	2016	UNR College of Engineering	Autonomous systems, intelligent control, machine learning, cyber-physical systems, unmanned aircraft	63	1349	Research in intelligent control and autonomous systems LiDAR-based applications and intelligent transportation systems CREDIT+: Advancing Data Analysis for Mission-Critical Applications ($1M DoD grant)	EBME; IEEE	LiDAR systems, UAS platforms, autonomous vehicle testbed		https://www.unr.edu/ebme/people/hao-xu
Computing	Hung Manh La	Associate Professor of Computer Science & Engineering	2014	Oklahoma State University (2011)	Robotics, autonomous systems, multi-robot control, infrastructure inspection, wearable computing			A Climbing Robot for Steel Bridge Inspection (J Intelligent Robotic Systems); Agile robotic inspection: bicycle-like approach (J Field Robotics 2024); Adaptive Flocking Control of Multiple UGVs by UAV	Advanced Robotics & Automation (ARA) Lab	Mobile robots, UAVs/drones, climbing robots, wearable sensors, bridge inspection platforms	Hands-on robotics; award-winning bridge inspection robots; field experience	https://ara.cse.unr.edu/
Computing	Javad Sattarvand	Associate Professor of Mining & Metallurgical Engineering		UNR Mackay School	Mining optimization, mining automation, AI applications in mining, autonomous mining systems	29	410	Research on autonomous mining systems and production planning AI applications in mine optimization Mining automation and robotics	Mining & Metallurgical Engineering; SME	Mining simulation software, autonomous systems		https://www.unr.edu/mackay/people/department-
Computing	Jill Heaton	Professor of Geography	2005		Remote sensing, species distribution, GIS, paleoclimate, ecosystem modeling	3	12	Tree-based ensemble algorithms for ecological classification via remote sensing Remote sensing for monitoring species distributions in arid regions Desert reptile ecology and conservation in southern Africa	Geography; Ozmen Institute; EECB Program	Hyperspectral imaging, GIS (ArcGIS), remote sensing tools		https://www.unr.edu/geography/people/jill-heaton
Computing	Josh Baker	Professor of Psychology		UNR College of Liberal Arts	Computational neuroscience, robotics, computer vision, machine learning, biomedical engineering	39	97	Computational neuroscience and brain modeling Machine learning for biomedical applications Computer vision and pattern recognition	Psychology; Institute for Neuroscience; INBRE Director	HPC clusters, neural network computing, robotics lab	INBRE Director; computational neuroscience + ML bridges Computing and Biology; leads NV biomedical research infrastructure	https://www.medicine.nevada.edu/baker-lab
Computing	Juli Petereit	Director of Nevada Bioinformatics Center		UNR	Bioinformatics, large-scale statistical analyses, network modeling, high-throughput omics data	31	149	High-throughput omics data analysis methods Network modeling in biomedical research Bioinformatics pipelines for genomic data	NV Bioinformatics Center; NIH INBRE Data Science Core	HPC clusters, bioinformatics software, genomic analysis pipelines	Directs the Nevada Bioinformatics Center; bridges Computing and Biology; NIH-funded data science infrastructure	https://www.unr.edu/bioinformatics/contact/juli-petereit
Computing	Jun Zhang	Assistant Professor of Mechanical Engineering	2018	UNR College of Engineering	Control theory, robotics, smart materials, artificial muscles, soft robots, assistive biomimetic systems	200	3384	Robotic Artificial Muscles: Current Progress and Future Perspectives Compliant, Large-Strain, and Self-Sensing Twisted String Actuators Three-Dimensional Hysteresis Modeling of Robotic Artificial Muscles	ME; IEEE; ASME	Smart materials testing, soft robotics fabrication, actuator testing		https://sites.google.com/view/unrsml/home
Computing	M. Sami Fadali	Professor of Electrical & Biomedical Engineering	1985	UNR College of Engineering	Fuzzy logic stability and control, state estimation, fault detection, power systems, autonomous systems	26	115	Digital Control Engineering: Analysis and Design (textbook, co-authored with Antonio Visioli) Random Signals and Applied Kalman Filtering (textbook) Research on distributed control for multi-agent systems	EBME; IEEE	Control systems lab, simulation tools		https://www.unr.edu/ebme/people/sami-fadali
Computing	Mehdi Etezadi-Amoli	Professor of Electrical & Biomedical Engineering	1983	UNR College of Engineering	Power systems analysis, protection, renewable energy, smart grids, electromagnetic transient simulation	2	1	Over 60 technical papers on power systems planning, protection, transient analysis, and renewable energy Research on power system protection and distribution systems Work on renewable energy and smart grid applications	EBME; IEEE	Power system simulation equipment, SCADA systems		https://www.unr.edu/ebme/people/mehdi-etezadi-amoli
Computing	Parikshit Maini	Assistant Professor of Computer Science & Engineering		UNR College of Engineering	Computer science research	8	45		CSE			https://www.parikshitmaini.com/
Computing	Sergiu Dascalu	Professor of Computer Science & Engineering	2002	Dalhousie University, Canada (2001)	Software engineering, personalized learning, data science, robotics education	68	280	Student-Centered Personalized Learning Framework (FIE 2023); 50+ journal articles, 200+ conference papers; TDVR: Tower Defense in VR	Software Engineering Lab (SOELA); Software Systems Lab	Software development environments, VR systems, robotic kits		https://www.cse.unr.edu/~ecc-admin/index.php
Computing	Sesh Commuri	Professor of Electrical & Biomedical Engineering	2016	UNR College of Engineering	Reconfigurable systems, sensor networks, robotics, embedded controls, Industry 4.0	12	144	Over 120 peer-reviewed journal and conference articles Architecting Networked Engineered Systems: Manufacturing Systems Design for Industry 4.0 Research on intelligent control of off-road equipment	EBME; IEEE	Sensor network testbeds, embedded control systems, robotics platforms		https://www.unr.edu/ebme/people/sesh-commuri
Computing	Shamik Sengupta	Professor of Computer Science & Engineering		UNR College of Engineering	Cybersecurity, wireless networks, cognitive radio, spectrum management, network security	75	544	Cognitive radio and dynamic spectrum access research Cybersecurity and network security Wireless network optimization	CSE; IEEE	Network security testbed, SDR equipment		https://www.cse.unr.edu/~shamik/
Computing	Xingang Fu	Assistant Professor of Electrical & Biomedical Engineering	2016	UNR College of Engineering	Neural network control, embedded AI, heterogeneous computing, power electronics, solar integration	32	197	AI/machine learning for power electronics control Solar inverter efficiency and reliability enhancement Neural network-based power converter control	EBME; IEEE	FPGA/DSP systems, power electronics testing		https://sites.google.com/view/dr-xingang-fu/contact
Computing	Yantao Shen	Professor of Electrical & Biomedical Engineering		UNR College of Engineering	Robotics, instrumentation and automation, sensors and actuators, electro-braille for blind assistance	42	278	Over 180 research articles in robotics, instrumentation, and automation Adaptive Electro-Braille project for visually impaired assistance Multiple Best Paper Awards at IEEE/ASME conferences	EBME; IEEE; ASME	Bioinstrumentation lab, robotics systems, sensor fabrication		https://www.unr.edu/ebme/people/yantao-shen
Physics	Aaron Covington	Professor of Physics; Assistant Research			Chemical physics, spectroscopy, photoionization, atomic physics, high-pressure spectroscopy	3	6	David A. Shirley Award at Advanced Light Source (2002); Spectroscopic characterization and photoionization research; Condensed matter spectroscopy (Raman, FTIR)	Chemical Physics Group	Raman spectrometers, FTIR, synchrotron beamlines, high-pressure cells, optical systems		https://www.unr.edu/physics/aaron-covington

Physics	Aditya Nair	Assistant Professor of Mechanical Engineering	2020	UNR College of Engineering	Computational fluid dynamics, unsteady aerodynamics, high-performance computing, data science, control	5	18	Phase-based analysis and control of low-Reynolds-number aeroelastic flows Data-Driven Unsteady Aeroelastic Modeling for Control Computational fluid dynamics investigations of vertical landing configurations	ME; Aerospace program	HPC clusters, CFD simulation tools		https://www.unr.edu/me/people/aditya-nair
Physics	Ahmad M. Itani	Foundation Professor of Civil & Environmental Engineering		UNR College of Engineering	Bridge engineering, seismic behavior of steel bridges, accelerated bridge construction			Seismic behavior and design of steel bridge systems Accelerated bridge construction research Structural dynamics of bridges under seismic loading	CEE Chair; Assoc VP Research	Shake table, structural testing systems		
Physics	Ang Liu	Assistant Professor of Mining & Metallurgical Engineering	2025	UNR Mackay School	Geomechanics, rock mechanics, mine ventilation, geo-energy, coupled processes in multiphase flow	6	246	Coupled processes in geomechanics and multiphase flow Rock mechanics and mine ventilation research Geo-energy systems modeling	Mining	Rock testing equipment, geomechanics lab		
Physics	Bernhard Bach	Professor of Physics; Associate Research	2007	College of William and Mary (1995)	Optics, diffractive/space flight optics, mathematical physics, HED diagnostics			A Student's Guide to Infinite Series and Sequences (Cambridge UP); Gamma-ray, X-ray and UV spectroscopy; Diffractive optics and instrumentation	Zebra Pulsed Power Lab	X-ray/UV spectrometers, diffractive optics, pulsed power diagnostics	Laser.	
Physics	Brad Parker Wham	Assistant Professor of Civil & Environmental Engineering		UNR College of Engineering	Infrastructure engineering, lifeline engineering, pipeline response to earthquake, wildfire, construction hazards	54	263	Pipeline response to earthquake hazards Wildfire and construction-induced hazards on critical infrastructure Lifeline engineering and infrastructure resilience	CEE	Structural testing, geotechnical testing equipment		https://www.bradwham.com/
Physics	Cahit Evrensel	Associate Professor of Mechanical Engineering		UNR College of Engineering	Thermal science, nuclear packaging, fluid mechanics, heat transfer	5	24	Nuclear packaging and thermal analysis Fluid mechanics and heat transfer research Thermal science applications in engineering	ME	Thermal testing equipment, fluid mechanics lab		
Physics	Christie Rowe	Professor & Director of Nevada Seismological Lab	2024		Earthquake geology, fault mechanics, rock alteration, field-based seismology			Rock record of earthquakes and geological fault characteristics Field-based approaches to understanding earthquake source mechanics Community outreach and earthquake preparedness in Nevada	Nevada Seismological Laboratory Director	Field mapping, structural geology tools, seismic monitoring		https://www.unr.edu/nevada-seismo-lab/christie-rowe
Physics	Dani Or	Nevada Engineering Distinguished Professor of Civil & Environmental Engineering	2023	UNR College of Engineering	Soil physics, hydrogeology, porous media, biophysical processes, landslides	132	1920	Soil physics and hydrology research (ETH Zurich) Porous media and biophysical processes Landslide mechanics and prediction	CEE; NAE member	Soil characterization equipment, hydrology instruments	National Academy of Engineering member; world-class soil physicist from ETH Zurich; brings exceptional prestige	
Physics	Daniel Trugman	Associate Professor of Physics; of Geophysics	2022		Seismology, earthquake rupture, ML in seismology, earthquake nucleation, induced seismicity	92	928	GrowClust: software for optimizing earthquake hypocenter locations Machine learning approaches to earthquake source characterization Earthquake early warning methodologies	Geological Sciences; Nevada Seismological Lab	Seismic waveform analysis, GrowClust software, ML platforms	ML in seismology; new faculty (2022); potentially accessible	https://www.unr.edu/nevada-seismo-lab/daniel-trugman
Physics	Elnaz Esmaeilzadeh Seylabi	Assistant Professor of Civil & Environmental Engineering	2019	UNR College of Engineering	Seismic engineering, machine learning, physics-informed ML, geostructural systems	29	100	Physics-Informed Machine Learning Framework for seismic engineering Geostructural systems modeling under seismic loading ML applications in earthquake engineering	CEE	Seismic sensors, HPC for ML, geotechnical testing		https://sites.google.com/view/elnazseylabi/home
Physics	Facundo Booman	Assistant Professor of Mechanical Engineering	2025	UNR College of Engineering	Turbulent boundary layers, Lagrangian dynamics of inertial particles, particle-laden flows, microgravity fluids	4	35	Droplet jump from a particle bed (Soft Matter, 2024) Tuning particle settling in fluids with magnetic fields (Experiments in Fluids, 2024) Path instabilities in settling of single spheres (Intl J Multiphase Flows, 2024)	ME; Aerospace	High-speed imaging, particle tracking velocimetry, acoustic measurement		
Physics	Floriana Petrone	Associate Professor of Civil & Environmental Engineering	2019	UNR College of Engineering	Earthquake engineering, performance-based seismic design, advanced earthquake simulation, ML	17	92	Performance-based seismic design research Advanced earthquake simulation and machine learning applications Structural dynamics under seismic loading	CEE; EERI	Shake table, seismic simulation software, structural sensors		https://www.unr.edu/cee/people/floriana-petrone
Physics	George Danko	Professor of Mining & Metallurgical Engineering	1995	UNR Mackay School	Mining ventilation, fluid flow modeling, MULTIFLUX software development, underground systems	5	10	Ventilation and climate simulation with the Multiflux code (2011) Underground ventilation system optimization Fluid flow modeling in mining environments	Mining; Graduate Director	Ventilation simulation, MULTIFLUX software		
Physics	Hamed Ebrahimian	Associate Professor of Civil & Environmental Engineering		UNR College of Engineering	Structural engineering, earthquake engineering, data assimilation, nonlinear structural modeling	57	657	Data assimilation for structural health monitoring Nonlinear structural modeling under seismic loading Earthquake engineering and structural dynamics	CEE	Structural testing, data acquisition systems		https://www.hamedebrahimian.com/
Physics	Hiroshi Sawada	Associate Professor of Physics	2012	Osaka University (est.)	HED physics, short-pulse laser-matter interactions, laser-driven particles, ICF, x-ray diagnostics	3	125	Ultrafast plasma dynamics in laser-irradiated nanowire arrays (Sci Reports); Magnetized fast isochoric laser heating (Nature Comms 2018); Ultrafast 2D imaging of fast electron transport (Rev Sci Instrum 2023)	HED Physics Group; Zebra Pulsed Power Lab	Zebra pulsed power machine, high-power lasers, x-ray cameras, plasma diagnostics	Assessible.	https://sawadalab.netlify.app/
Physics	James Hanna	Associate Professor of Mechanical Engineering		UNR College of Engineering	Classical mechanics, nonlinear dynamics, elastic structures, soft matter physics	18	63	Nonlinear dynamics of elastic rods and filaments Classical mechanics of soft structures Elastic instabilities and buckling phenomena	ME	High-speed imaging, mechanical testing	Interesting research.	https://cmag.neocities.org/
Physics	Jeongwon Park	Associate Professor of Electrical & Biomedical Engineering	2019	UNR College of Engineering	IoT sensors, nanotechnology-enabled flexible electronics, nanoelectronics, semiconductors, smart manufacturing	98	1597	Over 84 peer-reviewed journal articles on IoT and nanoelectronics Flexible hybrid electronics and IoT sensor applications Additively manufactured electronics and smart manufacturing	EBME; IEEE	Additive manufacturing, IoT sensor fabrication, flexible electronics testing		https://www.unr.edu/ebme/people/jeongwon-park

Physics	John Muntean	Associate Professor of Geological Sciences & Engineering	2005	UNR Mackay School	Economic geology, epithermal and Carlin-type gold deposits, ore formation	4	2	Carlin-type gold deposit formation research Epithermal and porphyry copper-gold systems Ore genesis and mineral exploration	Geological Sciences; Roberts Center	Fluid inclusion analysis, SEM-EDS, geochemical tools		https://nbmg.unr.edu/staff/johnmuntean.html
Physics	Jonathan Weinstein	Associate Professor of Physics; JT Endowed Prof.	2006	Harvard University (1998)	Atomic/molecular physics, cryogenic buffer-gas cooling, matrix isolation, quantum sensing, dark matter	5	37	Optical/spin-coherence properties of Rb atoms in solid neon (Phys Rev A 2021); High-purity solid parahydrogen (Rev Sci Instrum 2021); Spin coherence of alkali-metal atoms in solid parahydrogen (Phys Rev A 2019)	Weinstein Lab	Cryostats, laser systems, optical tables, vacuum chambers, magnetic shielding, spectrometers	Quantum sensing; near-absolute-zero spectroscopy; Harvard-trained; bridges physics+astronomy. Not active.	https://www.weinsteinlab.org/
Physics	Joshua Williams	Associate Professor of Physics; JT Endowed Prof.	2015	Auburn University (2012)	Atomic/molecular physics, photoelectron angular distributions, molecular ionization/dissociation	42	197	Imaging Polyatomic Molecules in 3D Using Molecular Frame PEADs (J Phys B); Interatomic Coulombic decay in small helium clusters; Molecular frame photoelectron angular distributions	Molecular Dynamics Group	Velocity map imaging, synchrotron beamlines, vacuum chambers, electron spectrometers		
Physics	Keri L. Ryan	Professor & Chair of Civil & Environmental Engineering		UNR College of Engineering	Seismic isolation, earthquake and structural engineering, shake table testing	16	153	International shake table testing program (US-Japan collaboration) Seismic isolation systems research Earthquake engineering and structural dynamics	CEE Chair; E.W. McKenzie Endowed	Shake table, seismic isolation test rigs, structural testing		https://www.unr.edu/cee/people/keri-ryan
Physics	Kristin Lewis	Teaching Assoc. Prof. & Associate Chair	2015	University of Nevada, Reno (2007)	Atmospheric science, physics education			Physics education and atmospheric science research; JEDI committee co-chair; Teaching innovation		Teaching lab equipment, atmospheric sensors		
Physics	Lei Cao	Associate Professor of Mechanical Engineering	2016	UNR College of Engineering	Materials science, phase-field modeling, microstructure evolution, titanium alloys, mechanical properties	31	3296	Twin boundary engineering through cyclic cross-phase-boundary thermomechanical processing Phase-Field Simulations of alpha Colony Formation in Titanium Alloys Understanding microstructure evolution in metallic alloys	ME; ASME	LAMMPS molecular dynamics, phase-field simulation, SEM	Receptive.	https://leicao-research.github.io/
Physics	Matteo Aureli	Associate Professor of Mechanical Engineering	2013	UNR College of Engineering	Fluid-structure interaction, vibration and dynamics, smart materials, multiphysics modeling	16	79	Finite Amplitude Underwater Flexural Vibrations of Cantilevers Harmonic oscillations of a lamina in viscous fluid near a solid surface Shape-morphing cantilever oscillations in multiphysics systems	ME Graduate Director; Multiphysics Lab	5W CW laser, high-speed camera (Phantom Miro), vibration shaker/controller		https://matteoaureli.com/lab.html
Physics	Mohamed Ben Idris	Assistant Professor of Electrical & Biomedical Engineering		UNR College of Engineering	Sustainable energy systems, power systems, microgrids, renewable energy integration	10	180	Sustainable energy system design and optimization Microgrid control and renewable integration Power system resilience research	EBME	Power system simulation, microgrid testbed		
Physics	Mohamed Moustafa	Professor of Civil & Environmental Engineering		UNR College of Engineering	Structural and earthquake engineering, seismic response, structural health monitoring	7	94	Seismic response of structures and bridges Structural health monitoring systems Earthquake engineering and resilient design	CEE	Shake table, structural testing systems, accelerometers		https://www.unr.edu/cee/people/mohamed-moustafa
Physics	Neil Lareau	Assistant Professor of Physics; (Physics & Atmos. Sci.)	2018	University of Utah (2014)	Wildfire plume dynamics, atmospheric dynamics, fire weather, boundary layer meteorology	39	335	Tracking Wildfires with Weather Radars (JGR Atmospheres 2022); Pyrotornadogenesis and wildfire plume behavior; Pyrocumulus development and boundary layer dynamics	Atmospheric Dynamics & Wildfire Group	Weather radars, radiosondes, LiDAR (atmospheric), weather stations, satellite data systems	Wildfire plume dynamics; part of $20M NSF project; atmospheric physics hands-on	https://www.unr.edu/physics/neil-lareau
Physics	Patrick Arnott	Clemons-Magee Endowed Prof.; Assoc. Prof.	2005	Washington State University (1988)	Atmospheric optics, aerosol physics/chemistry, instrumentation, radiative forcing, photoacoustic spectroscopy	19	205	Reno Aerosol Optics Study (RAOS); Photoacoustic spectroscopy instrumentation development; Aerosol optical properties and radiative forcing	LASSO (DRI); Atmospheric Sciences Program	Photoacoustic spectrometers, aerosol sampling instruments, nephelometers, optical benches		
Physics	Paul Neill	Professor of Physics; & Department Chair	2004	Queen's University Belfast (1984)	Particle/photon spectroscopy, atomic/molecular physics, astrophysical plasmas, HED diagnostics	31	527	Critical test of simulations of charge-exchange-induced x-ray emission (ApJ Letters); High-density plasma diagnostic spectroscopy; Laboratory and astrophysical plasma research	HED Plasma Research Group	X-ray spectrometers, plasma diagnostic instruments, vacuum chambers		https://www.unr.edu/physics/people/paul-neill
Physics	Petros Voulgaris	Professor & Chair of Mechanical Engineering	2020	UNR College of Engineering	Control systems, autonomous systems, multi-agent systems, robust and optimal control, aerospace engineering	65	166	Safe Velocity Regulation of Self-Driving Vehicles using Simplex Architecture Robust Vehicle Lane Keeping Control with Networked Proactive Adaptation Control of autonomous aerial vehicles and multi-agent systems	ME Chair; Founding Aerospace Program Director; UIUC (former)	Autonomous vehicle testbed, control systems lab		https://petros-lab-unr.github.io/
Physics	Poria Fajri	Associate Professor of Electrical & Biomedical Engineering	2015	UNR College of Engineering	Power electronics, electric and hybrid electric vehicles, motor drives, energy management, mechatronics	37	126	Frequency and voltage regulation of distribution grids using EV batteries Cyber-physical energy systems testbed development Power electronics and electric vehicle control	EBME; IEEE	Electric motor/dynamometer test bench, power electronics lab		
Physics	Raj R. Kallu	Associate Professor of Mining & Metallurgical Engineering		UNR Mackay School	Geomechanics, underground excavation design, rock fragmentation, numerical modeling of rock masses			Numerical modeling of complex rock masses Underground excavation design optimization Rock fragmentation mechanics	Mining	Rock mechanics testing, numerical modeling software		
Physics	Roberto Mancini	Professor of Physics; McMinn Endowed Research Prof.	1993	University of Buenos Aires (1983)	Atomic/radiation physics of HED plasmas, x-ray spectroscopy, Stark-broadened lines, radiation transport	35	128	Spectroscopic Determination of Dynamic Plasma Gradients (PRL 2002); Time-Resolved Spectroscopic Measurements in Microballoons (PRL 1989); Escape Factors for Stark-Broadened Line Profiles (J Phys B)	Plasma Spectroscopy & Atomic Physics Group	X-ray spectrometers, HED plasma facilities, radiation diagnostics, vacuum systems		https://www.unr.edu/physics/roberto-mancini
Physics	Ryan Tung	Associate Professor of Mechanical Engineering		UNR College of Engineering	Fluid mechanics, aerodynamics, aeroacoustics, dynamics, contact resonance AFM	6	11	Contact resonance atomic force microscopy research Aeroacoustics and aerodynamics studies Fluid mechanics and dynamics investigations	ME; ASME	Atomic force microscopy, wind tunnel, aeroacoustic measurement		

Physics	Simon Jowitt	Associate Professor of Geological Sciences & Engineering		UNR Mackay School	Economic geology, mineral resources, critical minerals, exploration geology	65	688	Critical mineral resource assessment Economic geology and mineral exploration Ore deposit formation and characterization	Geological Sciences; NV Bureau of Mines; Roberts Center Director	XRF, electron microprobe, petrographic microscopy		https://nbmg.unr.edu/staff/Jowitt.html
Physics	Steven Wesnousky	Foundation Professor of Geological Sciences	1989		Seismology, neotectonics, paleoseismology, earthquake mechanics, fault systems	8	54	Neotectonic framework and earthquake hazard of the western US Mechanics of earthquake rupture and seismic hazard quantification Paleoseismology and deformation of Earth's crust	Center for Neotectonic Studies; Nevada Seismological Lab	Seismic networks, paleoseismic analysis, tomographic imaging		http://neotectonics.seismo.unr.edu/CNS_Director.
Physics	Thomas White	Clemons-Magee Endowed Prof.; Assoc. Prof.	2017	University of Oxford (2015)	HED physics, laboratory astrophysics, extreme states of matter, laser-plasma interactions	58	1441	Heating gold to 19,000K without losing crystalline structure (Nature); Fourier heating in HED plasmas (Nature Comms); NSF CAREER; uses NIF, LCLS, European XFEL	High Energy Density Lab (HEDL); Lab Astrophysics Group	High-power lasers, X-ray free-electron lasers (XFEL), NIF access, plasma diagnostics, shock drivers	World-class pulsed power facility (Zebra); unique in US; Physics olympiad alignment. Active.	https://www.thomasgwhite.com/
Physics	Wendy Calvin	Foundation Professor & Chair of Geological Sciences	2001		Planetary science, Mars geology, mineral spectroscopy, surface composition	49	115	Mars surface mineralogy from orbital spectroscopy data Fifteen years of Mars rover science team contributions Layered deposits of Mars polar ice caps as climate records	Mars Exploration Rover science team; Mars Reconnaissance Orbiter	Spectroscopy, Mars rover data, remote sensing, mineralogical references	Mars rover scientist; planetary science; spectroscopy alignment with SIL; Astronomy bridge. Active.	
Physics	Xiyue Zhang	Assistant Professor & Associate Director of Atmospheric Sciences		UNR College of Science	Atmospheric sciences, climate dynamics, atmospheric physics	2	415	Climate dynamics and atmospheric physics research Atmospheric science observations and modeling Regional climate variability	Atmospheric Sciences; Physics	Weather stations, remote sensing, atmospheric modeling	Active	https://sites.google.com/view/xiyuezhang/home
Physics	Yafis Barlas	Assistant Prof. & Graduate Program Director	2019	UT Austin (2008)	Condensed matter theory, topological materials, 2D crystals, quantum geometry, superconductivity	18	63	Superfluid weight cross-over in singular flat bands (PNAS 2024); Counter-propagating Fractional Hall states in Dirac semi-metals (PRL 2018); Quantum parity Hall effect in trilayer graphene (PNAS 2019)	Quantum Materials Theory Group	HPC clusters, computational physics software (DFT, tight-binding codes)		https://www.unr.edu/physics/yafis-barlas
Physics	Yan Wang	Associate Professor of Mechanical Engineering	2016	UNR College of Engineering	Thermal science, heat transfer, phonon transport, nanoscale thermal transport, laser manufacturing	200	5296	Selective Phonon Stimulation Mechanism to Tune Thermal Transport Phonon Lateral Confinement Enabling Thermal Rectification in Asymmetric Nanostructures First principles calculations of lattice thermal conductivity of metals	ME; ASME	Laser manufacturing systems, computational thermal modeling		https://med.unr.edu/yan
Astronomy	Kelly Kosmo O'Neil	Teaching Assistant Professor of Physics & Astronomy	2023	UCLA (2023)	Galactic center dynamics, supermassive black holes, orbital dynamics, exoplanets	17	83	Relativistic redshift of S0-2 orbiting Galactic center SMBH (Science 2019, co-author); Primordial black holes (2023-2025); Orbital eccentricities of directly imaged companions	UCLA Galactic Center Group (affiliated)	Telescope access, adaptive optics data, astrometry software, spectroscopic data		
Astronomy	Melodi Rodrigue	Professor of Physics & Astronomy; Associate Teaching	2006	University of Nevada, Reno (1998)	Interacting galaxies, galaxy mergers, science education, brown dwarfs, exoplanets			Multi-Wavelength Investigation of Merging Galaxy HCG 95c (AJ 1995); Classroom technology and astronomy education; Great Basin Observatory undergraduate research	Great Basin Observatory; Fleischmann Planetarium (affiliated)	Telescopes, planetarium systems, spectroscopic data, HST archival data		
Astronomy	Richard M. Plotkin	Associate Professor of Physics & Astronomy; Acting Physics Chair	2011	University of Washington (2009)	Black hole accretion, multiwavelength X-ray astronomy, relativistic jets, AGN, quasars	2	186	Photometry+: Pipeline for Great Basin Observatory (A&C 2022); Radio Jets from Quiescent BH XRBs (MNRAS 2021); Decay of V404 Cygni: Disk-Jet Coupling (ApJ 2017)	UNR Black Holes Group; Great Basin Observatory	Telescopes (GBO 0.7m, MacLean 22"), CCD cameras, photometry pipelines, X-ray satellite data	Great Basin Observatory PI; direct astronomy olympiad alignment; accessible	https://www.unr.edu/physics/richard-plotkin
Chemistry	Ana de Bettencourt-Dias	Professor of Chemistry; & Chair; Foundation Prof.	2007	University of Cologne, Germany (1997)	Luminescent lanthanide complexes, inorganic chemistry, f-element chemistry, energy/diagnostics	55	134	Cyclic peptides for lanthanide binding (2025); Anharmonically Coupled Vibrational Scaffolds for Energy Dissipation (2025); Pairing Global Optimization with EXAFS to Characterize Lanthanide Structure (2024)	de Bettencourt-Dias Research Group	Fluorescence spectrometers, UV-Vis, X-ray crystallography, glovebox, NMR, CD spectrometer	Active	https://www.unr.edu/chemistry/ana-de-bettencourt-
Chemistry	Behrooz Abbasi	Associate Professor of Mining & Metallurgical Engineering		UNR Mackay School	Rare earth mineral recovery, flotation optimization, comminution optimization, tailing remediation			Rare earth mineral recovery through flotation optimization Comminution optimization research Tailing remediation and environmental mining	Mining; Graduate Director	Flotation cells, mineral processing equipment		https://www.unr.edu/mining/behrooz-abbasi
Chemistry	Benjamin T. King	Professor of Chemistry; Fuson Endowed		University of Colorado (2000)	Carborane chemistry, fluorine chemistry, weakly coordinating anions, macromolecular architecture	20	252	Dodecamethylcarba-closo-dodecaboranyl (JACS); Scholl reaction and 2D polymer research; Computational and X-ray crystallography	King Group	X-ray crystallography, NMR, glovebox (air-free), mass spectrometry, computational clusters	Active	https://kingchem.org/
Chemistry	Brandon Weeks	Professor of Chemical & Materials Engineering		UNR College of Engineering	Explosives safety, energetic materials, thermal sensitivity, sublimation kinetics	1	27	Effect of Porphyrin Doping on Thermodynamic Parameters of PETN Single Crystals Sublimation Kinetics and Diffusion Coefficients of TNT, PETN, and RDX in Air by TGA Device for Testing Thermal Impact Sensitivity of High Explosives	CME; Assoc. Dean Research & Graduate Affairs	TGA, calorimetry, thermal sensitivity testing		https://www.unr.edu/cme/people
Chemistry	Brian J. Frost	Assoc. Prof.; Asst. Dean, UNR Lake Tahoe	2002	Texas A&M University (1999)	Inorganic/organometallic chemistry, aqueous/biphasic catalysis, transition metal complexes	3	2	Inorganic/organometallic complex synthesis for catalysis; Green chemistry and aqueous-phase catalysis; NMR, UV-vis, mass spec, IR, X-ray crystallography methods	Frost Research Group	NMR, UV-Vis, FTIR, mass spectrometry, X-ray crystallography, Schlenk lines		https://www.unr.edu/chemistry/people/brian-frost
Chemistry	Carl Nesbitt	Associate Professor of Mining & Metallurgical Engineering		UNR Mackay School	Extractive metallurgy, mineral processing, carbon recovery, mercury removal, autoclave processes	3	9	Carbon recovery and mercury removal in mining Autoclave processes for mineral extraction Sustainability in extractive metallurgy	Mining; Newmont Endowed Chair	Autoclave systems, mineral processing lab		https://mnesbit7.wixsite.com/nesbitlab
Chemistry	Charles J. Coronella	Professor of Chemical & Materials Engineering	1993	UNR College of Engineering	Waste to energy conversion, biomass pretreatment, hydrothermal carbonization, chemical looping combustion	16	183	Near-Isothermal Hydrothermal Carbonization of Glucose (2025) Wet Air Oxidation of HTC Process Liquid (ACS Sustainable Chemistry & Engineering) Catalytic Hydrothermal Carbonization of Animal Manure characterization	CME; Fulbright	Hydrothermal carbonization reactors, biomass processing, TGA		https://packpages.unr.edu/coronella/

Chemistry	Christopher S. Jeffrey	Professor of Chemistry; Director Hitchcock Center	2010	University of Minnesota (2007)	Chemical ecology, organic synthesis, natural products, alkene functionalization, C-H activation	24	124	Structural/Compositional Dimensions of Phytochemical Diversity in Piper (J Ecol 2022); Natural products from Nevada and tropical plants; Siebens Endowed Chair (2025)	Jeffrey Research Group; Hitchcock Center for Chemical Ecology	LC-MS, GC-MS, NMR, HPLC, rotary evaporators, column chromatography, fume hoods	Hitchcock Center director; chemistry+biology bridge; uses SIL instruments	https://www.unr.edu/chemistry/christopher-jeffrey
Chemistry	David C. Cantu	Associate Professor of Chemical & Materials Engineering	2017	UNR College of Engineering	Molecular simulation, computational chemistry, rare earth element separations, enzymes, bioinformatics	39	418	Computational Prediction of All Lanthanide Aqua Ion Acidity Constants (Inorganic Chemistry) Chelator-Assisted Precipitation-Based Separation of Rare Earth Elements Nd and Dy Rare-Earth Separation Using Bacteria (ACS Environ Sci Technol	CME; ACS	Molecular simulation software, computational chemistry tools		https://www.unr.edu/cme/people/david-cantu
Chemistry	David Hanigan	Associate Professor of Civil & Environmental Engineering		UNR College of Engineering	Environmental engineering, water treatment, water quality, PFAS, disinfection byproducts, N-nitrosamines	31	632	Disinfection byproducts research in water treatment PFAS contamination and remediation studies N-nitrosamine formation and control in water treatment	CEE	LC-MS, GC-MS, water treatment pilot systems		https://www.dhanigan.com/
Chemistry	David M. Leitner	Professor of Chemistry; Foundation Prof.; Fuson Endowed	2000	University of Chicago	Theoretical/computational chemistry, energy flow in molecules, protein dynamics, nanoscale thermal transport	25	179	Vibrational energy flow in molecules on plasmonic nanoparticles (2024); Locating dynamic contributions to allostery via vibrational energy transfer (2023); Hydrogen bond dynamics in disordered proteins (2023)	Leitner Research Group	HPC clusters, molecular dynamics software (GROMACS, NAMD), spectral analysis tools	Active	https://leitnergroup.wixsite.com/dmlpage/david-leitner-1
Chemistry	Dev Chidambaram	Professor of Chemical & Materials Engineering	2009	UNR College of Engineering	Corrosion science, electrochemistry, molten salt research, high-temperature materials, environmental remediation	54	180	Duplex Mechanism for Chromate-Hydrated Oxide Interaction on Aluminum Alloys (J Electrochem Soc) Ti-48Al-2Cr-2Nb alloy in molten salts at 850C for aerospace Molybdate-based coating for depleted uranium protection	CME Assoc. Chair; Electrochemical Society	SEM, XRD, XPS, synchrotron spectroscopy, Raman spectroscopy	Direct SIL alignment: uses SEM, XRD, XPS, Raman; corrosion and electrochemistry map to Chemistry; $20M funding leader	https://www.unr.edu/cme/people/dev-chidambaram
Chemistry	Ehsan Vahidi	Assistant Professor of Mining & Metallurgical Engineering	2020	UNR Mackay School	Extractive metallurgy, sustainability, carbon footprinting, phytomining, corrosion resistance	48	668	Sustainability and carbon footprinting in metallurgical processes Phytomining research for metal extraction Corrosion resistance and metallurgical sustainability	Mining; SME; MIT (former)	Metallurgical processing, materials characterization		https://www.unr.edu/mining/ehsan-vahidi
Chemistry	Ilana J.P. Molesky	Assistant Professor of Chemistry	2025	UC Berkeley (2022)	Ultrafast spectroscopy, solid-state materials, light-matter interaction, electronic dynamics, XFELs	5	40	Carrier Cooling Bottleneck in Si Nanoparticles by XUV Transient Absorption; Stanford/SLAC postdoc on ultrafast dynamics; MIT BS; emerging materials research	Molesky Lab	Ultrafast laser systems, optical tables, XUV sources, vacuum chambers, cryostats		https://www.unr.edu/chemistry/ilana-molesky
Chemistry	Krista Carlson	Associate Professor of Chemical & Materials Engineering	2021	UNR College of Engineering	Sorbent synthesis, nuclear waste forms, high-temperature metallic glasses, mercury abatement	41	276	Mercury Abatement Research using carbon foam-based sorbents Carbon Foam-Based Sorbents for Gaseous Mercury and Iodine Capture Metallic Glass Development for Nuclear Waste Applications	CME	Electron microscopy, ceramic processing, high-temp testing, TGA/DSC		https://www.unr.edu/cme/people/krista-carlson
Chemistry	Laina M. Geary	Associate Professor of Chemistry	2014	University of Manitoba, Canada (2010)	Palladium catalysis, C-H functionalization, C-C and C-X bond formation, organometallic chemistry	5	3	Pd-catalyzed cross-coupling and C-H functionalization; Trichloroethylene derivative synthesis; Benzofuran and indole synthesis	Geary Lab	GC-MS, LC-MS, NMR, column chromatography, Schlenk lines, glovebox	Hands-on organic synthesis; uses SIL GC-MS/LC-MS for characterization	https://lgeary.wixsite.com/gearylab
Chemistry	Leslie T. Mushongera	Assistant Professor of Chemical & Materials Engineering		UNR College of Engineering	Computational materials science, phase-field modeling, crystal plasticity, superalloy design, fatigue prognosis	25	187	Understanding and Design of Metallic Alloys Guided by Phase-Field Simulations (npj Comp Materials) Phase-field Framework with Constraints for Ductile Fracture in Polycrystals (npj Comp Materials) Phase-field Simulations of Pearlitic Divergence in Fe-C-Mn Steels	CME	Crystal plasticity FEM, phase-field modeling software		https://www.lesliemushongera.com/
Chemistry	Louisa Hope-Weeks	Professor of Chemistry; Dean of College of Science	2023	University of Cambridge (2000)	Sol-gel chemistry, porous materials, heterogeneous catalysis, aerogels, energetic materials	3	21	Ferrite and photocatalyst materials research; Composite transition metal oxide aerogels; Porous materials and sol-gel synthesis	Hope-Weeks Research Group	TGA/DSC, powder XRD, BET surface area, SEM/TEM access, furnaces, sol-gel equipment		https://www.unr.edu/science/people/admin-faculty-
Chemistry	Manoranjan Misra	Professor of Chemical & Materials Engineering		UNR College of Engineering	Advanced materials for medical diagnostics, materials processing, electrochemistry, nanomaterials			Advanced materials for medical diagnostics applications Electrochemical processing and nanomaterial synthesis Materials engineering for biomedical applications	CME	Electrochemistry equipment, materials synthesis, SEM		https://www.unr.edu/cme/people
Chemistry	Mario Alpuche-Aviles	Associate Professor of Chemistry	2009	Mississippi State University (2005)	Electrochemistry, semiconductor nanoparticles, photoelectrochemistry, electrocatalysis, energy conversion	17	33	Single Nanoparticle Measurements in Electrocatalysis (Electrochimica Acta 2024); NSF CAREER ($650K, 2013); Semiconductor nanoparticle electrochemistry	Alpuche-Aviles Lab	Potentiostats, electrochemical cells, UV-Vis, photocurrent measurement, nanoparticle synthesis		https://www.unr.edu/chemistry/people/mario-alpuche
Chemistry	Matthew J. Tucker	Professor of Chemistry; Graduate Director	2011	University of Pennsylvania (2006)	Ultrafast spectroscopy, 2D IR spectroscopy, protein folding/binding, biophysics	4	67	Unraveling Hydration Shell Dynamics via 2D IR Spectroscopy (JACS 2024); The Long and Short of Coupling/Uncoupling via 2D IR (J Phys Chem B 2024); Amide I 2D IR of Proteins (Acc Chem Res)	Tucker Research Group; Hitchcock Center (co-director)	Femtosecond lasers, FTIR spectrometers, 2D IR setup, optical tables, cryostats	FTIR/spectroscopy directly uses SIL instruments; biophysics bridge. Toys list.	https://www.tuckergroupunr.org/
Chemistry	Ming Chen	Assistant Professor of Chemical & Materials Engineering		UNR College of Engineering	Advanced alloys, laser additive manufacturing, metallic materials, high-entropy alloys	130	5203	Laser additive manufacturing of metallic alloys High-entropy alloy design and characterization Advanced materials processing	CME	Laser AM systems, SEM, XRD, mechanical testing		https://www.unr.edu/cme/people/ming-chen
Chemistry	Pengbo Chu	Associate Professor of Mining & Metallurgical Engineering	2020	UNR Mackay School	Mineral processing, flotation science and engineering	32	445	Flotation science and engineering research Mineral processing optimization First MME faculty to receive NASA grant	Mining; NV Gold Mines Endowed	Flotation equipment, mineral characterization		https://sites.google.com/view/chu-group/home
Chemistry	Pradeep Menezes	Associate Professor of Mechanical Engineering	2015	UNR College of Engineering	Tribology, surface engineering, advanced manufacturing, composite materials, friction and wear	141	3551	Tribology for Scientists and Engineers (book) Tribology of Additively Manufactured Materials (Elsevier) Tribology of MXene Materials: Advances, Challenges, and Future Directions	ME; ASME; founding editor Tribology and Interfaces (Elsevier)	Tribology testing, surface characterization, SEM, profilometry	16,000 citations with h-index 60; top 2% worldwide; surface characterization aligns with SIL Raman and FTIR	https://sites.google.com/view/pradeepmenezes/home

Chemistry	Ravi Subramanian	Associate Professor of Chemical & Materials Engineering		UNR College of Engineering	Nanostructured materials for solar energy, photoactive materials, photovoltaics, clean fuel production	23	607	Meeting the Clean Energy Demand: Nanostructure Architectures for Solar Energy Conversion (J Phys Chem C) Materials and Processes for Solar Fuel Production (Springer, editor) Nanostructured photoactive materials for environmental remediation	CME; Deputy Director Ozmen Institute; Fulbright	Materials synthesis, photovoltaic testing, UV-Vis spectroscopy		https://www.unr.edu/cme/people/ravi-subramanian
Chemistry	Sage Hiibel	Associate Professor of Chemical & Materials Engineering		UNR College of Engineering	Renewable energy systems, membrane separations, water treatment, biological remediation, nutrient recovery	14	156	Membrane bioreactor systems for water treatment and reuse Biological remediation and nutrient recovery research Renewable and sustainable energy systems	CME; Director NV Center for Water Resiliency	Membrane separation equipment, bioreactors, water treatment systems		https://www.unr.edu/cme/people/sage-hiibel
Chemistry	Sarah A. Cummings	Teaching Prof.; Director Teaching Excellence	2007	Columbia University	Chemistry education, sustainable chemistry, systems thinking, curriculum development			Integration of sustainable chemistry in organic courses; Active learning strategies; Multiple awards (Tibbitts, Bible, Regents)		Teaching lab equipment, curriculum development tools		
Chemistry	Sean M. Casey	Assoc. Prof.; Assoc. Dean, College of Science		University of Minnesota (1993)	Semiconductor materials, plasma-enhanced CVD, surface chemistry, reactive species			Growth mechanisms of semiconductor materials; Hyperthermal beam deposition; WiSE program and NevadaTeach	Casey Research Group	CVD reactors, mass-selected ion beams, vacuum systems, surface analysis (XPS/AES)		https://www.unr.edu/science/people/admin-faculty-
Chemistry	Thomas W. Bell	Professor of Chemistry	1995	University College London (1980)	Medicinal chemistry, organic synthesis, molecular modeling, drug design	9	45	Isoform selectivities of 4-hydroxycoumarin imines as myosin II inhibitors (J Med Chem 2023); 114 publications, 14 patents; Supramolecular chemistry and chemosensors	Bell Research Group	NMR, LC-MS, molecular modeling software, organic synthesis equipment, fume hoods		https://www.unr.edu/chemistry/thomas-bell
Chemistry	Victor R. Vasquez	Professor & Chair of Chemical & Materials Engineering	1999	UNR College of Engineering	Thermodynamics, chemical process modeling, atomistic modeling, rare earth elements, lithium supply chains	10	22	System and Method for Energy Production from Sludge (US Patent) Uncertainty analysis of chemical process design and simulation Accelerated reactive carbonation for mineral liberation and REE separation	CME Chair	Atomistic modeling, computational chemistry tools		https://www.viticor.org/
Chemistry	Vincent J. Catalano	Professor of Chemistry; Senior Vice Provost	1993	UC Davis (1991)	Inorganic chemistry, NHC ligands, coinage metal chemistry (Ag/Au/Cu), luminescence	23	72	Luminescent Mechanochromism in Au(I)-Cu(I) NHC Complex (Inorg Chem); N-heterocyclic carbene coordination chemistry; Coinage metal luminescence	Catalano Research Group	X-ray crystallography, NMR, glovebox, fluorescence, UV-Vis		https://www.unr.edu/chemistry/faculty
Chemistry	Yftah Tal-Gan	Professor of Chemistry	2014	Hebrew University of Jerusalem	Chemical biology, bacterial quorum sensing, peptide chemistry, anti-bacterial strategies	26	199	Designing Cyclic CSP-Based Quorum-Sensing Modulators (ACS Infect Dis 2024); Targeting Peptide-Based QS for Gram-Positive Infections (Peptide Sci 2023); NIH/NSF funded peptide research	Tal-Gan Research Group	Peptide synthesizer, HPLC, LC-MS, MALDI-TOF, microplate readers, cell culture	Chemical biology; peptide synthesis; uses SIL LC-MS and MALDI; open to undergrads	https://www.unr.edu/chemistry/people/yftah-tal-gan
Chemistry	Yu (Frank) Yang	Associate Professor of Civil & Environmental Engineering		UNR College of Engineering	Water science, environmental engineering, emerging contaminants, water treatment	18	27	Emerging contaminants in water systems research Environmental chemistry and water treatment Water quality monitoring and remediation	CEE	LC-MS, water quality analyzers, environmental sampling	5,700 citations; water science aligns with Chemistry; LC-MS use connects to SIL instruments	https://www.yanglab.space/
Biology	Albert Gonzales	Professor of Physiology & Cell Biology		UNR School of Medicine	Cardiovascular disease, neuroscience, cancer biology, vascular physiology	20	169	Cardiovascular disease mechanisms and vascular physiology Neurovascular coupling research Cancer biology and cell signaling	Physiology & Cell Biology	Confocal microscopy, calcium imaging, electrophysiology	Spans cardiovascular, neuro, and cancer biology; broad interdisciplinary research profile	https://www.gonzaleslab.com/
Biology	Alexander van der Linden	Professor of Biology; Assoc. ; Associate Dean			Sleep biology, aging, circadian rhythms, metabolic regulation, neural circuits	6	36	SIK3 pathway in metabolic regulation of sleep; Circular RNAs in aging; C. elegans model organism	van der Linden Lab	C. elegans culture, fluorescence microscopy, molecular biology (PCR, qPCR), behavioral assays		https://www.vanderlindenlab.com
Biology	Amilton de Mello	Associate Professor of Agriculture, Veterinary & Rangeland Sciences		UNR CABNR	Meat science, food safety, molecular genetics, livestock production	13	35	Meat quality and food safety research Molecular genetics in livestock production Food science and safety	AVRS; CABNR	Meat science lab, molecular biology tools, food safety testing		https://naes.unr.edu/de-mello/
Biology	Andreia Nunes	Research Assistant Professor of Pharmacology		UNR School of Medicine	Muscular dystrophies (FSHD, DMD), immune response, biomarker discovery	5	150	FSHD and DMD muscular dystrophy research Immune response in muscular dystrophies Biomarker discovery for neuromuscular diseases	Pharmacology (research-track)	Immunoassays, molecular biology, mouse models		https://med.unr.edu/people/andreia-nunes
Biology	Anne Leonard	Professor of Biology; & Department Chair	2012	UC Davis	Plant-pollinator interactions, bee behavior, pollination biology, animal communication	18	227	Bumblebees and Nevada wildflowers (NSF $579K); Pollinator cognition and floral signaling; Bee behavior and multi-modal signaling	Leonard Lab	Bee observation equipment, field stations, spectrophotometers, behavioral recording		http://www.anneleonard.com/
Biology	Bahram Parvin	Professor & Chair of Electrical & Biomedical Engineering	2014	UNR College of Engineering	Tumor microenvironment reprogramming, bioimaging, ML for medical imaging, high-throughput drug screening			Over 150 publications and 10 patents in bioimaging and cancer biology BioSig3D: first system for high content screening of 3D cell culture Machine learning technologies for bioimaging and drug screening	EBME Chair; Lawrence Berkeley Natl Lab (former)	Bioimaging systems, 3D cell culture, high-throughput screening	Highest-cited engineering faculty (13,700+); bioimaging + ML bridges Computing and Biology; direct SIL alignment with fluorescence and spectroscopy instruments	https://www.unr.edu/ebme/people/bahram-parvin
Biology	Ben Ma	Assistant Professor of Civil & Environmental Engineering		UNR College of Engineering	Environmental engineering, water treatment, pathogen control, UV disinfection, biofouling, water reuse	26	967	UV disinfection systems for water treatment Pathogen control in water reuse systems Biofouling mitigation research	CEE	UV disinfection equipment, microbiology lab, water quality analyzers	Active	https://www.enve-micro.com
Biology	Benjamin Sullivan	Associate Professor of Natural Resources & Environmental Science		UNR CABNR	Applied soil ecology, soil microbiology, microbiome, nutrient cycling	13	188	Soil microbiome and nutrient cycling research Applied soil ecology in arid environments Microbial ecology of soils	NRES; CABNR	Soil sampling, 16S sequencing, qPCR, soil chemistry analyzers		https://sullivanlab.weebly.com/

Biology	Brian Perrino	Professor of Physiology & Cell Biology		UNR School of Medicine	Smooth muscle biology, signal transduction, calcineurin signaling			Smooth muscle signal transduction research Calcineurin signaling in smooth muscle Molecular mechanisms of muscle contraction	Physiology & Cell Biology	Molecular biology tools, protein analysis		https://med.unr.edu/people/brian-perrino
Biology	Caroline Cobine	Associate Professor of Physiology & Cell Biology		UNR School of Medicine	Smooth muscle biology, lower esophageal sphincter function, GI motility	18	134	Lower esophageal sphincter smooth muscle function GI motility and smooth muscle biology Neural regulation of gastrointestinal tract	Physiology & Cell Biology	Electrophysiology, tissue imaging		https://med.unr.edu/people/caroline-cobine
Biology	Charis Himeda	Associate Professor of Pharmacology			FSHD, DUX4 epigenetic regulation, CRISPR inhibition, muscular dystrophy therapeutics	4	15	FSHD therapeutic strategies: what will it take to get to clinic? Targeted epigenetic repression by CRISPR/dSaCas9 suppresses DUX4 Novel drug targets and epigenetic mechanisms in FSHD	RENOGENYX; FSH Society Young Investigator	CRISPR systems, epigenetic analysis, molecular cloning		https://med.unr.edu/people/charis-himeda
Biology	Cherie Singer	Associate Professor of Pharmacology		U of Washington	Asthma, airway inflammation, microRNA biomarkers, neuropeptide Y, calcium channels	7	19	Gene network analysis for microRNA biomarkers in asthma α7 integrin overexpression in smooth muscle allergen response Estrogen neuroprotection and neuronal cell death	Respiratory Research; Associate Dean for Student Affairs	Gene expression analysis, cell culture, microRNA techniques		https://med.unr.edu/people/cherie-singer
Biology	Chris Feldman	Assoc. Prof.; Curator of Vertebrates			Herpetology, reptile/amphibian evolution, molecular evolution, tetrodotoxin resistance	23	92	Tetrodotoxin-resistant sodium channels in snakes (PNAS); Molecular evolution and conservation; Museum curation	Feldman Lab; Museum of Natural History	Molecular biology tools (PCR, sequencing), field equipment, specimen collections		https://www.unr.edu/biology/chris-feldman
Biology	Christopher Von Bartheld	Professor of Physiology & Cell Biology		UNR School of Medicine	Neuroscience, neuroanatomy, neuron counting methods, brain cell quantification	30	735	Systematic review of neuron numbers in the human brain Stereological methods for neuron counting Neuroanatomy and brain structure research	Physiology & Cell Biology	Stereology equipment, histology, microscopy	Active	https://www.unr.edu/neuroscience/people/christopher-
Biology	Claudia Ruckert	Assistant Professor of Biochemistry & Molecular Biology		UNR CABNR	Virology, arbovirology, mosquito-borne viruses, antiviral immune responses	14	83	Mosquito-borne virus research and arbovirology Antiviral immune responses to arboviral infections Vector-borne disease biology	BMB; CABNR	BSL-2/3 virology tools, PCR, cell culture, sequencing		https://www.unr.edu/bmb/directory/ruckert-
Biology	David AuCoin	Professor & Chair of Microbiology and Immunology			Infectious disease diagnostics, Ebola, melioidosis, Burkholderia, monoclonal antibodies	40	356	Antigen detection assay for early POC diagnosis of Zaire ebolavirus Laboratory diagnosis of melioidosis Immunoassay using dendritic Au-Pt nanoparticles for Burkholderia	Diagnostics Discovery Lab; InMAD	Diagnostic assays, monoclonal antibody generation, nanoparticle tech	Department Chair; infectious disease diagnostics; nanoparticle technology	https://med.unr.edu/ddl/about/faculty
Biology	Dean Burkin	Professor of Pharmacology		U of Colorado	Muscular dystrophy, integrins, laminin, extracellular matrix signaling	10	113	Spatial proteomics reveals laminin-111 restores adhesion signaling Vemurafenib improving muscle histopathology in LAMA2-CMD Gene therapy for limb-girdle muscular dystrophy 2C/R5	Wellstone Cooperative Research Center for MD	Confocal microscopy, protein expression, mouse models		https://med.unr.edu/faculty-and-staff/dean-burkin
Biology	Dennis Mathew	Professor of Biology; Assoc. & Associate Chair	2014	UMass Amherst (2006)	Neuroscience of olfaction, insect neuroscience, Drosophila olfaction, neuromodulation	12	52	Insect olfaction and neural circuits (NSF 2024); Drosophila olfactory neuromodulation; NIH ENDURE training program director	Mathew Lab; Integrative Neuroscience Program	Drosophila culture, electrophysiology rigs, fluorescence microscopy, behavioral arenas, optogenetics	Drosophila accessible model; NIH training program for undergrads; neuro olympiad prep. Github.	https://mathewlab.com/home/dennis-mathew/
Biology	Dorothy Hudig	Professor of Microbiology & Immunology		UNR School of Medicine	Lymphocyte-mediated cytotoxicity, NK cell mechanisms, innate immunity	13	70	Natural killer cell cytotoxicity mechanisms Lymphocyte-mediated immune defense Innate immunity and cell-mediated killing	Microbiology & Immunology	Flow cytometry, cell killing assays, cell culture		https://www.unr.edu/molecular-biosciences/people/
Biology	Dylan Kosma	Assistant Professor of Biochemistry & Molecular Biology	2009		Plant lipid polymers, cutin and suberin, wound healing genetics, CRISPR gene editing	21	422	Transcription factors regulating wound suberin deposition in potato Plant cutin and suberin lipid polymer formation mechanisms CRISPR-based identification of wound healing gene function	Biochemistry & Molecular Biology; Hitchcock Center	CRISPR equipment, analytical chromatography, greenhouse	CRISPR gene editing; Hitchcock Center affiliate; plant biochemistry	https://www.unr.edu/bmb/directory/kosma-dylan
Biology	Elena Pravosudova	Professor of Biology; Teaching & Associate Chair	2007	Ohio State University (1999)	Behavioral ecology, ornithology, evolution, peer instruction	12	150	Educational research and innovations; Tibbitts Distinguished Teacher Award (2020); Biology Peer Instruction co-founder		Teaching lab equipment, ornithology field tools		
Biology	Elizabeth Pringle	Assoc. Prof.; McMinn Endowed Research Prof.	2016	Stanford University (2011)	Evolutionary ecology, plant-animal interactions, chemical ecology, dryland ecosystems	17	52	Remote sensing reveals importance of habitat on aphid biocontrol (2024); Evolutionary ecology of mutualisms; Chemical ecology of dryland plants	The Pringle Lab	Field equipment, remote sensing data, chemical analysis tools, ecological software	Active	https://www.multimutualism.org/
Biology	Floris van Breugel	Assistant Professor of Mechanical Engineering		UNR College of Engineering	Insect flight biomechanics, neural control of movement, bio-inspired robotics, neuroethology	35	313	Insect flight dynamics and neural control Bio-inspired aerial robotics Olfactory navigation in flying insects	ME; Institute for Neuroscience	High-speed cameras, wind tunnel, motion capture, insect flight arena	Insect flight + neuroethology bridges Biology and Computing; bio-inspired robotics is interdisciplinary	https://www.florisvanbreugel.com/
Biology	Gale Craviso	Professor of Pharmacology		UNR School of Medicine	Electric field effects on cells, chromaffin cells, catecholamine secretion, neuroendocrinology	9	22	Electric field stimulation of chromaffin cells Catecholamine secretion and neuroendocrine mechanisms Pulsed electric field effects on biological cells	Pharmacology	Electrophysiology, amperometry, cell culture		https://med.unr.edu/people/gale-craviso
Biology	Gideon Caplovitz	Professor of Psychology		UNR College of Liberal Arts	Visual perception, motion perception, cognitive neuroscience, visual processing	21	190	Visual motion perception and neural mechanisms Illusions and visual processing research Cognitive neuroscience of perception	Psychology; Cognitive & Brain Sciences	Eye tracking, psychophysics displays, EEG		https://www.unr.edu/neuroscience/people/gideon-

Biology	Gillian Moritz	Associate Professor of Physiology & Cell Biology		UNR School of Medicine	Sensory systems, visual system anatomy, comparative neuroanatomy	2	4	Comparative neuroanatomy of visual systems Sensory system structure and function Gross anatomy and neuroanatomy research	Physiology & Cell Biology (tenure-track)	Histology, microscopy, anatomical imaging		https://med.unr.edu/faculty-and-staff/gillian-moritz
Biology	Heather Burkin	Associate Professor of Pharmacology		UNR School of Medicine	Preterm labor, uterine contractions, bioprinted tissue models, reproductive pharmacology	4	22	Preterm labor mechanisms and uterine contractions Bioprinted tissue models for drug testing Reproductive pharmacology research	Pharmacology (tenure-track)	3D bioprinting, tissue culture, muscle physiology rigs	Bioprinted tissue models is innovative, bridging engineering and pharmacology	https://med.unr.edu/directory/?d=594
Biology	Iain Buxton	Professor of Pharmacology			Cardiovascular pharmacology, preterm labor, uterine myometrium, purinergic signaling	9	105	Proteomic network analysis of human uterine smooth muscle Novel identification of mechanosensitive Piezo1 channel in myometrium β3 adrenergic receptor signaling in human myometrium	Cardiovascular Research Labs	Patch clamp electrophysiology, proteomics, calcium imaging		https://med.unr.edu/pharmacology/research/lab-buxton
Biology	Irina Derkatch	Professor of Pharmacology		UNR School of Medicine	Prion biology, protein aggregation, yeast genetics, amyloid formation	3		Prion propagation and protein misfolding research Yeast prion genetics and amyloid biology Mechanisms of protein aggregation diseases	Pharmacology	Yeast genetics tools, fluorescence microscopy, molecular biology		
Biology	Jack Hayes	Professor of Biology; (Emeritus ~2026)		UC Riverside	Evolutionary physiology, physiological ecology, allometry, body condition, metabolic rates	1	39	Comparative physiology and metabolic rates; 72+ publications; Allometry and body condition		Metabolic chambers, respirometry equipment, physiological measurement tools		
Biology	Jeffrey G. Baguley	Associate Professor of Biology; Teaching	2006	UT Austin (2004)	Marine benthic ecology, deep-sea ecology, meiofauna, Deepwater Horizon impacts, polar ecology	7	12	Meiofauna communities in Gulf of Mexico; Deep-sea ecology and ecotoxicology; Deepwater Horizon impact	Marine Ecology Research Group	Microscopy (stereoscopy, compound), sediment corers, field sampling, museum specimens		https://www.unr.edu/biology/jeff-baguley
Biology	Jennifer Hollander	Associate Professor of Biology; Teaching	2007		Human anatomy/physiology, plant-animal interactions, seed dispersal	6	68	2025 Tibbitts Distinguished Teacher Award; Nevada Regents Teaching Award (2019); Advanced Human Dissection Team		Dissection/anatomy lab, microscopy, herbarium access		
Biology	Jennifer Hoy	Assoc. Prof.; Director Institute for Neuroscience		University of Oregon	Visual neuroscience, prey capture behavior, neural circuits, sensory systems	9	41	Neural circuits underlying prey-capture in mice; Binocular vision in predatory behavior; NIH RO1 (~$2M/5yr)	Hoy Lab; Institute for Neuroscience	Electrophysiology, optogenetics, 2-photon microscopy, behavioral tracking, mouse colony		https://www.unr.edu/biology/jennifer-hoy
Biology	Jenny Ouyang	Assoc. Prof.; EECB Graduate Program Director	2016	Princeton University (2012)	Urban ecology, artificial light pollution, physiological adaptation, circadian biology	37	244	Effects of artificial lighting on birds; NSF CAREER ($1.2M) for light pollution; Endocrinology and animal behavior	Ouyang Lab	Light sensors, hormone assay equipment, field monitoring stations, spectroradiometers	NSF CAREER; accessible field biology; bridges biology+physics via light/circadian	https://www.jennyouyang.com/
Biology	Jessie Krause	Associate Professor of Biology; Teaching	2018	UC Davis	Comparative physiology, endocrinology, stress/reproduction, behavioral endocrinology, Arctic biology	5	40	Arctic ecosystem research on biological systems; Stress and reproduction in songbirds; Climate change impacts on physiology		Hormone assay equipment, field sampling tools, physiological measurement		
Biology	Jihwan Yoon	Assistant Professor of Electrical & Biomedical Engineering		UNR College of Engineering	Radiofrequency and microwave bioelectromagnetics, RF neuromodulation, wearable therapeutic devices			NSF CAREER Award for wearable tech treating depression RF-based neuromodulation devices for brain and nervous system stimulation Development of therapeutic wearable technology for mental health	EBME; IEEE	RF/microwave testing, wearable device fabrication		https://www.unr.edu/ebme/people/jihwan-yoon
Biology	John Cushman	Foundation Professor of Biochemistry & Molecular Biology	2000		CAM photosynthesis, drought stress tolerance, molecular genetics, plant biotechnology	29	367	Crassulacean acid metabolism: a plastic adaptation to arid environments The agave genome reveals core features of CAM Engineering CAM for sustainable food and biofuel production	Biochemistry & Molecular Biology; AAAS Fellow (2023)	Real-time PCR, gene expression analysis, greenhouse, spectrophotometry	AAAS Fellow; Foundation Professor; CAM photosynthesis; plant biotechnology. Active.	https://www.unr.edu/bmb/directory/cushman-john
Biology	John Malysz	Research Associate Professor of Physiology & Cell Biology		UNR School of Medicine	GI motility, electrical pacemaker slow wave generation, peristaltic motor activity	12	150	GI pacemaker slow wave generation mechanisms Peristaltic motor activity regulation Smooth muscle ion channel physiology	Physiology & Cell Biology (research-track)	Electrophysiology, organ bath systems		
Biology	Justin Legleiter	Professor & Chair of Biochemistry & Molecular Biology		UNR CABNR	Protein structure-function, neurological diseases, Alzheimer's research, amyloid aggregation	30	116	Protein misfolding and amyloid aggregation in Alzheimer's Atomic force microscopy of protein structure Neurodegeneration mechanisms	BMB Chair; CABNR	Atomic force microscopy, protein characterization, biophysics tools	AFM protein characterization aligns with SIL instruments; Alzheimer's research is high-impact translational work	https://justinlegleiter.wixsite.com/mysite
Biology	Karen Hannigan	Research Assistant Professor of Pharmacology		UNR School of Medicine	Calcium signaling in smooth muscle and cardiac cells, cardiovascular disease	11	8	Calcium signaling mechanisms in smooth muscle Cardiac cell calcium dynamics Cardiovascular disease signaling pathways	Pharmacology (research-track)	Calcium imaging, electrophysiology, cell culture		https://med.unr.edu/faculty-and-staff/karen-hannigan
Biology	Kelley Stewart	Professor of Natural Resources & Environmental Science		UNR CABNR	Wildlife ecology, conservation genetics, biodiversity, mammalian ecology	37	214	Conservation genetics of western wildlife Biodiversity and mammalian population ecology Genetic tools for wildlife management	NRES; CABNR	Genetics lab, DNA extraction/sequencing, GPS telemetry		https://naes.unr.edu/kelley-stewart/
Biology	Kenton Sanders	Professor of Physiology & Cell Biology		UNR School of Medicine	Interstitial cells of Cajal, GI pacemaker mechanisms, smooth muscle electrophysiology			Interstitial cells of Cajal as pacemakers in the GI tract Smooth muscle electrophysiology of the gut Neural regulation of GI motility	Physiology & Cell Biology	Electrophysiology, confocal imaging, Ca2+ imaging	Pioneered interstitial cells of Cajal research; one of the most highly cited GI physiologists worldwide	https://www.unr.edu/neuroscience/people/kenton-

Biology	Krishna Pagilla	Foundation Professor of Civil & Environmental Engineering		UNR College of Engineering	Water treatment, wastewater, water reuse, biological treatment, nutrient removal	50	573	Biological wastewater treatment and nutrient removal Water reuse systems and advanced treatment Nevada Water Innovation Institute research	CEE; ASCE Fellow; NV Water Innovation Institute Director	Wastewater treatment pilot systems, biological reactors, water quality analyzers		https://www.unr.edu/cee/people/krishna-pagilla
Biology	Lai Wen	Assistant Professor of Pharmacology		UNR School of Medicine	Cell adhesion mechanisms, inflammation, signal transduction	16	623	Cell adhesion and inflammatory signaling research Molecular mechanisms of inflammation Signal transduction in immune cells	Pharmacology	Cell culture, flow cytometry, molecular biology		https://www.wenlaboratory.org/
Biology	Lee Dyer	Foundation Professor of Biology; EECB Director	2010	University of Colorado Boulder (1994)	Chemical ecology, tropical biodiversity, plant-herbivore interactions, conservation biology	77	1192	Phytochemical diversity drives tropical plant-insect diversity (PNAS 2015); Trait-mediated trophic cascade for hummingbirds (Sci Advances 2015); New dimensions of tropical diversity (2014)	EECB Program; Hitchcock Center (affiliated)	GC-MS, LC-MS (chemical ecology), field equipment, specimen collections, tropical stations	Foundation Professor; Chemical ecology uses GC-MS/LC-MS at SIL; Hitchcock Center	https://www.unr.edu/biology/lee-dyer
Biology	Liming Huang	Research Associate Professor of Microbiology & Immunology		UNR School of Medicine	Fluorescent molecule design, fluorescent biosensors, carbon nanomaterials, electrochemical sensors	4	7	Fluorescent chemosensor and biosensor design Carbon nanomaterials for sensing applications Electrochemical sensor development	Microbiology & Immunology (research-track)	Fluorescence spectroscopy, electrochemistry, nanomaterial synthesis	Fluorescent biosensor design aligns with SIL fluorescence instruments; bridges Chemistry and Biology	https://www.researchgate.net/lab/Liming-Huang-Lab
Biology	Lora Richards	Assoc. Prof.; Co-director Hitchcock Center		University of Utah (2006)	Chemical ecology, plant chemistry, herbivore communities, metabolomics	29	124	Toxins & Secret Messages: Chemical Language of Plants; Host immunity and herbivore communities; 16+ years chemical ecology	Richards Lab; Hitchcock Center for Chemical Ecology	LC-MS (metabolomics), GC-MS, HPLC, chemical extraction equipment, field collection		https://www.lorarichards.com/
Biology	Lucia Notterpek	Professor of Physiology & Cell Biology		UNR School of Medicine	Peripheral neuropathies, neurobiology, myelin biology, translational neuroscience	3	14	Peripheral neuropathy mechanisms and myelin disorders Neurobiology of Schwann cells and axonal degeneration Translational neuroscience research	Physiology & Cell Biology; Senior Assoc Dean Research	Electron microscopy, cell culture, Western blot	Senior Associate Dean for Research at UNR Med; peripheral neuropathy is important translational neuroscience	https://www.unr.edu/neuroscience/people/lucia-notterpek
Biology	Mae Gustin	Foundation Professor of Environmental Geochemistry	1994		Atmospheric mercury, biogeochemical cycles, inorganic contaminants, tree ring analysis	28	330	Understanding mercury fate and transport in atmospheric systems Tree rings as archives of atmospheric mercury concentrations Biogeochemistry of mercury movement in plants and soils	Natural Resources & Environmental Science; AAAS Fellow (2023)	Mercury analysis, tree ring lab, atmospheric monitoring	AAAS Fellow; Foundation Professor; mercury biogeochemistry; atmospheric chemistry	https://www.unr.edu/nres/people/gustin-mae
Biology	Marian Berryhill	Professor of Psychology		UNR College of Liberal Arts	Cognitive neuroscience, working memory, attention, transcranial stimulation	36	127	Working memory and attention in the brain Transcranial direct current stimulation effects on cognition Neural basis of memory and attention	Psychology Assoc. Chair; Institute for Neuroscience	tDCS, EEG, fMRI, behavioral testing		https://www.unr.edu/neuroscience/people/marian-
Biology	Marjorie Matocq	Professor of Biology; (NRES; EECB-affiliated)		UC Berkeley (2000)	Population genetics, evolutionary genetics, phylogeography, hybrid zones, mating systems	28	179	Long-lived species and climate change adaptation; Woodrat population genetics; Western North American mammal diversity	Matocq Lab (NRES)	Molecular genetics tools (PCR, sequencing), field trapping, tissue banking		https://naes.unr.edu/matocq/
Biology	Mary Burtnick	Professor of Microbiology and Immunology			Melioidosis vaccine development, virulence factors, T6SS, immunoassays	17	125	Subunit vaccines providing high-level protection against melioidosis Melioidosis patient survival correlates with IFN-γ T cells Rapid ELISAs for serological screening of melioidosis	Melioidosis Research Consortium	Bacterial culture, vaccine development, immunological assays		https://med.unr.edu/directory/mary-n-burtnick
Biology	Maryam Raeeszadeh-Sarmazdeh	Associate Professor of Chemical & Materials Engineering	2019	UNR College of Engineering	Protein engineering, synthetic biology, directed evolution, enzyme inhibitors for neurodegeneration and cancer	22	152	Protein Language Models for Low-N Engineering of Metalloproteinase Inhibitors (IEEE) Engineering TIMP-1 for Fine Discrimination Between MMP-3 and MMP-10 Tissue Inhibitors of Matrix Metalloproteinases for Brain Cancer	CME; NIH	Protein expression systems, directed evolution platforms, molecular characterization	Protein engineering for cancer and neurodegeneration is core Biology; NSF CAREER + NIH funded; bridges Chemistry and Biology	https://sarmazdehlab.wixsite.com/sarmazdehlab
Biology	Matt Forister	Foundation Prof.; McMinn Endowed Research Prof.	2008		Plant-insect ecology, butterfly ecology, diversification, diet breadth evolution	7	54	Rapid butterfly declines across the US (Science 2025); Monitoring, global change, specialization, extinction; Phytochemical and insect community diversity	Great Basin Bug Lab	Field collection equipment, specimen collections, molecular tools, ecological software	Very high impact (Science 2025 paper); accessible fieldwork; biodiversity+conservation. Active	https://sites.google.com/site/greatbasinbuglab/
Biology	Melinda Yerka	Associate Professor of Plant Breeding	2017		Sorghum breeding, plant genetics, drought tolerance, crop development	12	80	200 nutritious drought-tolerant sorghum varieties for food and biofuel MAGIC population design for sorghum Low-water sorghum as sustainable crop alternative	Agriculture, Veterinary & Rangeland Sciences; Yerka Seeds	Greenhouse, seed analysis, genetic screening, field plots		https://naes.unr.edu/yerka/
Biology	Michael Crognale	Professor of Psychology		UNR College of Liberal Arts	Color vision, visual neuroscience, perception, electrophysiology of vision	31	29	Color vision and visual electrophysiology research Neural mechanisms of color perception Visual evoked potentials and perception	Psychology; Cognitive & Brain Sciences	VEP recording, color calibrated displays, electrophysiology		https://labs.psych.unr.edu/crognale_lab/
Biology	Nathan Grainger	Research Assistant Professor of Physiology & Cell Biology		UNR School of Medicine	Cardiovascular disease, vascular regulation, cardiac sinoatrial node physiology	10	63	Cardiovascular vascular regulation research Cardiac sinoatrial node physiology Vascular smooth muscle signaling	Physiology & Cell Biology (research-track)	Patch clamp, calcium imaging, confocal microscopy		https://graingerlab.science/
Biology	Nicole Procacci	Research Assistant Professor of Pharmacology		UNR School of Medicine	Motor and secretory control, peripheral neuropathy treatment, myelin restoration	4	17	Peripheral neuropathy treatment approaches Myelin restoration and nerve regeneration Motor and secretory neural control	Pharmacology (research-track)	Electrophysiology, nerve conduction, histology		https://med.unr.edu/pharmacology/our-team
Biology	Normand LeBlanc	Professor of Pharmacology			Smooth muscle electrophysiology, calcium-activated chloride channels, cardiac physiology	29	138	Ca2+-activated Cl- channels in vascular smooth muscle β-adrenergic receptor antagonism of cholinergic stimulation Biophysical properties of CaCCs in vasoconstriction	Cardiovascular Research; Vice Chair of Pharmacology	Patch clamp, confocal/super-resolution microscopy, TIRFM		https://med.unr.edu/pharmacology/research/lab-leblanc

Biology	Pamela Sandstrom	Associate Professor of Biology; Teaching	2002	University of Nevada, Reno (2007)	Biology education, molecular genetics, biochemistry, health sciences education			2024 Bible Teaching Excellence Award; Biology Peer Instruction co-founder; Nevada Regents Teaching Award (2021)		Teaching lab equipment, molecular biology tools		
Biology	Paul Brett	Professor of Microbiology and Immunology			Burkholderia pseudomallei, melioidosis vaccine development, bacterial pathogenesis	36	1208	Burkholderia pseudomallei and mallei vaccines: clinical trial potential Reverse vaccinology nanoglycoconjugate vaccine design Complex subunit and glycoconjugate vaccines	Mahidol University Thailand (adjunct); Oxford UK collab	Vaccine formulation, bacterial culture, immunoassays		https://med.unr.edu/people/paul-brett
Biology	Paul Sumby	Professor of Microbiology & Immunology		UNR School of Medicine	Group A Streptococcus genetics, genomics, transcriptomics, bacterial pathogenesis	5	17	Group A Streptococcus molecular genetics and virulence Bacterial transcriptomics and gene regulation Genomic analysis of pathogenic bacteria	Microbiology & Immunology	Next-gen sequencing, RNA-seq, molecular microbiology tools		https://med.unr.edu/people/paul-sumby
Biology	Paul Verburg	Associate Professor of Natural Resources & Environmental Science		UNR CABNR	Soil science, soil chemistry, nutrient cycling, biogeochemistry	6	66	Soil chemistry and nutrient dynamics Biogeochemistry of arid and semi-arid ecosystems Carbon and nitrogen cycling in soils	NRES; CABNR	Elemental analyzers, soil chemistry instruments, IC/ICP		https://www.unr.edu/nres/people/verburg-paul
Biology	Peter Jones	Associate Professor of Pharmacology		Emory	FSHD, epigenetic mechanisms, DNA methylation, muscle development and disease	109	3763	The genetics and epigenetics of FSHD Epigenetic diagnostics for FSHD1 and FSHD2 FSHD animal models and novel epigenetic diagnostic cassettes	Wellstone Center; Mick Hitchcock Endowed Chair of Medical Biochemistry	Epigenetic analysis, DNA methylation assays, animal models	Exceptionally high impact (15k citations); epigenetics; muscular dystrophy center	https://med.unr.edu/directory/research-labs/jones-lab
Biology	Peter Weisberg	Professor of Natural Resources & Environmental Science		UNR CABNR	Landscape ecology, fire ecology, forest management, spatial ecology	20	288	Landscape ecology and fire dynamics in western forests Spatial ecology and forest management Fire ecology and ecosystem resilience	NRES; CABNR	GIS/remote sensing, LiDAR, spatial analysis tools		https://naes.unr.edu/weisberg/
Biology	Robert Harvey	Professor & Chair of Pharmacology		Northwestern (1987)	Cardiac arrhythmias, sympathetic signaling, autonomic neurotransmitters, heart failure	21	146	Sympathetic control of cardiac arrhythmias and sudden death Neurotransmitter signaling in isolated cardiac myocytes β-adrenergic and muscarinic receptor signaling in heart	Pharmacology Department	Patch clamp electrophysiology, isolated cell preparations, cardiac tissue		https://med.unr.edu/pharmacology/research/lab-harvey
Biology	Robert Renden	Professor of Physiology & Cell Biology		UNR School of Medicine	Neuroscience, synaptic transmission, neurotransmitter release, presynaptic mechanisms	13	87	Synaptic transmission and neurotransmitter release Presynaptic mechanisms of neural communication Neuroscience of brain function	Physiology & Cell Biology; Institute for Neuroscience	Patch clamp, two-photon microscopy, electrophysiology		https://www.unr.edu/neuroscience/people/robert-renden
Biology	Ruben Dagda	Associate Professor of Pharmacology		Iowa	Mitochondrial dysfunction, Parkinson's disease, neurotoxicology, neuroprotection	21	187	BDNF protects neurons by stimulating mitochondrial function via PKA Mitochondrial foundations of Parkinson's disease Role of PINK1 and PKA in mitochondrial function	Molecular Biosciences Graduate Program	Cell culture, mitochondrial imaging, fluorescence microscopy		https://med.unr.edu/people/ruben-dagda
Biology	Sal Baker	Professor of Physiology & Cell Biology		UNR School of Medicine	Smooth muscle biology, GI motility, gastrointestinal physiology			Smooth muscle biology and GI motility research Gastrointestinal physiology studies Intestinal smooth muscle function	Physiology & Cell Biology	Electrophysiology equipment, tissue bath systems		https://med.unr.edu/faculty-and-staff/sal-baker
Biology	Sang Don Koh	Professor of Physiology & Cell Biology		UNR School of Medicine	Spinal cord injury, bladder function, smooth muscle, ion channels	14	66	Spinal cord injury and bladder dysfunction research Ion channel physiology in smooth muscle Neural control of urinary function	Physiology & Cell Biology	Patch clamp electrophysiology, ion channel recording		https://med.unr.edu/faculty-and-staff/research-labs/koh-lab
Biology	Sarah M. Haigh	Associate Professor of Psychology		UNR College of Liberal Arts	Sensory processing, neurological responses to stimuli, schizophrenia, autism, migraine, TBI	39	256	Sensory processing and neural excitability in schizophrenia Visual perception and migraine research Autism spectrum sensory processing studies	Psychology; Cognitive & Brain Sciences; Institute for Neuroscience	EEG, fMRI, psychophysics equipment, eye tracking	Directs the Cognitive & Brain Science PhD program; bridges neuroscience and computing with sensory processing research. Active.	https://sarahmhaigh.github.io
Biology	Seungil Ro	Professor of Physiology & Cell Biology		UNR School of Medicine	Diabetes, smooth muscle biology, microRNA, translational research	44	1006	Diabetes and smooth muscle dysfunction research MicroRNA regulation in GI physiology Translational diabetes research	Physiology & Cell Biology	Molecular biology, microRNA analysis, mouse models		https://www.unr.edu/molecular-biosciences/people/
Biology	Shailesh Agarwal	Assistant Professor of Pharmacology		UNR School of Medicine	Nanoscale cell signaling in heart muscle, cardiovascular disease, cAMP signaling, mitochondria			Nanoscale cAMP signaling in cardiac myocytes Mitochondrial function in cardiovascular disease Cell signaling compartmentalization in heart	Pharmacology (tenure-track)	Super-resolution microscopy, FRET imaging, patch clamp		https://med.unr.edu/pharmacology/research/lab-agarwal
Biology	Steve Vander Wall	Professor of Biology; Biology Graduate Director	1990s		Food hoarding, seed dispersal, plant-animal interactions, community ecology, forest ecology	17	420	Bird and rodent seed dispersal in Sierra Nevada; Food hoarding behavior; 2024 AAAS Fellow		Field stations, radio telemetry, seed tracking equipment, specimen collections		https://www.unr.edu/biology/steve-vander-wall
Biology	Steven Frese	Assistant Professor of Nutrition & Microbiology	2020		Gut microbiome, food biotechnology, nutritional science, infant health, probiotic development	44	1514	Breastfeeding, immune development, and infant microbiome composition Biotechnology approaches to improving infant nutrition and health Microbiome-based interventions for malnourished children	Nutrition; School of Medicine	DNA sequencing, microbiome analysis, fermentation equipment		https://naes.unr.edu/frese/
Biology	Subhash Verma	Professor of Microbiology & Immunology		UNR School of Medicine	Viral replication, human herpesviruses, SARS-CoV-2, KSHV genome replication	58	305	KSHV genome replication and latency mechanisms SARS-CoV-2 replication and pathogenesis Human herpesvirus molecular biology	Microbiology & Immunology; CMB Graduate Program Director (research-track)	BSL-2/3 virology, PCR, sequencing, cell culture	9,172 citations; directs Cell & Molecular Biology grad program; virology from KSHV to SARS-CoV-2	https://med.unr.edu/faculty-and-staff/subhash-verma

Biology	Sudeep Chandra	Professor of Biology; Director Global Water Center		UC Davis (2003)	Lake ecology, limnology, mountain lake dynamics, aquatic conservation, fisheries	72	1802	Lake metabolism and mountain lake systems; Castle Lake Research Station director; NSF Ecosystem Sciences Program Director	Aquatic Ecosystems Analysis Lab; Global Water Center; Ozmen Institute	Water quality probes, CTD profilers, boats, field sampling equipment, isotope analysis		http:// www.aquaticecosystemslab.org/
Biology	Sung Jin Hwang	Research Associate Professor of Physiology & Cell Biology		UNR School of Medicine	Gastrointestinal physiology, smooth muscle biology, interstitial cells of Cajal	14	133	Interstitial cell biology in the GI tract Smooth muscle electrophysiology GI motility regulation	Physiology & Cell Biology (research-track)	Electrophysiology, confocal imaging, immunohistochemistry		https:// med.unr.edu/people/sung-jin-hwang
Biology	Susan Liebman	Professor of Pharmacology		Rochester	Yeast prion biology, protein misfolding, neurodegeneration, Alzheimer's, Parkinson's	16	69	Prions in yeast: mechanisms and models for neurodegeneration Role of chaperone Hsp104 in prion propagation Heterologous cross-seeding mimics cross-species prion conversion	Pharmacology Department; $13M+ career funding	Yeast genetics, molecular biology, protein biochemistry	Very high impact (11k citations); yeast prion biology; protein misfolding	https:// med.unr.edu/people/susan-liebman
Biology	Takako Jones	Associate Professor of Pharmacology		UNR School of Medicine	Muscular dystrophy, muscle biology, extracellular matrix, translational research	15	98	Muscular dystrophy mechanisms and treatment Extracellular matrix in muscle disease Translational muscle biology research	Pharmacology	Mouse models, histology, molecular biology		https:// med.unr.edu/people/takako-jones
Biology	Thomas Gould	Professor of Physiology & Cell Biology		UNR School of Medicine	Neuroscience, GI motility, neural circuit analysis	15	154	Neuroscience and neural circuit research GI motility and neural regulation Neural control of organ function	Physiology & Cell Biology; Institute for Neuroscience	Electrophysiology, optogenetics, confocal microscopy		https:// med.unr.edu/faculty-and-staff/research-labs/gould-lab
Biology	Thomas Kozel	Professor of Microbiology & Immunology		UNR School of Medicine	Translational mycology, fungal infection diagnostics, Cryptococcus, innate immunity to fungi			Fungal infection diagnostic development Cryptococcal pathogenesis and immunity Translational mycology research	Microbiology & Immunology	Immunoassay development, fungal culture, diagnostic tools		https:// www.medicine.nevada.edu/ddl/about/faculty
Biology	Todd Steury	Professor & Chair of Natural Resources & Environmental Science		UNR CABNR	Wildlife ecology, statistical ecology, population dynamics, conservation biology	11	140	Wildlife population dynamics and statistical ecology Conservation biology in arid ecosystems Ecological modeling and analysis	NRES Chair; CABNR	GPS telemetry, camera traps, statistical modeling software		https:// www.unr.edu/nres/people/steury-todd
Biology	Vincent Lombardi	Associate Professor of Microbiology & Immunology		UNR School of Medicine	Innate immunity, gut-brain axis, neuroimmune disease, mucosal immunology	19	224	Gut-brain axis and neuroimmune interactions Innate immunity in mucosal tissues Neuroimmune mechanisms of disease	Microbiology & Immunology	Flow cytometry, ELISA, gnotobiotic models	Gut-brain axis is a hot interdisciplinary research area bridging immunology and neuroscience	https:// med.unr.edu/directory/vincent-lombardi
Biology	Violeta Mutafova-Yambolieva	Professor of Physiology & Cell Biology		UNR School of Medicine	Purinergic signaling, smooth muscle, neurotransmission, vascular biology	10	70	Purinergic signaling in smooth muscle Neurotransmission and vascular biology Purine nucleotide metabolism	Physiology & Cell Biology Vice Chair	HPLC for purine analysis, electrophysiology		https:// www.unr.edu/neuroscience/people/violeta-
Biology	William Courchesne	Associate Professor of Microbiology & Immunology	1989	UNR School of Medicine	Yeast cell biology, quiescence to proliferation transition, calcium influx, cell cycle reentry			Quiescence to proliferation transition in yeast Calcium influx and cell cycle reentry mechanisms Molecular yeast cell biology	Microbiology & Immunology; Institutional Biosafety Committee Chair	Yeast genetics, fluorescence microscopy, molecular biology		https:// med.unr.edu/people/william-courchesne
Biology	Xiaoshan Zhu	Professor of Electrical & Biomedical Engineering	2009	UNR College of Engineering	Nanomaterials for biosensing and imaging, MEMS-integrated sensory systems, health-care monitoring			Novel TB test development using quantum dots (NSF Phase 2) Biosensing research utilizing manganese-doped quantum dots Smartphone-based health diagnostic technology	EBME; Biomedical Engineering	MEMS fabrication, fluorescence spectroscopy, nanomaterial synthesis	Biosensing + nanomaterials directly aligns with SIL fluorescence and MALDI instruments; health diagnostics bridge to Biology. Active.	https:// sites.google.com/view/the-zhu-lab/home
Biology	Yifei Jin	Assistant Professor of Mechanical Engineering	2019	UNR College of Engineering	3D bioprinting, additive manufacturing, soft materials, hydrogels, stimuli-responsive materials			Multiscale embedded printing of engineered human tissue and organ equivalents (PNAS, 2024) Self-Supporting Nanoclay as Internal Scaffold for Direct Printing of Soft Hydrogel Structures Research on 3D bioprinting of uterine and heart structures	ME; Biomedical	3D bioprinting systems, materials characterization	3D bioprinting of human tissues is cutting-edge Biology+Engineering; published in PNAS; direct relevance to biomedical research. Bookable toys.	https:// yifeijin.org/people-2/
Biology	Zeb Hogan	Research Assoc. Prof.; Wonders of Mekong Director	2006	UC Davis (2004)	Aquatic conservation, freshwater fish ecology, large fish, migration, endangered species	42	841	Mekong giant catfish conservation; Freshwater fish biodiversity; Host of Nat Geo "Monster Fish"	Aquatic Ecosystems Analysis Lab; Wonders of the Mekong	Field sampling equipment, acoustic telemetry, boats, water quality probes		https:// www.unr.edu/biology/zeb-hogan

science	name	type	people	overview	total	impact	publications	toys	access	reasoning	web	web
Mathematics	Applied Mathematics Group	Research Group	Eric Olson, Mark Pinsky, Deena Schmidt	Navier-Stokes equations, dynamical systems, mathematical biology, and applied probability research across fluid mechanics, networks, and bioremediation	30	1564	Discrete data assimilation in the Lorenz and 2D Navier-Stokes equations Monotonicity criteria for energy-period function in planar Hamiltonian systems Stability criteria for linear systems with partly uncertain periodic	HPC clusters, simulation software (MATLAB, Mathematica), network analysis tools	Join through graduate program or faculty mentorship		https://www.unr.edu/math/people	http://fractal.math.unr.edu/~ejolson/applied/
Mathematics	TechWise Program	Program	Deena Schmidt (Faculty Director), Google partnership	Google-funded program at UNR connecting mathematics, computing, and data science for students from diverse backgrounds	4	31		Computational resources, data science tools, Google Cloud access	Application-based; open to undergrads	Google-backed program; data science + math bridge; accessible to motivated students	https://www.unr.edu/nevada-today/news/2022-01-19-	https://www.unr.edu/nevada-today/
Computing	Advanced Robotics & Automation (ARA) Lab	Lab	Hung Manh La (Director)	Autonomous robotic systems for civil infrastructure inspection; climbing robots, multi-robot control, UAV flocking; 2025 ASCE Pankow Award	70	690	A climbing robot for steel bridge inspection A practical climbing robot for steel bridge inspection Visual and 3D mapping for steel bridge inspection using a climbing robot	Magnetic-wheel climbing robots, UAVs/drones, GPR, seismic/acoustic arrays, laser scanners, IMU, Intel NUC, ROS	Project-based; inquire through lab	Hands-on robotics; award-winning bridge inspection robots; field deployment experience	https://ara.cse.unr.edu/	https://ara.cse.unr.edu/?page_id=25
Computing	BRAIN Core (Bridge to Research in AI and Neurocomputing)	Core Facility	Alireza Tavakkoli (Director)	UNR core providing computational support for biomedical research: machine learning, AI, statistical analysis, VR/AR platforms for neuroscience investigators.	200	2048	Machine learning applications in neuroscience research AI-driven biomedical data analysis pipelines Computational methods for neuroimaging analysis	HPC infrastructure, cloud AI systems, ML/DL frameworks, statistical tools, VR/AR equipment	Free consultation; fee-based equipment access; available to COBRE and biomedical researchers		https://www.unr.edu/neuroscience/center/core-	https://www.cse.unr.edu/~alit/
Computing	CATER (Center for Advanced Transportation Education and Research)	Research Center	Zong Tian (Founding Director)	Develops efficient, sustainable transportation systems through applied research in traffic engineering, intelligent transportation systems, and freeway operations since 2010.	4	88	Impact of on-ramp traffic flow arrival on queue length at metered ramps Model of transition delay for unaccommodated pedestrian timing Large-scale transportation network modeling for regional systems	Video image systems, traffic signal modeling software, freeway simulation tools, network analysis platforms	Through Civil & Environmental Engineering; NCAR network		https://catalog.unr.edu/content.php?catoid=45&n	https://www.unr.edu/cee/people/zong-tian
Computing	Computer Vision Laboratory (CVL)	Lab	George Bebis (Director), Alireza Tavakkoli	Pure and applied research in computer vision since 1998: image segmentation, object recognition, 3D reconstruction, face detection, biometrics, tracking	239	2248	Face detection and verification using genetic search Low-resolution face recognition with single sample per person Face detection in the near-IR spectrum	GPU clusters, image capture/display devices, robotic peripherals, Pronghorn HPC access	Open applications for research assistants year-round; undergrads welcome	Major CV lab; open to undergrads; GPU access; directly relevant to Computing olympiad	https://cvl.cse.unr.edu/	https://www.cse.unr.edu/CVL/index.php
Computing	Electrical Power & Renewable Energy Systems Lab (EPRESL)	Research Lab	Mehdi Etezadi-Amoli, M. Sami Fadali, Yantao Shen	Power systems modeling, renewable energy integration, smart grid, electric vehicle charging infrastructure, and power electronics research.	68	393	Optimal scheduling of electric vehicle charging stations with renewable energy Smart grid integration of distributed generation systems Power system stability analysis with high renewable penetration	Power systems simulators, SCADA systems, PV arrays, battery storage testbeds, power electronics test benches, real-time digital	Through Electrical & Biomedical Engineering; graduate research assistantships		https://www.unr.edu/ebme/research/power-	https://www.unr.edu/ebme/people/mehdi-
Computing	Evolutionary Computing Systems Lab (ECSL)	Lab	Sushil Louis (Director)	Evolutionary computing, genetic algorithms, AI for RTS games, Case-Injected Genetic Algorithms (CIGAR)	25	99	Playing to learn: case-injected genetic algorithms for computer games Trap avoidance in strategic game playing with case injected genetic algorithms Co-evolving influence map tree based strategy game players	Computing systems, ML frameworks, game engines, optimization software	Through graduate programs and research assistantships		https://ecsl.cse.unr.edu/	https://www.cse.unr.edu/~sushil/
Computing	Full Spectrum Cybersecurity Lab	Lab	Faculty-directed (multi-PI)	Hands-on cybersecurity sandbox: create, attack, and defend servers and networks; IoT security, smart grid, cloud security	20	450	Analysis and prevention of security vulnerabilities in a smart city Guarding the grid: IoT and IIoT security vulnerabilities in smart power Knowledge base information security	Sandbox environment (20 stations), staging servers, VMs, network security testing, power/control systems	Open to students through cybersecurity programs; LME 321		https://fsc.cse.unr.edu/	https://www.unr.edu/cybersecurity
Computing	HPCVIS Lab	Lab	Frederick Harris Jr. (Co-director), Phillip Goodman	Graphics, scientific visualization, virtual reality applications, and high-performance computation including immersive VR environments	87	344	VFire: immersive wildfire simulation and visualization Open-source tools for immersive environmental visualization A Unity framework for multi-user VR experiences	GPU clusters, VR headsets (Oculus, HTC Vive), Unity VR framework, visualization hardware	Faculty-led; undergrad/grad involvement through CSE department		https://www.unr.edu/cyberinfrastructure/high-performance-computing/	https://www.cse.unr.edu/~fredh/
Computing	Intelligent Data & Systems (IDS) Lab	Lab	Feng Yan (Director)	ML systems, federated learning, deep learning optimization, serverless computing; leads ML component of $20M NSF Wildfire project	77	2275	BatchCrypt: efficient homomorphic encryption for cross-silo federated learning TiFL: a tier-based federated learning system Bridging big data machine learning and computer systems	GPU/TPU clusters, cloud computing infrastructure, distributed systems	Through CSE department; grad and undergrad research	ML systems directly relevant to Computing olympiad and AI research goals	https://www.cse.unr.edu/~fyan/	https://www.cse.unr.edu/~fyan/research.h
Computing	Machine Perception Lab	Lab	Emily Hand (Director)	Wearable devices for social interaction support (autism, visual impairment); bridges human and computer vision using ML	28	120	Improving face recognition using artistic interpretations of prominent features Transformation of face recognition technology and representation for faces Empirical study of face recognition using machine learning	Wearable prototyping, CV systems, ML frameworks, GPU clusters	Open to grad and undergrad students; part of CVL community	ML/vision with social impact; NASA background PI; open to undergrads	https://www.machineperceptionlab.com/	https://www.machineperceptionlab.com/about
Computing	Nevada Autonomous (UAS Test Site)	Research Center	NCAR management, FAA-certified operations team	Manages Nevada's FAA-designated UAS test site enabling beyond-visual-line-of-sight drone testing across 1000 sq mi. Reno-Las Vegas corridor for autonomous systems research.	12	180	FAA Civil UAS operations waiver for Nevada autonomous test site Public attitudes on using drone technology for visual surveillance Beyond-visual-line-of-sight corridor operations analysis	UAS testing infrastructure (1000 sq mi), FAA-compliant ground control, beyond-VLOS corridor, 300-lb payload capacity aircraft	Through NCAR (ncar@unr.edu); FAA Certificate of Authorization required		https://www.unr.edu/ncar/programs/nevada-	https://www.unr.edu/nevada-today/
Computing	Pronghorn HPC Cluster	Infrastructure	UNR Cyberinfrastructure group	Shared high-performance computing: 108 CPU nodes (3,456 cores), 44 NVIDIA P100 GPUs, 3 V100 GPUs; hosted at Switch data center	0	0		3,456 CPU cores, 24.8 TiB memory, 44 Tesla P100 GPUs, 3 Tesla V100 GPUs, high-speed interconnect	UNR employees: email hpc@unr.edu; external: through NCAR (paid)	Central GPU/HPC resource for ML and computational research across all sciences	https://www.unr.edu/cyberinfrastructure/high-	https://github.com/UNR-HPC/pronghorn
Computing	Robotics Research Lab (RRL) / Social Robotics Lab	Lab	David Feil-Seifer (Director)	Socially assistive robotics for autism therapy, elder care, post-stroke rehab; human-robot interaction and socially-aware navigation	40	1500	Modeling engagement in long-term in-home socially assistive robot interventions Toward socially assistive robotics for augmenting interventions for autism Defining socially assistive robotics	Humanoid robots (InMoov), VICON motion capture (8 cameras, sub-mm accuracy, 15x7x5m volume), UAS platforms	REU Site for undergrads; has mentored 100+ undergrads; very accessible	Exceptional undergrad mentoring track record; motion capture system; directly relevant to Computing	https://rrl.cse.unr.edu/	https://www.cse.unr.edu/~dave/
Computing	Software Engineering Lab (SOELA)	Lab	Sergiu Dascalu (Director)	Requirements specification, software architectures, UI design, model/data interoperability, simulation environments	68	280	Software tool for naval surface warfare simulation and training CrownVision: metrics visualization for project management A new workflow to interact with and visualize big data for web applications	Software development frameworks, VR/AR tools, simulation environments	Faculty-directed; grad/undergrad through department		https://www.unr.edu/cse/people/sergiu-dascalu	https://www.unr.edu/cse/research/data-and-
Physics	Atmospheric Dynamics & Wildfire Group	Research Group	Neil Lareau	Wildfire plume dynamics, pyrotornadogenesis, boundary layer meteorology, fire weather; part of $20M NSF wildfire project	39	335	The Carr Fire Vortex: a case of pyrotornadogenesis? Plume dynamics drive extreme long-range spotting Fire-generated tornadic vortices	Weather radars, radiosondes, LiDAR (atmospheric), weather stations, satellite data systems	Through Physics/Atmospheric Sciences program		https://www.unr.edu/physics/neil-lareau	https://www.unr.edu/atmospheric-

Physics	CTEMPS (Center for Transformative Environmental Monitoring Programs)	Research Center	Scott Tyler (Director), Christine Hatch	NSF-funded center pioneering distributed temperature sensing (DTS) fiber optic technology for environmental monitoring of rivers, glaciers, caves, and subsurface systems.	31	203	Environmental temperature sensing using Raman spectra DTS fiber-optic methods Distributed temperature sensing for hydrology and environmental monitoring Fiber-optic DTS methods for monitoring glacial melt processes	Fiber-optic DTS instruments, environmental sensors, weather stations, data loggers, field deployment systems	Equipment lending program for researchers; NSF-supported; ctemps.org	Pioneered DTS technology for environmental science; global impact	https://ctemps.org/	https://www.unr.edu/geology/scott-tyler
Physics	Center for Civil Engineering Earthquake Research (CCEER)	Research Center	Sherif Elfass (Director), Ian Buckle, Saiid Saiidi, Patrick Laplace	Nation's premier university earthquake engineering lab with 5 shake tables including the largest Laminar Soil Box (350 tons). Research in structural, geotechnical, and bridge earthquake engineering since 1984.	16	113	Seismic performance of reinforced concrete bridge columns under combined loading Laminar soil box testing of soil-foundation-structure interaction Shake table testing of bridge systems with rocking foundations	5 shake tables, Laminar Soil Box (350 tons), strong floor, reaction wall, 300-kip actuators, data acquisition systems, SAGE building	Graduate research through Civil & Environmental Engineering; DOE/NNSA/FHWA supported	Unique facility in US university system; 5 shake tables; world-class earthquake engineering	https://www.unr.edu/cceer	https://www.unr.edu/cceer/about
Physics	Center for Neotectonic Studies	Research Center	Steven Wesnousky (Director), Rich Koehler, Annie Kell	Earthquake geology, paleoseismology, and seismic hazard assessment for the western US. Studies active faulting, tsunami hazards, and geomorphic processes.	30	190	Displacement and geometrical characteristics of earthquake surface ruptures Active faults in the Walker Lane and implications for seismic hazard The Cascadia subduction zone megathrust earthquake	Paleoseismic trenching equipment, ground-penetrating radar, lidar, total stations, seismometers, cosmogenic dating	Through Geological Sciences & Engineering; field-based research	Premier earthquake geology group; high-impact paleoseismology research	http://neotectonics.seismo.unr.edu/CNS_Director	http://neotectonics.seismo.unr.edu/CNS_Wesn
Physics	Condensed Matter Theory Group	Research Group	Yafis Barlas and others	2D crystals, graphene, moiré quantum matter, superconductivity, topological materials, correlated electron systems	30	1400	Quantum parity Hall effect in Bernal-stacked trilayer graphene Topological flat bands in multilayer graphene moiré Hofstadter moiré butterfly in twisted trilayer graphene	HPC clusters, DFT codes, tight-binding models; collaborations with LBNL, SLAC, Sandia	Grad/undergrad through Physics dept		https://www.unr.edu/physics/yafis-barlas	https://www.unr.edu/physics/people
Physics	Derevianko Group	Research Group	Andrei Derevianko (AAAS Fellow, Simons Fellow)	Novel atomic clock development; precision tests of fundamental symmetries; ytterbium optical lattice clocks; nuclear clocks; parity-non-conservation	64	614	Hunting for topological dark matter with atomic clocks Physics of optical lattice clocks Search for new physics with atoms and molecules	Precision laser systems, optical lattice clock apparatus, high-precision spectroscopy	Grad/undergrad through Physics dept	World-leading precision measurement; AAAS Fellow; atomic clocks are cutting-edge physics	https://www.derevianko group.com/	https://www.unr.edu/physics/andrei-
Physics	Great Basin Center for Geothermal Energy	Research Center	Bridget Ayling (Director), James Faulds, Nicholas Hinz	DOE-funded geothermal research center advancing exploration, characterization, and development of geothermal resources in the Great Basin and globally.	99	1152	Controls on Great Basin geothermal systems Geothermal play fairway analysis of the Great Basin Machine learning approaches for geothermal resource assessment	Geothermal well logging equipment, geophysical instruments, thermal modeling software, GIS systems, field vehicles	Through Geological Sciences & Engineering; DOE-funded positions	DOE-funded; unique geothermal expertise leveraging Nevada's geologic setting	https://gbcge.org/	https://www.unr.edu/nevada-today/
Physics	High Energy Density Lab (HEDL)	Research Group	Thomas White	Laboratory astrophysics; extreme states of matter; equation of state in planetary cores; uses NIF, LCLS, European XFEL	58	1441	Accessing extreme equation of state on National Ignition Facility Probing matter at Gbar pressures at the NIF Developing absolute shock wave equation of state at NIF	Access to National Ignition Facility, LCLS X-ray laser, European XFEL, DFT/MD computing	Through White research group; DOE funded	Access to world-leading national facilities (NIF, LCLS); bridges physics + astronomy	https://www.thomasgwhite.com/	https://www.unr.edu/physics/research/
Physics	LASSO (Lab for Aerosol Science, Spectroscopy & Optics)	Lab	Hans Moosmüller (NSHE Regents Researcher), Patrick Arnott	Aerosol light absorption/scattering/extinction measurement; air quality; dust; biomass combustion; DRI-based	59	511	Aerosol light absorption and its measurement: a review Absorption Ångström coefficient of brown carbon aerosols Towards aerosol light absorption with aethalometer	Photoacoustic spectrometers, cavity ring-down spectroscopy, aerosol samplers, nephelometers, optical benches	Research collaboration; DRI-based; UNR Atmospheric Sciences affiliated		https://www.dri.edu/labs/lasso/	https://www.unr.edu/physics/patrick-
Physics	ME Manufacturing & Multiphysics Labs	Research Lab	Mechanical Engineering faculty (Yiliang Liao, Matteo Aureli)	Advanced manufacturing processes (laser machining, additive manufacturing) and multiphysics simulation research. Part of NCAR infrastructure.	40	1600	Ultrafast laser processing of functional materials Multiphysics modeling of fluid-structure interactions Additive manufacturing parameter optimization for metallic alloys	CNC machines, 3D printers (metal/polymer), laser machining systems, tensile testing, optical profilometer, thermal cameras	Through Mechanical Engineering; NCAR access for external users		https://matteoaureli.com/lab.html	https://www.unr.edu/me/people/yiliang-liao
Physics	Nevada Geodetic Laboratory	Research Lab	Geoffrey Blewitt (Director), Corné Kreemer, William Hammond	World-leading GPS/geodesy research operating 414-station GPS network. Real-time crustal deformation, tectonic strain analysis, and sea-level change monitoring since 2004.	75	1240	Harnessing the GPS data explosion for interdisciplinary science A global strain rate model including plate boundary zones GPS measurement of present-day crustal deformation rates in the Basin and Range	414-station GPS network, GNSS processing systems, InSAR data, high-performance computing, geodetic data archives	Public data at geodesy.unr.edu; research through Nevada Bureau of Mines & Geology	World-leading GPS geodesy lab; highest-impact geoscience group at UNR; open data. Data.	https://geodesy.unr.edu/	https://geodesy.unr.edu/about.php
Physics	Nevada State Climate Office	Research Center	Steph McAfee (State Climatologist), staff climatologists	Official state climate office providing climate data, monitoring, research, and outreach for Nevada. Housed at UNR, focusing on drought, temperature trends, and climate variability.			Nevada climate assessment and monitoring report Drought characterization in the western Great Basin Temperature and precipitation trends across Nevada	Weather station network, climate databases, remote sensing data, GIS software, statistical analysis tools	Public resource; research collaboration through Geography dept		https://www.unr.edu/nevada-climate-office	https://www.unr.edu/nevada-climate-
Physics	Tscherbul Group	Research Group	Timur Tscherbul (NSF CAREER)	Ultracold quantum dynamics: molecular collisions, ultracold chemistry, photosynthetic light-harvesting, Rydberg aggregates, superfluid helium nanodroplets	72	313	Hyperfine-to-rotational energy transfer in ultracold collisions A rigorous adiabatic approach to ultracold collisions A Feshbach resonance in triplet ground-state molecules	Quantum computing resources, HPC (coupled-channel methods, quantum master equations, Monte Carlo)	Grad/undergrad through Physics dept		https://tscherbul.github.io/	https://www.unr.edu/physics/people/
Physics	Weinstein Lab	Lab	Jonathan Weinstein	Cryogenic cooling and laser spectroscopy of atoms/molecules trapped in solid neon and parahydrogen at near-absolute-zero; quantum sensing; dark matter searches	5	37	Spin coherence and optical properties in solid parahydrogen Enhanced spin coherence of rubidium atoms in solid neon Ultralong spin coherence times via dynamical decoupling	Cryostats, laser systems, optical tables, vacuum chambers, magnetic shielding, spectrometers, matrix isolation apparatus	Grad/undergrad through Physics dept	Quantum sensing and fundamental physics; Harvard-trained PI; bridges physics and astronomy	https://www.weinsteinlab.org/	https://www.unr.edu/physics/jonathan-
Physics	Zebra Pulsed Power Lab (ZPPL)	Lab	Thomas White (lead), Roberto Mancini, Hiroshi Sawada, Bernhard Bach	Highest-power pulsed electrical device at any US university; Z-pinch physics, HED science, laser-plasma interactions; 153,000 sq ft SAGE facility	96	1694	Two-terawatt Zebra Z-pinch at Nevada terawatt facility Staged Z-pinch experiments at 1MA Zebra generator Measurement of ionization state of wire-array z-pinch	Zebra (2 TW pulsed power), Leopard laser (50 TW, 300 fs), Falcon optical compressor, X-ray diagnostics, streak cameras,	Grad/undergrad through Physics dept; DOE/NNSA supported; SAGE building in Stead, NV	World-class pulsed power facility; unique in US university system; Core diagnostics.	https://www.unr.edu/physics/research/zppl	https://www.unr.edu/physics/research/
Astronomy	Fleischmann Planetarium & Science Center	Outreach	UNR (public facility)	Historic planetarium (opened 1963); 360-degree digital projection; interactive exhibits (meteorites, black hole demo, Earth/moon globes); teacher training	0	0		Digital planetarium projection system, interactive exhibits, meteorite collection	Open to public ($3 admission); Fri-Sun hours; school field trips		https://www.unr.edu/planetarium/	https://www.unr.edu/planetarium/about
Astronomy	Great Basin Observatory (GBO)	Observatory	Richard Plotkin (UNR PI), multi-institutional	0.7-meter robotic remotely-operated telescope in Great Basin National Park (elev. 6,825 ft, 70+ mi from light pollution); NSF-funded astrophysics and K-12 education	69	335	User-guided photometric pipeline for GBO robotic telescope Photometry development for GBO telescope Robotic telescope black hole accretion observations	0.7m research-grade robotic telescope, CCD cameras, photometry pipeline, remote operation software	UNR faculty/student access as partner institution; K-12 outreach programs	Research-grade remote telescope; direct astronomy olympiad alignment; accessible for observations	https://greatbasinobservatory.org/	https://www.unr.edu/physics/richard-
Astronomy	MacLean Observatory	Observatory	Department of Physics	Twin-dome facility at Redfield Campus: 22-inch MacLean telescope (largest educational telescope in Nevada) and 14-inch Jones telescope; digital remote control	5	80		22" MacLean telescope, 14" Jones telescope, CCD imaging, digital remote control	Open to UNR students, community colleges, K-12, and public (clear Friday evenings)	Very accessible; public viewing nights; hands-on telescope experience for astronomy olympiad	https://www.unr.edu/physics/people	https://www.unr.edu/planetarium

Chemistry	Borotto Research Group	Lab	Nicholas Borotto	Mass spectrometry-based detection of biomolecules; protein structure characterization; post-translational modification analysis	36	168	The path forward for protein footprinting covalent labeling and mass spec Investigating therapeutic protein structure with diethylpyrocarbonate labeling Covalent labeling with diethylpyrocarbonate for studying protein	timsTOF Pro2, ion mobility MS, tandem MS, IRMPD, CO laser-equipped MS, LC separations	Open to grad and undergrad students	Advanced mass spec directly relevant to SIL instruments and proteomics. Organized. Toys list.	https://www.unr.edu/chemistry/nicholas-borotto	https://nborotto.wixsite.com/borottolab/instrument
Chemistry	Environmental Engineering Lab	Research Lab	Krishna Pagilla, Eric Marchand (Civil & Environmental Engineering)	Water/wastewater treatment, environmental remediation, soil contamination analysis, and sustainable water resource management research.	58	651	Nutrient removal in membrane bioreactors for wastewater treatment Soil remediation strategies for mining-impacted sites Water reuse optimization for arid climate regions	ICP-OES, TOC analyzer, membrane bioreactors, pilot-scale treatment systems, soil columns, analytical balances	Through Civil & Environmental Engineering; graduate research		https://www.unr.edu/cee/research/research-	https://www.unr.edu/cee/people/krishna-
Chemistry	Fluid Inclusions Lab	Research Lab	Matthieu Harlaux (Director, Economic Geology)	Microthermometric and microanalytical study of fluid inclusions in minerals for understanding hydrothermal ore deposit genesis, geothermal systems, and geological processes.	24	209	Geochemical signature of magmatic-hydrothermal fluids in rare-metal granites Sequential LA-ICP-MS for fluid inclusion trace element analysis Fluid inclusion constraints on ore-forming hydrothermal systems	Linkam heating/freezing stages (-200 to +600°C and up to 1400°C), Raman spectrometer, Olympus BX51 microscope with NIR	Through Geological Sciences; courses GEOL 713/776; contact mharlaux@unr.edu	Toys list.	https://nbmg.unr.edu/staff/Harlaux.html	https://www.unr.edu/geology/research/labs-and-
Chemistry	Geary Lab	Lab	Laina Geary	Palladium-catalyzed cross-coupling, C-H functionalization, Pauson-Khand cycloadditions, alcohol substitution for C-C/C-X bonds	5	3	Recent advances in the Pauson-Khand reaction Nitrous oxide-promoted Pauson-Khand cycloadditions Regioselectivity in Pauson-Khand reaction of enynes	GC-MS, LC-MS, NMR, column chromatography, Schlenk lines, glovebox	Open to grad and undergrad students	Hands-on organic synthesis; uses SIL GC-MS/LC-MS for characterization	https://lgeary.wixsite.com/gearylab	https://www.unr.edu/chemistry/laina-
Chemistry	Hitchcock Center for Chemical Ecology	Center	Christopher Jeffrey (Director), Lora Richards, Matthew Tucker (Co-directors)	20+ faculty across chemistry, biology, ecology, agriculture; studies plant defensive chemistry, insect diversity, bio-prospecting in Great Basin; $3M NSF NRT grant	59	328	Phytochemical diversity drives plant-insect community diversity Structure revision of scytonemin imine and its relationship to scytonemin Isolation of new neolignans and unusual meroterpenoid	Shared analytical instruments (LC-MS, GC-MS, NMR), synthesis labs, field collection equipment, metabolomics pipelines	NRT fellowships (25+ supported); open to grad and undergrad; interdisciplinary training	Premier interdisciplinary center bridging chemistry + biology; NRT fellowships; uses SIL instruments	https://www.unr.edu/chemical-ecology	https://www.unr.edu/chemistry/christophe
Chemistry	King Group	Lab	Benjamin King	Carborane chemistry, fluorine chemistry, weakly coordinating anions, PAH synthesis, Scholl reactions	3	156	Investigation of the mechanism of the intramolecular Scholl reaction Controlling the Scholl reaction for PAH synthesis A perfluorinated nanosphere: perfluoro-deca-B-methyl-para-carborane	X-ray crystallography, NMR, glovebox (air-free), mass spectrometry, computational clusters	Open to grad and undergrad students		https://kingchem.org/	https://www.unr.edu/chemistry/benjamin-
Chemistry	Leitner Research Group	Lab	David Leitner	Theoretical/computational: energy flow in biomolecules, thermal transport, protein dynamics, terahertz spectroscopy	25	179	Energy flow in proteins Heat transport in molecules and reaction kinetics: energy flow and localization Energy transport across interfaces in biomolecular systems	HPC clusters, GROMACS, NAMD, spectral analysis software	Open to grad students (computational focus)		https://leitnergroup.wixsite.com/dmlpage/david-	https://www.unr.edu/chemistry/david-m-
Chemistry	Materials Characterization Nevada (MCNV)	Shared Facility	Dev Chidambaram, Krista Carlson (contacts)	Advanced materials characterization facility housing world's second lab-based EXAFS system. XRD, XRF, thermal analysis, surface characterization for materials science and engineering research.	95	456	EXAFS characterization of novel battery electrode materials XRD phase analysis of advanced ceramic composites Thermal stability of nanostructured alloys via DSC/TGA	Lab-based EXAFS system, XRD, XRF, DSC/TGA, surface area analyzers, nanoindenter, profilometer	Fee-for-service through NCAR; open to UNR, external, and industry users	World's second lab EXAFS; complements SIL chemistry capabilities. Toys list.	https://www.unr.edu/ncar/infrastructure/mcnv	https://www.unr.edu/cme/people/dev-
Chemistry	Microbeam & SEM Laboratory (Mackay School)	Core Facility	Joel DesOrmeau (Manager), Krista Carlson	State-of-the-art microanalysis facility with NSF-funded field-emission EPMA (first in US), FE-SEM, and LA-ICP-MS for geological, materials, and environmental sample analysis at micrometer resolution.	65	356	GHR1 Zircon: a new natural reference material for microbeam U-Pb geochronology Rapid exhumation of Papua New Guinea UHP terrane via eclogite retrogression High-precision microanalysis of mineral phases in metamorphic rocks	JEOL JXA iHP200F FE-EPMA (first in US), JEOL JSM 7100FT FE-SEM, LA-ICP-MS, 5 WDS spectrometers,	Fee-for-service; open to UNR, other universities, and industry; contact jdesormeau@unr.edu	First FE-EPMA installed in US; NSF-funded; critical analytical infrastructure. Toys list.	https://www.unr.edu/mackay/about/microbeam-	https://nbmg.unr.edu/microbeamlab/
Chemistry	Mineral Processing Lab	Research Lab	Pengbo Chu, Barbara Wold (Mining & Metallurgical Engineering)	Ore processing, flotation chemistry, hydrometallurgy, and mineral characterization research supporting Nevada's mining industry. Part of Mackay School.	32	445	Flotation optimization for complex sulfide ores Hydrometallurgical recovery of rare earth elements Mineral liberation analysis using automated mineralogy	Flotation cells, ball mills, jaw crushers, leaching reactors, mineral liberation analyzer, assay furnaces	Through Mining & Metallurgical Engineering; industry partnerships	Toys list.	https://sites.google.com/view/chu-group/home	https://www.unr.edu/mining/pengbo-
Chemistry	Shared Instruments Laboratory	Core Facility	SIL / Chemistry Department	Liquid-state and solid-state NMR for structural characterization of organic, inorganic, and materials samples	0	0		3x Agilent Varian liquid-state NMR (including 400-MR with auto-switchable probe: 1H, 19F, 13C, 31P), Tecmag Discovery 400 MHz solid-	Through SIL; training required; iLab scheduling	NMR is essential for organic/inorganic chemistry characterization; complement to SIL instruments. Instructions	https://www.unr.edu/chemistry/research/shared-	http://argenta2.chem.unr.edu/
Chemistry	Nevada Institute for Sustainability	Research Center	Dev Chidambaram (Director), Vasquez (Associate Professor)	Multidisciplinary team within College of Engineering addressing sustainable energy production, battery technology, nuclear materials, and renewable energy research.	54	180	Battery technology and energy storage system optimization Sustainable energy production strategies for Nevada Materials science for next-generation renewable energy	Energy research labs, battery testing equipment, materials characterization, renewable energy demonstration systems	Engineering and science faculty; sustainability program students		https://www.unr.edu/cme/people/dev-chidambara	https://www.unr.edu/ncar
Chemistry	Ralph Roberts Center for Economic Geology (CREG)	Research Center	John Muntean (Director, Arthur Brant Endowed Chair)	30-year partnership between Mackay School, Nevada Bureau of Mines & Geology, mining industry, and USGS. Focuses on Carlin-type gold deposit genesis and mineral exploration methods.	4	2	Diversity of Carlin-Style Gold Deposits: applications to exploration The Carlin gold system: exploration applications in Nevada Geochemistry of metasedimentary rock-hosted Au-(Ag-Sb-W) deposits	Analytical lab access (SEM, EPMA, LA-ICP-MS at Mackay), field exploration equipment, mining industry partnerships, USGS	Graduate student thesis funding; mining industry internships; contact jmuntean@unr.edu	Premier economic geology center; 30-year mining industry partnership. Toys list.	https://nbmg.unr.edu/CREG/faculty.html	https://nbmg.unr.edu/creg/analytical.html
Chemistry	Shared Instruments Laboratory	Core Facility	Stephen Spain (Director)	Free-access core facility with mass spec, vibrational spectroscopy, thermal analysis, optical spectroscopy, chromatography; Room 016 Chemistry Building				LC-MS (Waters + Agilent TOF), MALDI-8020, Raman microscope, 2x FTIR (Nicolet 380 + Bruker Tensor 27), GC-MS, DSC, TGA,	Free for UNR students/ faculty after training; 24/7 card access; consumables user-provided; external users: fee-based through	You already have 24/7 access; comprehensive instrument suite across spectroscopy, MS, and thermal analysis	https://www.unr.edu/chemistry/research/shared-	http://argenta2.chem.unr.edu/
Chemistry	Tal-Gan Research Group	Lab	Yftah Tal-Gan	Peptide-based tools for bacterial quorum sensing; therapeutic implications for bacterial pathogenesis and virulence	35	1900	Highly potent inhibitors of quorum sensing in Staphylococcus aureus Targeting peptide-based quorum sensing systems for Gram-positive treatment Designing cyclic competence-stimulating peptide analogs with pan-group inhibition	Peptide synthesizer, HPLC, LC-MS, MALDI-TOF, microplate readers, cell culture, molecular microbiology setup	Open to undergrad, grad, and postdocs; NIH/NSF funded	Chemical biology; peptide synthesis; uses SIL LC-MS and MALDI; open to undergrads	https://talgan.wixsite.com/researchgroup	https://www.unr.edu/chemistry/yftah-tal-
Chemistry	Tucker Research Group	Lab	Matthew Tucker	Ultrafast 2D IR spectroscopy of protein folding, binding, and function; also studies magnetism and energy dissipation in desert lichen molecules	6	80	Nonequilibrium dynamics of helix reorganization observed by transient 2D IR Site-specific coupling of hydration water and protein flexibility via 2D-IR 2D IR spectroscopy of histidine: probing side-chain structure and dynamics	Femtosecond lasers, FTIR spectrometers, 2D IR setup, optical tables, cryostats	Open to grad and undergrad students	FTIR/spectroscopy directly relevant to SIL instruments; biophysics bridge. Toys list.	https://www.tuckergroupunr.org/	https://www.tuckergroupunr.org/equipment
Chemistry	X-ray Crystallography Facility	Core Facility	SIL / Chemistry Department	Single-crystal X-ray diffraction for molecular structure determination	0	0		Bruker Smart APEX diffractometer	Through SIL; sample submission; specialist operation		https://www.unr.edu/chemistry/shared-instrumenta	https://www.unr.edu/chemistry/research/

Chemistry	Yang Research Group	Lab	Ying Yang (NSF CAREER)	Dynamic polymers with reversible bonds for recyclable plastics, self-healing materials, 3D printing, wearable sensors, soft robotics	87	2082	A stretchable mechanically robust polymer with shape-memory-assisted healing Intrinsic self-healing polymers with high E-modulus based on dynamic urea bonds Dynamic polymers with recyclable polydithioacetal composition	Polymer synthesis equipment, 3D printers, mechanical testing, spectroscopic characterization	Open to grad and undergrad; 2025 Foundation Early Career Innovator	New faculty building lab; dynamic materials are hands-on; likely open to motivated students	https://www.yanglab.space/	https://www.unr.edu/chemistry/people/
Chemistry	de Bettencourt-Dias Group	Lab	Ana de Bettencourt-Dias	Luminescent lanthanide complexes; f-element chemistry for energy efficiency, diagnostics, therapy	28	2400	Luminescent lanthanide complexes with dual activity for photodynamic therapy Lanthanide luminescence for biomedical analyses and imaging Ligand design in lanthanide complexes for luminescence therapy and sensing	Fluorescence spectrometers, UV-Vis, X-ray crystallography, glovebox, NMR, CD spectrometer	Open to grad and undergrad students		https://www.unr.edu/chemistry/ana-de-bettencourt-	
Biology	Aquatic Ecosystems Analysis Lab	Lab	Sudeep Chandra, Zeb Hogan	Limnology, aquatic conservation, invasive species management, freshwater fish ecology; Castle Lake station (est. 1959)	114	2643	Lake Tahoe invasive species management and control Aquatic ecosystems conservation and food web ecology Invasive species impacts on aquatic ecosystem health	Water quality probes, CTD profilers, boats, field sampling, isotope analysis	Open to undergrad/grad; Castle Lake has trained 700+ undergrads since 2007		http://www.aquaticecosystemslab.org/	https://www.unr.edu/biology/sudeep-
Biology	Cardiovascular COBRE	Research Center	Scott Earley (PI/ Director, Professor of Pharmacology)	NIH COBRE-funded cardiovascular research center ($11M, 5-year). First cardiovascular basic science center in Nevada. Studies molecular signal transduction in cardiovascular system, TRP channels.	4	175	TRP channel signaling in vascular smooth muscle function Molecular mechanisms of cerebral blood flow regulation Ion channel regulation of cardiovascular system	Electrophysiology rigs, vascular biology labs, patch clamp systems, calcium imaging, pressure myography	UNR pharmacology and cardiovascular researchers; COBRE pilot funding	First cardiovascular COBRE in Nevada; $11M NIH funding	https://med.unr.edu/pharmacology/research	https://www.unr.edu/nevada-today/
Biology	Cellular and Molecular Imaging Core	Core Facility	Alexander van der Linden (Director), Andrew Yanez (Training Coordinator)	Part of Center for Integrative Neuroscience COBRE. Advanced microscopy, DNA/RNA/protein analysis, and cellular imaging on Fleischmann 3rd floor.	8	40	Confocal microscopy applications in neuroscience Two-photon imaging of neural tissue architecture Tissue clearing and 3D imaging techniques for brain analysis	Leica SP8 confocal with LIGHTNING, Scientifica two-photon, Leica THUNDER 3D, Typhoon Phosphoimager, PCR robot,	UNR researchers; iLab reservation system; training available	Toys list.	https://www.unr.edu/neuroscience/center/core-	https://www.vanderlindenlab.com
Biology	Center for Integrative Neuroscience (COBRE)	Research Center	70+ affiliated faculty across 13 departments	NIH COBRE-funded center (P30 Phase 3, $5.3M) since 2012. Operates molecular imaging, neuroimaging, and VR/AI cores. Supports interdisciplinary PhD in Integrative Neuroscience.	220	8800	Multi-modal neuroimaging studies of brain network organization Neural circuit mechanisms of sensory processing Computational neuroscience approaches to brain function	Confocal microscope (Leica SP8), two-photon microscope, THUNDER 3D imaging, VR/AR systems, fMRI, EEG, $9M invested in	70+ faculty in 13 departments; pilot project funding; PhD program	Major NIH-funded neuroscience center; $9M equipment investment; Phase 3 COBRE. Toys list.	https://www.unr.edu/neuroscience/center	https://www.unr.edu/neuroscience
Biology	Center for Molecular Medicine (UNR Med)	Research Center	Multiple department heads (Microbiology, Pharmacology, Pathology, Physiology, Cell Biology)	107,000 sq ft research facility housing wet labs, rodent vivarium, BSL3 select agent lab. Programs in cancer biology, infectious disease, cardiovascular disease, and neurosciences.	120	4800	Cell biology signaling mechanisms in cardiovascular disease Neuro-immune disease pathogenesis and biomarkers Infectious disease mechanisms in BSL3 containment studies	BSL3 select agent lab, rodent vivarium, wet labs, medical office suites, cell culture facilities, microscopy	UNR Med faculty and students; partnership with Renown Health	Major UNR Med research hub; BSL3 capability; 107,000 sq ft	https://med.unr.edu/research/core-facilities-centers	https://www.unr.edu/around-campus/
Biology	Global Water Center	Center	Sudeep Chandra (Director)	Multidisciplinary center for water sustainability; research on every continent; translates science into actionable solutions	72	1802	Lake Tahoe invasive species management and control Water sustainability and aquatic ecosystems conservation Global freshwater solutions for sustainability	Water quality instruments, drone technology, monitoring equipment, predictive modeling	Open to faculty/students across disciplines; NGO partnerships		https://www.unr.edu/water-center	https://www.unr.edu/water-center/people/
Biology	Great Basin Bug Lab	Lab	Matt Forister	Plant-insect ecology, butterfly ecology, diet breadth evolution, phenology, conservation; Science 2025 paper on US butterfly declines	7	54	Increasing neonicotinoid use and butterfly fauna decline Dramatic decline in western butterfly populations linked to pesticides Climate change is a stressor of insect populations worldwide	Field collection equipment, specimen collections, molecular tools, genomic sequencing, ecological software	Open to PhD, Master's, and undergrad students; field + lab research	Very high impact PI (Science 2025); accessible fieldwork; biodiversity + conservation	https://sites.google.com/site/greatbasinbuglab/	https://www.unr.edu/biology/matt-
Biology	Hitchcock Center for Chemical Ecology	Center	Christopher Jeffrey, Lora Richards, Lee Dyer, Matt Forister, Angela Smilanich	See Chemistry listing — 20+ faculty studying plant-insect chemical ecology across both departments	183	1737	Phytochemical diversity drives plant-insect community diversity Structure revision of scytonemin imine and its relationship to scytonemin Isolation of new neolignans and unusual meroterpenoid	See Chemistry listing	NRT fellowships; open to grad and undergrad		https://www.unr.edu/chemical-ecology	https://www.unr.edu/nevada-today/
Biology	Hoy Lab	Lab	Jennifer Hoy	Visual neuroscience, prey capture neural circuits, sensory systems; NIH RO1 (~$2M/5yr)	9	41	Defined cell types in superior colliculus prey capture Layer-specific refinement of visual cortex after eye opening Identification of a brainstem circuit regulating visual state	Electrophysiology, optogenetics, 2-photon microscopy, behavioral tracking, mouse colony	Through Institute for Neuroscience		https://www.unr.edu/biology/jennifer-hoy	http://www.hoylab.com/publications.html
Biology	Institute for Neuroscience	Institute	Jennifer Hoy (Director), Dennis Mathew, multiple faculty; 70+ affiliated faculty	Multidisciplinary neuroscience center; $25M NIH COBRE funding across 3 phases; 40 active grad students; doctoral program	21	93	Defined cell types in superior colliculus prey capture Layer-specific refinement of visual cortex after eye opening Identification of a brainstem circuit regulating visual state	HD EEG, 2x 3T MRI (at Renown), NIRS, Leica SP8 confocal, Scientifica 2-photon microscope, Leica THUNDER 3D, VR/AR	Grad students in Integrative Neuroscience Program; core facilities available to affiliates	Major neuroscience institute; 2-photon and confocal microscopy; Biology olympiad neuro alignment	https://www.unr.edu/neuroscience	https://www.unr.edu/neuroscience/
Biology	Mathew Lab	Lab	Dennis Mathew	Neuroscience of olfaction in Drosophila; neural circuit function; neuromodulation; NIH ENDURE training program	12	52	Internal state affects local neuron function in olfactory behavior Behavior responses to chemical and optogenetic Drosophila stimuli Differential contributions of olfactory receptor neurons to circuit	Drosophila culture, electrophysiology rigs, fluorescence microscopy, behavioral arenas, optogenetics	Open to grad and undergrad; runs NIH training program; co-directs Integrative Neuroscience	Drosophila accessible model; NIH training program for undergrads; neuro olympiad prep. Github.	https://mathewlab.com/home/dennis-mathew/	https://med.unr.edu/people/dennis-
Biology	Museum of Natural History	Museum	Chris Feldman (Curator of Vertebrates), multiple curators	Irreplaceable specimen collections of Nevada/Great Basin plants and animals; among the oldest and most extensive Nevada collections	23	92		Plant herbarium, insect collections (local + tropical), vertebrate collections (mammals, birds, herps)	Open to researchers and students; undergrad research opportunities in collections		https://www.unr.edu/natural-history	https://www.unr.edu/biology/chris-
Biology	Nevada Agricultural Experiment Station	Research Center	Chris Pritsos (Associate Dean), CABNR department heads	Land-grant research facility since 1887 (Hatch Act). Six field stations throughout Nevada for agriculture, ranching, environmental stewardship, and natural resource management.			Sustainable agricultural practices for arid Nevada climate Rangeland conservation and livestock management Water-efficient crop production in the Great Basin	Six field stations statewide, Valley Road Field Lab, agricultural equipment, soil analysis, greenhouses, growth chambers	CABNR faculty and students; Nevada agricultural community; field stations available	Historic land-grant research; statewide network of field stations	https://naes.unr.edu/	https://www.unr.edu/cabnr
Biology	Nevada INBRE (IDeA Network of Biomedical Research Excellence)	Research Network	NIH-funded program directors, multi-institution faculty	NIH program building biomedical research infrastructure across Nevada. Supports early-stage investigators, research labs, data science core, and undergraduate pipeline programs at 5+ institutions.	180	3600	Bioinformatics and computational biology methodologies for Nevada researchers Biomedical research capacity building in EPSCoR states Undergraduate research pipeline programs in biomedical sciences	Bioinformatics servers, computational platforms, data science infrastructure, research funding	Faculty and students at participating Nevada institutions; undergraduate fellowships	Statewide biomedical research infrastructure; NIH-funded. Core Service Awards. Core list.	https://www.unr.edu/nevada-inbre	https://med.unr.edu/inbre
Biology	Ouyang Lab	Lab	Jenny Ouyang	Urban ecology, artificial light pollution effects on birds, circadian biology, physiological adaptation; NSF CAREER $1.2M	37	244	Restless roosts: light pollution affects behavior and sleep physiology Effects of dim artificial light on circadian clock genes in songbird Birds of a feather flock together: social context and light pollution	Light sensors, hormone assay equipment, field monitoring stations, spectroradiometers	Open to grad and undergrad students	NSF CAREER; accessible field biology; bridges biology + physics via light/circadian	https://www.jennyouyang.com/	https://www.unr.edu/biology/jenny-

Biology	Parchman Lab	Lab	Thomas Parchman	Population genomics: adaptive divergence, hybridization, speciation in conifers, tropical Piper, birds, and fish	57	429	Genome divergence and diversification in geographic mosaic of coevolution Coevolution, local adaptation, and ecological speciation in birds Resource stability and geographic isolation drive genome divergence	Molecular biology lab, genomic sequencing, computational resources for genome-scale data	Open to grad and undergrad students		https://parchmanlab.com/	https://www.unr.edu/biology/thomas-
Biology	Tahoe Center for Environmental Sciences	Research Center	Sudeep Chandra (UNR Director), UC Davis partnership	LEED Platinum collaborative facility between UC Davis and UNR on Lake Tahoe campus. Environmental research, public engagement, and workforce training for Tahoe Basin ecology and sustainability.	72	1802	Lake Tahoe watershed and aquatic ecosystem dynamics Plastic debris characterization in freshwater environments Algal community composition and environmental indicators	Limnology lab, water sampling systems, aquatic culture facilities, Lake Tahoe field monitoring stations, GIS systems	UC Davis and UNR researchers; external collaborators; Lake Tahoe field sites	LEED Platinum; UC Davis partnership; Lake Tahoe living laboratory	https://www.unr.edu/tahoe-institute	https://www.unr.edu/biology/sudeep-
Biology	Voyles Lab	Lab	Jamie Voyles (McMinn Research Professor)	Emerging infectious diseases in amphibians; chytridiomycosis; conservation biology; field research in Panama since 2003	31	260	Pathogenesis of chytridiomycosis causing catastrophic amphibian declines Amphibian fungal panzootic causes catastrophic ongoing biodiversity loss Shifts in disease dynamics in tropical amphibian assemblage	Pathogen research facilities, physiological analysis, field stations (Panama: Smithsonian SCBI)	Open to grad and undergrad; international field opportunities		https://voyleslab.weebly.com/	https://www.unr.edu/biology/faculty/
Core Facility	Cell Analysis Core (Flow Cytometry)	Core Facility	School of Medicine	Flow cytometry and cell sorting; spatial molecular imaging; cell cycle analysis, proliferation assays	0	0		WOLF microfluidics sorter, Cytoflex 3-laser cytometer, CosMx spatial molecular imager, FlowJo	Fee-based; iLab scheduling; School of Medicine	Toys list.	https://med.unr.edu/research/core-facilities-	
Core Facility	Core Analytical Laboratory (CAL)	Core Facility	College of Agriculture + USDA ARS	Analytical chemistry for agricultural, environmental, and plant science; elemental analysis, chromatography	0	0		ICP-AES, ICP-MS, GC (MS/FID/ECD), elemental analyzer, flow injection, UV-Vis, microwave digestion, particle size	Students can train to operate independently; staff can analyze for fees; open to all researchers	Toys list.	https://naes.unr.edu/cal/default.aspx	
Core Facility	Davidson Cleanroom / Nanofab Lab	Core Facility	Russ Renzas (Director)	Class 1,000 semiconductor/photonic fabrication; lithography, deposition, etch, metrology, wire bonding; 2,937 sq ft				Lithography systems, deposition tools, etching equipment, metrology instruments, wire bonder	UNR researchers; external: industrial rates; Pennington Engineering Bldg; opened April 2025	Nanofabrication for quantum/photonic devices; Physics olympiad alignment. Toys list.	https://www.unr.edu/engineering/research/	
Core Facility	Electron Microscopy Facility (EMMF)	Core Facility	Yufeng Zheng (Director)	Scanning and transmission electron microscopy for metals, ceramics, polymers, composites, biological samples	4	1352		Talos F200S S/TEM with EDS, Scios 2 FIB/SEM, FEI Nano600 NanoSEM, sample prep instruments	Contact EMMF; fee-based; iLab scheduling	Electron microscopy gives nanoscale imaging across all sciences; complements SIL optical methods. Toys list.	https://www.unr.edu/ncar/infrastructure/emmf	
Core Facility	Neuroimaging Core	Core Facility	Center for Integrative Neuroscience	MRI/fMRI and neurophysiology; human and small animal imaging	0	0		2x 3T MRI (Philips Ingenia + Siemens Skyra at Renown), 2x HD EEG, NIRS (TechEn)	Available to affiliated faculty/students; Renown South Meadows location	Toys list.	https://www.unr.edu/neuroscience/center/core-	
Core Facility	Nevada Bioinformatics Center	Core Facility	Hans Vasquez-Gross (Director)	Bioinformatics services: NGS analysis, single-cell RNA-seq, transcriptomics, genome resequencing, proteomics data, workshops	0	0	Genomic surveillance of Nevada SARS-CoV-2 variants Plant genomics and water-use efficiency in crassulacean metabolism Cellular paraspeckle component in viral processivity	HPC clusters, bioinformatics software, NGS analysis pipelines, computational workstations	Pay-per-service; workshops open to all; iLab scheduling	Toys list.	https://www.unr.edu/bioinformatics	
Core Facility	Nevada Genomics Center	Core Facility	Andrew Gorzalski (Director)	DNA sequencing core: Illumina NGS, 10X single-cell, Sanger, fragment analysis; serves 132 Nevada labs + 250 labs nationally/internationally	26	341		Illumina NextSeq 2000, MiniSeq, Sanger sequencing, fragment analysis, BSL-2 lab	Pay-per-service; open to all researchers (academic + industry); iLab scheduling	Genomics is central to Biology olympiad; accessible pay-per-service model. Toys list.	https://www.unr.edu/genomics	
Core Facility	Nevada Proteomics Center	Core Facility	David Quilici (Director)	Mass spectral proteomic analysis; protein identification, quantitation, and PTM analysis from blood, saliva, tissue	8	47	Quantitative proteomics unveils alterations in neuropathic nerves Mass spectrometry analysis for clinical diagnostics Proteomics of physiological pathways and molecular interactions	Bruker timsTOF Pro2, ThermoFisher Eclipse MS, Evosep One LC, gel/IP/solution workflows	Pay-per-service; open to all researchers (academic + industry); iLab scheduling	Proteomics bridges chemistry + biology; advanced mass spec complements SIL instruments. Toys list.	https://www.unr.edu/proteomics	https://www.unr.edu/proteomics/contact/
Core Facility	Nevada Stable Isotope Lab (NSIL)	Core Facility	Simon Poulson	C, O, H, N, S stable isotope analysis for geological, environmental, and biological samples	25	144	Environmental low temperature geochemistry via stable isotopes Stable isotope analysis of carbon, oxygen, hydrogen, nitrogen, sulfur Isotope geochemistry research applications in geosciences	Stable isotope ratio mass spectrometers, sample preparation systems	Fee-based through NCAR; ncar@unr.edu	Toys list.	https://www.unr.edu/ncar/infrastructure/nsil	
External	Biosciences Entrepreneurial Lab (BEL)	Incubator	NCAR	Turnkey wet lab incubator (2,200 sq ft) for biotech startups; fully furnished with standard lab equipment	0	0		Complete wet lab infrastructure, standard lab equipment, shared analytical instruments	Through NCAR incubation program; monthly membership or dedicated bench rental	Toys list.	https://www.unr.edu/ncar/infrastructure/bel	
External	DeLaMare Makerspace	Maker Space	DeLaMare Science & Engineering Library	3D printing, laser cutting, PCB milling, soldering, prototyping; free for campus community	0	0		3D printers/scanners, laser cutter, vinyl cutter, PCB mill, soldering stations, sewing stations, hand tools	Free for all UNR community; training required; 180 min/day reservation; (775) 682-7385	Free access; 3D printing and prototyping for any science project; very accessible. Toys list.	https://library.unr.edu/places/delamare/themakerspa	
External	Desert Research Institute (DRI)	Research Institute	500+ research faculty/staff; Reno + Las Vegas campuses	Nonprofit sister institution to UNR (NSHE); 40+ specialized labs; atmospheric, earth/ecosystem, hydrologic sciences; >$50M annual research	800	35000	Mineral dust aerosol size distribution and Earth system impacts Dust emissions characterization and atmospheric visibility impact Wildfire smoke air quality and aerosol composition analysis	Atmospheric science instruments, hydrologic measurement, environmental monitoring, LASSO optical/aerosol	Research collaboration with UNR; some facilities accessible through partnerships; Dandini Research Park, 6 mi from	Major research institution adjacent to UNR; atmospheric science labs directly relevant to Physics	https://www.dri.edu/	
External	Innevation Center	Innovation Hub	UNR-affiliated	Startup incubator with co-working, makerspace, conference rooms, mentorship; access to NCAR facilities	0	0		Makerspace tools, co-working spaces, conference rooms	Free for UNR students/faculty/staff; community: paid membership; 450 Sinclair St	Toys list.	https://www.unr.edu/innevation	
External	NCAR (National Center for Atmospheric Research)	National Research Center	NSF-sponsored (headquarters: Boulder, CO)	Climate science, meteorology, atmospheric chemistry, space weather; no permanent Reno facility but collaborative partnerships with UNR on drought/climate	0	0	NCEP/NCAR 40-year reanalysis project for global climate record Indicators of global climate change and temperature emissions AMOC Atlantic meridional overturning circulation tipping point	NCAR-Wyoming Supercomputing Center, Mauna Loa Observatory, field campaign instruments	Collaborative partnerships; no direct Reno presence; data and models freely available		https://ncar.ucar.edu/	

External	NNSS (Nevada National Security Site)	National Lab	DOE/NNSA (110 mi NW of Las Vegas)	US Department of Energy national laboratory; nuclear security, environmental management, emergency response	0	0		DOE research equipment, national security infrastructure	12-week paid summer internships (70+ students/yr); apply at nnss.gov/careers/student-programs		https://nnss.gov/	
External	Nevada Center for Applied Research (NCAR)	Applied Research Center	UNR (600+ companies engaged since 2013)	Coordinates access to all 40+ UNR shared research facilities; supports startups; $201M venture capital raised by affiliates	0	0		Access to all UNR core facilities, Biosciences Entrepreneurial Lab (wet lab incubator), Mechanical Engineering Manufacturing	External companies/ researchers: coordinate through ncar@unr.edu, (775) 784-4781; paid access; iLab platform for scheduling	Gateway to all UNR equipment for paid access; NCAR is the main contact for external users	https://www.unr.edu/ncar	
External	Nevada Seismological Laboratory	Research Lab	Christie Rowe (Director)	186-station seismic network across Nevada/E. Sierra; 105 fire watch cameras; Earthquake Early Warning; 50+ years of data			Rapid seismic monitoring deployment for earthquake sequence Nevada earthquake detection and monitoring network Earthquake early warning system for Nevada seismic monitoring	Seismometers (186-station network), fire watch cameras (105), EEW systems, seismic data archives	Research collaboration; data publicly available; Mackay School of Earth Sciences		https://www.unr.edu/nevada-seismo-lab	http://www.seismo.unr.edu/

science	professor	title	year	school	subject	total	impact	publications	affiliations	toys	reasoning	w	w
Mathematics	Deena Schmidt	Associate Professor of Mathematics & Statistics		Cornell University (2007)	Mathematical biology, applied probability, networks (neuro/epi), population genetics	4	31	Sleep-wake dynamics in mammalian development (2024); Stochastic models in neuroscience and epidemiology; NSF and Simons Foundation funded network research	Mathematical Biology Group; Integrative Neuroscience Program;	Computational biology software, network analysis tools, HPC	TechWise Program director; math+computing+data science bridge; Google partnership; accessible to	www	www
Mathematics	Minggen Lu	Professor of Health Analytics & Biostatistics	2010		Biostatistics, health analytics, nonparametric regression, survival analysis, longitudinal data			Advanced statistical methods for high-dimensional biomedical data Panel count data analysis in epidemiology Machine learning approaches to health analytics and prediction	School of Public Health	Statistical computing, HPC, ML platforms, biostatistical tools	Health analytics+ML; bridges math+computing; high-dimensional data methods		www
Mathematics	Thomas Quint	Foundation Professor of Mathematics & Statistics; 2025	1996	Stanford University (1988)	Game theory, mathematical economics, graph theory, Shapley values	2	20	Measures of Powerlessness in Simple Games; Evaluating Presidential Elections Using Game Theory; ~30 journal articles and 2 books on game theory		Computational software			www
Computing	Alireza Tavakkoli	Associate Professor of Computer Science & Engineering	2023	University of Nevada, Reno (2009)	Computer vision, robot vision, VR/AR/MR, biomedical imaging, remote sensing	200	2048	Guided/Unguided Conditional Diffusion for 3D Point Cloud Generation (2025); Bayesian Graphical Models for Joint Degeneration Analysis (2024); Feature Representation for Retinal Disease Detection	Computer Vision Laboratory (CVL); Integrative Neuroscience Faculty	GPU clusters, VR/AR headsets, 3D scanners, medical imaging systems, cameras	BRAIN Core ML as a service	www	www
Computing	Anikita Shukla	Assistant Professor of Computer Science & Engineering	2024	IIIT-Delhi (2021)	Computer vision, deep learning, geometric methods, AI for science, AI for wildlife conservation	30	533	XPRIZE Rainforest winner ($5M, 1st place); Polynomial Implicit Neural Representations (CVPR 2023); Tiger re-identification; Geometric priors for ICF generative models (NeurIPS 2021 workshop)		GPU clusters, cameras, imaging sensors	AI for Science focus; postdoc at ASU; XPRIZE Rainforest team; open to undergrads; directly relevant to ML research	www	www
Computing	David Feil-Seifer	Associate Professor of Computer Science & Engineering	2013	USC Viterbi (2011)	Socially assistive robotics, human-robot interaction, robotics for ASD intervention	45	167	Defining Socially Assistive Robotics (seminal work); Toward SAR for Augmenting ASD Interventions; Benchmarks for evaluating socially assistive robots	Social Robotics Lab	Social robots (NAO, Pepper-type), sensors, actuators, eye-tracking, ML hardware	Exceptional undergrad mentoring (100+ undergrads); motion capture system; accessible	www	www
Computing	Emily Hand	Assistant Professor of Computer Science & Engineering	2018	University of Maryland (2018)	Computer vision, human perception, ML, facial recognition, assistive technology	28	120	Transformation of face recognition technology; Wearable devices for facial recognition; Prior positions at NASA Ames, JPL, Naval Research Lab	Machine Perception Lab; Integrative Neuroscience Faculty	GPU clusters, cameras, wearable prototyping, ML infrastructure	ML/vision with social impact; NASA background; open to undergrads	www	www
Computing	Feng Yan	Assistant Professor of Computer Science & Engineering	2016	College of William and Mary (2016)	ML systems, deep learning, neural architecture search, federated learning, serverless computing	77	2275	Noctua: Automated Fine-grained Consistency Analysis (EuroSys 2024); Gradient similarity for large-scale training (NeurIPS 2021); Gradient compression for DNN training (SOSP 2021)	Intelligent Data & Systems (IDS) Lab; $20M NSF Wildfire project	GPU/TPU clusters, cloud computing infrastructure, distributed systems	ML systems; leads ML on $20M NSF wildfire project; directly relevant to Computing olympiad	www	www
Computing	George Bebis	Foundation Professor of Computer Science & Engineering		University of Central Florida (1996)	Computer vision, image processing, pattern recognition, machine learning, evolutionary computing	39	200	On-road vehicle detection: a review (934 citations); Vision-based hand pose estimation: A review (682 citations); 3D point cloud classification and mammography analysis	Computer Vision Laboratory (CVL)	GPU clusters, cameras, image sensors, ML infrastructure	Major CV lab; GPU access; open to undergrads; directly relevant to Computing olympiad	www	www
Computing	Hao Xu	Associate Professor of Electrical & Biomedical Engineering	2016	UNR College of Engineering	Autonomous systems, intelligent control, machine learning, cyber-physical systems, unmanned aircraft	63	1349	Research in intelligent control and autonomous systems LiDAR-based applications and intelligent transportation systems CREDIT+: Advancing Data Analysis for Mission-	EBME; IEEE	LiDAR systems, UAS platforms, autonomous vehicle testbed		www	www
Computing	Hung Manh La	Associate Professor of Computer Science & Engineering	2014	Oklahoma State University (2011)	Robotics, autonomous systems, multi-robot control, infrastructure inspection, wearable computing			A Climbing Robot for Steel Bridge Inspection (J Intelligent Robotic Systems); Agile robotic inspection: bicycle-like approach (J Field Robotics 2024); Adaptive Flocking Control of Multiple UGVs by UAV	Advanced Robotics & Automation (ARA) Lab	Mobile robots, UAVs/drones, climbing robots, wearable sensors, bridge inspection platforms	Hands-on robotics; award-winning bridge inspection robots; field experience	www	www
Computing	Jun Zhang	Assistant Professor of Mechanical Engineering	2018	UNR College of Engineering	Control theory, robotics, smart materials, artificial muscles, soft robots, assistive biomimetic systems	200	3384	Robotic Artificial Muscles: Current Progress and Future Perspectives Compliant, Large-Strain, and Self-Sensing Twisted String Actuators Three-Dimensional Hysteresis Modeling of Robotic	ME; IEEE; ASME	Smart materials testing, soft robotics fabrication, actuator testing		www	www
Physics	Daniel Trugman	Associate Professor of Physics; of Geophysics	2022		Seismology, earthquake rupture, ML in seismology, earthquake nucleation, induced seismicity	92	928	GrowClust: software for optimizing earthquake hypocenter locations Machine learning approaches to earthquake source characterization Earthquake early warning methodologies	Geological Sciences; Nevada Seismological Lab	Seismic waveform analysis, GrowClust software, ML platforms	ML in seismology; new faculty (2022); potentially accessible	www	www

Physics	Hiroshi Sawada	Associate Professor of Physics	2012	Osaka University (est.)	HED physics, short-pulse laser-matter interactions, laser-driven particles, ICF, x-ray diagnostics	3	125	Ultrafast plasma dynamics in laser-irradiated nanowire arrays (Sci Reports); Magnetized fast isochoric laser heating (Nature Comms 2018); Ultrafast 2D imaging of fast electron transport (Rev Sci Instrum 2023)	HED Physics Group; Zebra Pulsed Power Lab	Zebra pulsed power machine, high-power lasers, x-ray cameras, plasma diagnostics	Assessible.	www	www
Physics	James Hanna	Associate Professor of Mechanical Engineering		UNR College of Engineering	Classical mechanics, nonlinear dynamics, elastic structures, soft matter physics	18	63	Nonlinear dynamics of elastic rods and filaments Classical mechanics of soft structures Elastic instabilities and buckling phenomena	ME	High-speed imaging, mechanical testing	Interesting research.	www	www
Physics	Jeongwon Park	Associate Professor of Electrical & Biomedical Engineering	2019	UNR College of Engineering	IoT sensors, nanotechnology-enabled flexible electronics, nanoelectronics, semiconductors, smart manufacturing	98	1597	Over 84 peer-reviewed journal articles on IoT and nanoelectronics Flexible hybrid electronics and IoT sensor applications Additively manufactured electronics and smart	EBME; IEEE	Additive manufacturing, IoT sensor fabrication, flexible electronics testing		www	www
Physics	Lei Cao	Associate Professor of Mechanical Engineering	2016	UNR College of Engineering	Materials science, phase-field modeling, microstructure evolution, titanium alloys, mechanical properties	31	3296	Twin boundary engineering through cyclic cross-phase-boundary thermomechanical processing Phase-Field Simulations of alpha Colony Formation in Titanium Alloys Understanding microstructure evolution in metallic	ME; ASME	LAMMPS molecular dynamics, phase-field simulation, SEM	Receptive.	www	www
Physics	Matteo Aureli	Associate Professor of Mechanical Engineering	2013	UNR College of Engineering	Fluid-structure interaction, vibration and dynamics, smart materials, multiphysics modeling	16	79	Finite Amplitude Underwater Flexural Vibrations of Cantilevers Harmonic oscillations of a lamina in viscous fluid near a solid surface Shape-morphing cantilever oscillations in	ME Graduate Director; Multiphysics Lab	5W CW laser, high-speed camera (Phantom Miro), vibration shaker/ controller		www	www
Physics	Roberto Mancini	Professor of Physics; McMinn Endowed Research Prof.	1993	University of Buenos Aires (1983)	Atomic/radiation physics of HED plasmas, x-ray spectroscopy, Stark-broadened lines, radiation transport	35	128	Spectroscopic Determination of Dynamic Plasma Gradients (PRL 2002); Time-Resolved Spectroscopic Measurements in Microballoons (PRL 1989); Escape Factors for Stark-Broadened Line Profiles (J Phys B)	Plasma Spectroscopy & Atomic Physics Group	X-ray spectrometers, HED plasma facilities, radiation diagnostics, vacuum systems		www	www
Physics	Thomas White	Clemons-Magee Endowed Prof.; Assoc. Prof.	2017	University of Oxford (2015)	HED physics, laboratory astrophysics, extreme states of matter, laser-plasma interactions	58	1441	Heating gold to 19,000K without losing crystalline structure (Nature); Fourier heating in HED plasmas (Nature Comms); NSF CAREER; uses NIF, LCLS, European XFEL	High Energy Density Lab (HEDL); Lab Astrophysics Group	High-power lasers, X-ray free-electron lasers (XFEL), NIF access, plasma diagnostics, shock drivers	World-class pulsed power facility (Zebra); unique in US; Physics olympiad alignment. Active.	www	www
Physics	Wendy Calvin	Foundation Professor & Chair of Geological Sciences	2001		Planetary science, Mars geology, mineral spectroscopy, surface composition	49	115	Mars surface mineralogy from orbital spectroscopy data Fifteen years of Mars rover science team contributions Layered deposits of Mars polar ice caps as climate	Mars Exploration Rover science team; Mars Reconnaissance Orbiter	Spectroscopy, Mars rover data, remote sensing, mineralogical references	Mars rover scientist; planetary science; spectroscopy alignment with SIL; Astronomy bridge. Active.		www
Physics	Yan Wang	Associate Professor of Mechanical Engineering	2016	UNR College of Engineering	Thermal science, heat transfer, phonon transport, nanoscale thermal transport, laser manufacturing	200	5296	Selective Phonon Stimulation Mechanism to Tune Thermal Transport Phonon Lateral Confinement Enabling Thermal Rectification in Asymmetric Nanostructures First principles calculations of lattice thermal	ME; ASME	Laser manufacturing systems, computational thermal modeling		www	www
Astronomy	Kelly Kosmo O'Neil	Teaching Assistant Professor of Physics & Astronomy	2023	UCLA (2023)	Galactic center dynamics, supermassive black holes, orbital dynamics, exoplanets	17	83	Relativistic redshift of S0-2 orbiting Galactic center SMBH (Science 2019, co-author); Primordial black holes (2023-2025); Orbital eccentricities of directly imaged companions	UCLA Galactic Center Group (affiliated)	Telescope access, adaptive optics data, astrometry software, spectroscopic data			www
Astronomy	Richard M. Plotkin	Associate Professor of Physics & Astronomy; Acting Physics Chair	2011	University of Washington (2009)	Black hole accretion, multiwavelength X-ray astronomy, relativistic jets, AGN, quasars	2	186	Photometry+: Pipeline for Great Basin Observatory (A&C 2022); Radio Jets from Quiescent BH XRBs (MNRAS 2021); Decay of V404 Cygni: Disk-Jet Coupling (ApJ 2017)	UNR Black Holes Group; Great Basin Observatory	Telescopes (GBO 0.7m, MacLean 22"), CCD cameras, photometry pipelines, X-ray satellite data	Great Basin Observatory PI; direct astronomy olympiad alignment; accessible	www	www
Chemistry	Benjamin T. King	Professor of Chemistry; Fuson Endowed		University of Colorado (2000)	Carborane chemistry, fluorine chemistry, weakly coordinating anions, macromolecular architecture	20	252	Dodecamethylcarba-closo-dodecaboranyl (JACS); Scholl reaction and 2D polymer research; Computational and X-ray crystallography	King Group	X-ray crystallography, NMR, glovebox (air-free), mass spectrometry, computational clusters	Active	www	www
Chemistry	David Hanigan	Associate Professor of Civil & Environmental Engineering		UNR College of Engineering	Environmental engineering, water treatment, water quality, PFAS, disinfection byproducts, N-nitrosamines	31	632	Disinfection byproducts research in water treatment PFAS contamination and remediation studies N-nitrosamine formation and control in water treatment	CEE	LC-MS, GC-MS, water treatment pilot systems		www	www

Chemistry	Dev Chidambaram	Professor of Chemical & Materials Engineering	2009	UNR College of Engineering	Corrosion science, electrochemistry, molten salt research, high-temperature materials, environmental remediation	54	180	Duplex Mechanism for Chromate-Hydrated Oxide Interaction on Aluminum Alloys (J Electrochem Soc) Ti-48Al-2Cr-2Nb alloy in molten salts at 850C for aerospace Molybdate-based coating for depleted uranium	CME Assoc. Chair; Electrochemical Society	SEM, XRD, XPS, synchrotron spectroscopy, Raman spectroscopy	Direct SIL alignment: uses SEM, XRD, XPS, Raman; corrosion and electrochemistry map to Chemistry; $20M funding	www	www
Chemistry	Matthew J. Tucker	Professor of Chemistry; Graduate Director	2011	University of Pennsylvania (2006)	Ultrafast spectroscopy, 2D IR spectroscopy, protein folding/ binding, biophysics	4	67	Unraveling Hydration Shell Dynamics via 2D IR Spectroscopy (JACS 2024); The Long and Short of Coupling/Uncoupling via 2D IR (J Phys Chem B 2024); Amide I 2D IR of Proteins (Acc Chem Res)	Tucker Research Group; Hitchcock Center (co-director)	Femtosecond lasers, FTIR spectrometers, 2D IR setup, optical tables, cryostats	FTIR/spectroscopy directly uses SIL instruments; biophysics bridge. Toys list.	www	www
Chemistry	Ming Chen	Assistant Professor of Chemical & Materials Engineering		UNR College of Engineering	Advanced alloys, laser additive manufacturing, metallic materials, high-entropy alloys	130	5203	Laser additive manufacturing of metallic alloys High-entropy alloy design and characterization Advanced materials processing	CME	Laser AM systems, SEM, XRD, mechanical testing		www	www
Chemistry	Pradeep Menezes	Associate Professor of Mechanical Engineering	2015	UNR College of Engineering	Tribology, surface engineering, advanced manufacturing, composite materials, friction and wear	141	3551	Tribology for Scientists and Engineers (book) Tribology of Additively Manufactured Materials (Elsevier) Tribology of MXene Materials: Advances, Challenges, and Future Directions	ME; ASME; founding editor Tribology and Interfaces (Elsevier)	Tribology testing, surface characterization, SEM, profilometry	16,000 citations with h-index 60; top 2% worldwide; surface characterization aligns with SIL Raman and FTIR	www	www
Chemistry	Yftah Tal-Gan	Professor of Chemistry	2014	Hebrew University of Jerusalem	Chemical biology, bacterial quorum sensing, peptide chemistry, anti-bacterial strategies	26	199	Designing Cyclic CSP-Based Quorum-Sensing Modulators (ACS Infect Dis 2024); Targeting Peptide-Based QS for Gram-Positive Infections (Peptide Sci 2023); NIH/NSF funded peptide research	Tal-Gan Research Group	Peptide synthesizer, HPLC, LC-MS, MALDI-TOF, microplate readers, cell culture	Chemical biology; peptide synthesis; uses SIL LC-MS and MALDI; open to undergrads	www	www
Chemistry	Yu (Frank) Yang	Associate Professor of Civil & Environmental Engineering		UNR College of Engineering	Water science, environmental engineering, emerging contaminants, water treatment	18	27	Emerging contaminants in water systems research Environmental chemistry and water treatment Water quality monitoring and remediation	CEE	LC-MS, water quality analyzers, environmental sampling	5,700 citations; water science aligns with Chemistry; LC-MS use connects to SIL instruments	www	www
Biology	Ben Ma	Assistant Professor of Civil & Environmental Engineering		UNR College of Engineering	Environmental engineering, water treatment, pathogen control, UV disinfection, biofouling, water reuse	26	967	UV disinfection systems for water treatment Pathogen control in water reuse systems Biofouling mitigation research	CEE	UV disinfection equipment, microbiology lab, water quality analyzers	Active	www	www
Biology	Christopher Von Bartheld	Professor of Physiology & Cell Biology		UNR School of Medicine	Neuroscience, neuroanatomy, neuron counting methods, brain cell quantification	30	735	Systematic review of neuron numbers in the human brain Stereological methods for neuron counting Neuroanatomy and brain structure research	Physiology & Cell Biology	Stereology equipment, histology, microscopy	Active	www	www
Biology	Dennis Mathew	Professor of Biology; Assoc. & Associate Chair	2014	UMass Amherst (2006)	Neuroscience of olfaction, insect neuroscience, Drosophila olfaction, neuromodulation	12	52	Insect olfaction and neural circuits (NSF 2024); Drosophila olfactory neuromodulation; NIH ENDURE training program director	Mathew Lab; Integrative Neuroscience Program	Drosophila culture, electrophysiology rigs, fluorescence microscopy, behavioral arenas, optogenetics	Drosophila accessible model; NIH training program for undergrads; neuro olympiad prep. Github.	www	www
Biology	Elizabeth Pringle	Assoc. Prof.; McMinn Endowed Research Prof.	2016	Stanford University (2011)	Evolutionary ecology, plant-animal interactions, chemical ecology, dryland ecosystems	17	52	Remote sensing reveals importance of habitat on aphid biocontrol (2024); Evolutionary ecology of mutualisms; Chemical ecology of dryland plants	The Pringle Lab	Field equipment, remote sensing data, chemical analysis tools, ecological software	Active	www	www
Biology	Floris van Breugel	Assistant Professor of Mechanical Engineering		UNR College of Engineering	Insect flight biomechanics, neural control of movement, bio-inspired robotics, neuroethology	35	313	Insect flight dynamics and neural control Bio-inspired aerial robotics Olfactory navigation in flying insects	ME; Institute for Neuroscience	High-speed cameras, wind tunnel, motion capture, insect flight arena	Insect flight + neuroethology bridges Biology and Computing; bio-inspired robotics is interdisciplinary	www	www
Biology	Jennifer Hoy	Assoc. Prof.; Director Institute for Neuroscience		University of Oregon	Visual neuroscience, prey capture behavior, neural circuits, sensory systems	9	41	Neural circuits underlying prey-capture in mice; Binocular vision in predatory behavior; NIH RO1 (~$2M/5yr)	Hoy Lab; Institute for Neuroscience	Electrophysiology, optogenetics, 2-photon microscopy, behavioral tracking, mouse colony		www	www
Biology	Jenny Ouyang	Assoc. Prof.; EECB Graduate Program Director	2016	Princeton University (2012)	Urban ecology, artificial light pollution, physiological adaptation, circadian biology	37	244	Effects of artificial lighting on birds; NSF CAREER ($1.2M) for light pollution; Endocrinology and animal behavior	Ouyang Lab	Light sensors, hormone assay equipment, field monitoring stations, spectroradiometers	NSF CAREER; accessible field biology; bridges biology+physics via light/ circadian	www	www

Biology	John Cushman	Foundation Professor of Biochemistry & Molecular Biology	2000		CAM photosynthesis, drought stress tolerance, molecular genetics, plant biotechnology	29	367	Crassulacean acid metabolism: a plastic adaptation to arid environments The agave genome reveals core features of CAM Engineering CAM for sustainable food and biofuel production	Biochemistry & Molecular Biology; AAAS Fellow (2023)	Real-time PCR, gene expression analysis, greenhouse, spectrophotometry	AAAS Fellow; Foundation Professor; CAM photosynthesis; plant biotechnology. Active.	www www
Biology	Krishna Pagilla	Foundation Professor of Civil & Environmental Engineering		UNR College of Engineering	Water treatment, wastewater, water reuse, biological treatment, nutrient removal	50	573	Biological wastewater treatment and nutrient removal Water reuse systems and advanced treatment Nevada Water Innovation Institute research	CEE; ASCE Fellow; NV Water Innovation Institute Director	Wastewater treatment pilot systems, biological reactors, water quality analyzers		www www
Biology	Lee Dyer	Foundation Professor of Biology; EECB Director	2010	University of Colorado Boulder (1994)	Chemical ecology, tropical biodiversity, plant-herbivore interactions, conservation biology	77	1192	Phytochemical diversity drives tropical plant-insect diversity (PNAS 2015); Trait-mediated trophic cascade for hummingbirds (Sci Advances 2015); New dimensions of tropical diversity (2014)	EECB Program; Hitchcock Center (affiliated)	GC-MS, LC-MS (chemical ecology), field equipment, specimen collections, tropical stations	Foundation Professor; Chemical ecology uses GC-MS/LC-MS at SIL; Hitchcock Center	www www
Biology	Maryam Raeeszadeh-Sarmazdeh	Associate Professor of Chemical & Materials Engineering	2019	UNR College of Engineering	Protein engineering, synthetic biology, directed evolution, enzyme inhibitors for neurodegeneration and cancer	22	152	Protein Language Models for Low-N Engineering of Metalloproteinase Inhibitors (IEEE) Engineering TIMP-1 for Fine Discrimination Between MMP-3 and MMP-10 Tissue Inhibitors of Matrix Metalloproteinases for	CME; NIH	Protein expression systems, directed evolution platforms, molecular characterization	Protein engineering for cancer and neurodegeneration is core Biology; NSF CAREER + NIH funded; bridges	www www
Biology	Matt Forister	Foundation Prof.; McMinn Endowed Research Prof.	2008		Plant-insect ecology, butterfly ecology, diversification, diet breadth evolution	7	54	Rapid butterfly declines across the US (Science 2025); Monitoring, global change, specialization, extinction; Phytochemical and insect community diversity	Great Basin Bug Lab	Field collection equipment, specimen collections, molecular tools, ecological software	Very high impact (Science 2025 paper); accessible fieldwork; biodiversity+conservation. Active	www www
Biology	Peter Jones	Associate Professor of Pharmacology		Emory	FSHD, epigenetic mechanisms, DNA methylation, muscle development and disease	109	3763	The genetics and epigenetics of FSHD Epigenetic diagnostics for FSHD1 and FSHD2 FSHD animal models and novel epigenetic diagnostic cassettes	Wellstone Center; Mick Hitchcock Endowed Chair of Medical Biochemistry	Epigenetic analysis, DNA methylation assays, animal models	Exceptionally high impact (15k citations); epigenetics; muscular dystrophy center	www www
Biology	Sarah M. Haigh	Associate Professor of Psychology		UNR College of Liberal Arts	Sensory processing, neurological responses to stimuli, schizophrenia, autism, migraine, TBI	39	256	Sensory processing and neural excitability in schizophrenia Visual perception and migraine research Autism spectrum sensory processing studies	Psychology; Cognitive & Brain Sciences; Institute for Neuroscience	EEG, fMRI, psychophysics equipment, eye tracking	Directs the Cognitive & Brain Science PhD program; bridges neuroscience and computing with sensory	www www
Biology	Seungil Ro	Professor of Physiology & Cell Biology		UNR School of Medicine	Diabetes, smooth muscle biology, microRNA, translational research	44	1006	Diabetes and smooth muscle dysfunction research MicroRNA regulation in GI physiology Translational diabetes research	Physiology & Cell Biology	Molecular biology, microRNA analysis, mouse models		www www
Biology	Steven Frese	Assistant Professor of Nutrition & Microbiology	2020		Gut microbiome, food biotechnology, nutritional science, infant health, probiotic development	44	1514	Breastfeeding, immune development, and infant microbiome composition Biotechnology approaches to improving infant nutrition and health Microbiome-based interventions for malnourished	Nutrition; School of Medicine	DNA sequencing, microbiome analysis, fermentation equipment		www www
Biology	Sudeep Chandra	Professor of Biology; Director Global Water Center		UC Davis (2003)	Lake ecology, limnology, mountain lake dynamics, aquatic conservation, fisheries	72	1802	Lake metabolism and mountain lake systems; Castle Lake Research Station director; NSF Ecosystem Sciences Program Director	Aquatic Ecosystems Analysis Lab; Global Water Center; Ozmen	Water quality probes, CTD profilers, boats, field sampling equipment, isotope analysis		www www
Biology	Xiaoshan Zhu	Professor of Electrical & Biomedical Engineering	2009	UNR College of Engineering	Nanomaterials for biosensing and imaging, MEMS-integrated sensory systems, health-care monitoring			Novel TB test development using quantum dots (NSF Phase 2) Biosensing research utilizing manganese-doped quantum dots Smartphone-based health diagnostic technology	EBME; Biomedical Engineering	MEMS fabrication, fluorescence spectroscopy, nanomaterial synthesis	Biosensing + nanomaterials directly aligns with SIL fluorescence and MALDI instruments; health	www www
Biology	Yifei Jin	Assistant Professor of Mechanical Engineering	2019	UNR College of Engineering	3D bioprinting, additive manufacturing, soft materials, hydrogels, stimuli-responsive materials			Multiscale embedded printing of engineered human tissue and organ equivalents (PNAS, 2024) Self-Supporting Nanoclay as Internal Scaffold for Direct Printing of Soft Hydrogel Structures Research on 3D bioprinting of uterine and heart	ME; Biomedical	3D bioprinting systems, materials characterization	3D bioprinting of human tissues is cutting-edge Biology+Engineering; published in PNAS; direct relevance to biomedical	www www

science	name	type	people	overview	total	impact	publications	toys	access	reasoning	w	w	w
Computing	Advanced Robotics & Automation (ARA) Lab	Lab	Hung Manh La (Director)	Autonomous robotic systems for civil infrastructure inspection; climbing robots, multi-robot control, UAV flocking; 2025 ASCE Pankow Award	70	690	A climbing robot for steel bridge inspection A practical climbing robot for steel bridge inspection Visual and 3D mapping for steel bridge	Magnetic-wheel climbing robots, UAVs/drones, GPR, seismic/acoustic arrays, laser scanners, IMU, Intel NUC, ROS	Project-based; inquire through lab	Hands-on robotics; award-winning bridge inspection robots; field deployment	www	www	www
Computing	BRAIN Core (Bridge to Research in AI and Neurocomputing)	Core Facility	Alireza Tavakkoli (Director)	UNR core providing computational support for biomedical research: machine learning, AI, statistical analysis, VR/AR platforms for neuroscience investigators.	200	2048	Machine learning applications in neuroscience research AI-driven biomedical data analysis pipelines Computational methods for	HPC infrastructure, cloud AI systems, ML/DL frameworks, statistical tools, VR/AR equipment	Free consultation; fee-based equipment access;		www	www	www
Computing	Computer Vision Laboratory (CVL)	Lab	George Bebis (Director), Alireza Tavakkoli	Pure and applied research in computer vision since 1998: image segmentation, object recognition, 3D reconstruction, face detection, biometrics, tracking	239	2248	Face detection and verification using genetic search Low-resolution face recognition with single sample per person Face detection in the near-IR spectrum	GPU clusters, image capture/ display devices, robotic peripherals, Pronghorn HPC access	Open applications for research assistants year-round;	Major CV lab; open to undergrads; GPU access; directly relevant to	www	www	www
Computing	Electrical Power & Renewable Energy Systems Lab (EPRESL)	Research Lab	Mehdi Etezadi-Amoli, M. Sami Fadali, Yantao Shen	Power systems modeling, renewable energy integration, smart grid, electric vehicle charging infrastructure, and power electronics research.	68	393	Optimal scheduling of electric vehicle charging stations with renewable energy Smart grid integration of distributed generation systems	Power systems simulators, SCADA systems, PV arrays, battery storage testbeds, power electronics test benches, real-time digital simulators	Through Electrical & Biomedical Engineering; graduate		www	www	www
Computing	Evolutionary Computing Systems Lab (ECSL)	Lab	Sushil Louis (Director)	Evolutionary computing, genetic algorithms, AI for RTS games, Case-Injected Genetic Algorithms (CIGAR)	25	99	Playing to learn: case-injected genetic algorithms for computer games Trap avoidance in strategic game playing with case injected genetic algorithms	Computing systems, ML frameworks, game engines, optimization software	Through graduate programs and research assistantships		www	www	www
Computing	HPCVIS Lab	Lab	Frederick Harris Jr. (Co-director), Phillip Goodman	Graphics, scientific visualization, virtual reality applications, and high-performance computation including immersive VR environments	87	344	VFire: immersive wildfire simulation and visualization Open-source tools for immersive environmental visualization A Unity framework for multi-user VR	GPU clusters, VR headsets (Oculus, HTC Vive), Unity VR framework, visualization hardware	Faculty-led; undergrad/grad involvement through CSE department		www	www	www
Computing	Machine Perception Lab	Lab	Emily Hand (Director)	Wearable devices for social interaction support (autism, visual impairment); bridges human and computer vision using ML	28	120	Improving face recognition using artistic interpretations of prominent features Transformation of face recognition technology and representation for faces Empirical study of face recognition	Wearable prototyping, CV systems, ML frameworks, GPU clusters	Open to grad and undergrad students; part of CVL community	ML/vision with social impact; NASA background PI; open to undergrads	www	www	www
Computing	Pronghorn HPC Cluster	Infrastructure	UNR Cyberinfrastructure group	Shared high-performance computing: 108 CPU nodes (3,456 cores), 44 NVIDIA P100 GPUs, 3 V100 GPUs; hosted at Switch data center	0	0		3,456 CPU cores, 24.8 TiB memory, 44 Tesla P100 GPUs, 3 Tesla V100 GPUs, high-speed interconnect	UNR employees: email hpc@unr.edu; external:	Central GPU/HPC resource for ML and computational research across all sciences	www	www	www
Physics	CTEMPS (Center for Transformative Environmental Monitoring Programs)	Research Center	Scott Tyler (Director), Christine Hatch	NSF-funded center pioneering distributed temperature sensing (DTS) fiber optic technology for environmental monitoring of rivers, glaciers, caves, and subsurface systems.	31	203	Environmental temperature sensing using Raman spectra DTS fiber-optic methods Distributed temperature sensing for hydrology and environmental	Fiber-optic DTS instruments, environmental sensors, weather stations, data loggers, field deployment systems	Equipment lending program for researchers; NSF-supported;	Pioneered DTS technology for environmental science; global impact	www	www	www
Physics	Center for Civil Engineering Earthquake Research (CCEER)	Research Center	Sherif Elfass (Director), Ian Buckle, Saiid Saiidi, Patrick Laplace	Nation's premier university earthquake engineering lab with 5 shake tables including the largest Laminar Soil Box (350 tons). Research in structural, geotechnical, and bridge earthquake	16	113	Seismic performance of reinforced concrete bridge columns under combined loading Laminar soil box testing of soil-foundation-structure interaction	5 shake tables, Laminar Soil Box (350 tons), strong floor, reaction wall, 300-kip actuators, data acquisition systems, SAGE building (153,000 sq ft)	Graduate research through Civil & Environmental Engineering;	Unique facility in US university system; 5 shake tables; world-class earthquake	www	www	www
Physics	Derevianko Group	Research Group	Andrei Derevianko (AAAS Fellow, Simons Fellow)	Novel atomic clock development; precision tests of fundamental symmetries; ytterbium optical lattice clocks; nuclear clocks; parity-non-conservation	64	614	Hunting for topological dark matter with atomic clocks Physics of optical lattice clocks Search for new physics with atoms and molecules	Precision laser systems, optical lattice clock apparatus, high-precision spectroscopy	Grad/undergrad through Physics dept	World-leading precision measurement; AAAS Fellow; atomic clocks are	www	www	www
Physics	Great Basin Center for Geothermal Energy	Research Center	Bridget Ayling (Director), James Faulds, Nicholas Hinz	DOE-funded geothermal research center advancing exploration, characterization, and development of geothermal resources in the Great Basin and globally.	99	1152	Controls on Great Basin geothermal systems Geothermal play fairway analysis of the Great Basin Machine learning approaches for	Geothermal well logging equipment, geophysical instruments, thermal modeling software, GIS systems, field vehicles	Through Geological Sciences & Engineering; DOE-funded	DOE-funded; unique geothermal expertise leveraging Nevada's geologic	www	www	www

Physics	High Energy Density Lab (HEDL)	Research Group	Thomas White	Laboratory astrophysics; extreme states of matter; equation of state in planetary cores; uses NIF, LCLS, European XFEL	58	1441	Accessing extreme equation of state on National Ignition Facility Probing matter at Gbar pressures at the NIF Developing absolute shock wave	Access to National Ignition Facility, LCLS X-ray laser, European XFEL, DFT/MD computing	Through White research group; DOE funded	Access to world-leading national facilities (NIF, LCLS); bridges physics +	www	www	www
Physics	Nevada Geodetic Laboratory	Research Lab	Geoffrey Blewitt (Director), Corné Kreemer, William Hammond	World-leading GPS/geodesy research operating 414-station GPS network. Real-time crustal deformation, tectonic strain analysis, and sea-level change monitoring since 2004.	75	1240	Harnessing the GPS data explosion for interdisciplinary science A global strain rate model including plate boundary zones GPS measurement of present-day	414-station GPS network, GNSS processing systems, InSAR data, high-performance computing, geodetic data archives	Public data at geodesy.unr.edu; research through Nevada Bureau of	World-leading GPS geodesy lab; highest-impact geoscience group at UNR; open	www	www	www
Physics	Tscherbul Group	Research Group	Timur Tscherbul (NSF CAREER)	Ultracold quantum dynamics: molecular collisions, ultracold chemistry, photosynthetic light-harvesting, Rydberg aggregates, superfluid helium nanodroplets	72	313	Hyperfine-to-rotational energy transfer in ultracold collisions A rigorous adiabatic approach to ultracold collisions A Feshbach resonance in triplet ground-	Quantum computing resources, HPC (coupled-channel methods, quantum master equations, Monte Carlo)	Grad/undergrad through Physics dept		www	www	www
Physics	Weinstein Lab	Lab	Jonathan Weinstein	Cryogenic cooling and laser spectroscopy of atoms/molecules trapped in solid neon and parahydrogen at near-absolute-zero; quantum sensing; dark matter searches	5	37	Spin coherence and optical properties in solid parahydrogen Enhanced spin coherence of rubidium atoms in solid neon Ultralong spin coherence times via	Cryostats, laser systems, optical tables, vacuum chambers, magnetic shielding, spectrometers, matrix isolation apparatus	Grad/undergrad through Physics dept	Quantum sensing and fundamental physics; Harvard-trained PI; bridges physics and	www	www	www
Physics	Zebra Pulsed Power Lab (ZPPL)	Lab	Thomas White (lead), Roberto Mancini, Hiroshi Sawada, Bernhard Bach	Highest-power pulsed electrical device at any US university; Z-pinch physics, HED science, laser-plasma interactions; 153,000 sq ft SAGE facility	96	1694	Two-terawatt Zebra Z-pinch at Nevada terawatt facility Staged Z-pinch experiments at 1MA Zebra generator Measurement of ionization state of wire-	Zebra (2 TW pulsed power), Leopard laser (50 TW, 300 fs), Falcon optical compressor, X-ray diagnostics, streak cameras, ICCD cameras, 800-	Grad/undergrad through Physics dept; DOE/ NNSA supported;	World-class pulsed power facility; unique in US university system; Core diagnostics.	www	www	www
Astronomy	Great Basin Observatory (GBO)	Observatory	Richard Plotkin (UNR PI), multi-institutional	0.7-meter robotic remotely-operated telescope in Great Basin National Park (elev. 6,825 ft, 70+ mi from light pollution); NSF-funded astrophysics and K-12 education	69	335	User-guided photometric pipeline for GBO robotic telescope Photometry development for GBO telescope Robotic telescope black hole accretion	0.7m research-grade robotic telescope, CCD cameras, photometry pipeline, remote operation software	UNR faculty/ student access as partner institution; K-12 outreach	Research-grade remote telescope; direct astronomy olympiad alignment;	www	www	www
Chemistry	Borotto Research Group	Lab	Nicholas Borotto	Mass spectrometry-based detection of biomolecules; protein structure characterization; post-translational modification analysis	36	168	The path forward for protein footprinting covalent labeling and mass spec Investigating therapeutic protein structure with diethylpyrocarbonate labeling	timsTOF Pro2, ion mobility MS, tandem MS, IRMPD, CO laser-equipped MS, LC separations	Open to grad and undergrad students	Advanced mass spec directly relevant to SIL instruments and proteomics.	www	www	www
Chemistry	Fluid Inclusions Lab	Research Lab	Matthieu Harlaux (Director, Economic Geology)	Microthermometric and microanalytical study of fluid inclusions in minerals for understanding hydrothermal ore deposit genesis, geothermal systems, and geological processes.	24	209	Geochemical signature of magmatic-hydrothermal fluids in rare-metal granites Sequential LA-ICP-MS for fluid inclusion trace element analysis	Linkam heating/freezing stages (-200 to +600°C and up to 1400°C), Raman spectrometer, Olympus BX51 microscope with NIR video, LA-ICP-MS	Through Geological Sciences; courses GEOL 713/776;	Toys list.	www	www	www
Chemistry	Geary Lab	Lab	Laina Geary	Palladium-catalyzed cross-coupling, C-H functionalization, Pauson-Khand cycloadditions, alcohol substitution for C-C/C-X bonds	5	3	Recent advances in the Pauson-Khand reaction Nitrous oxide-promoted Pauson-Khand cycloadditions Regioselectivity in Pauson-Khand	GC-MS, LC-MS, NMR, column chromatography, Schlenk lines, glovebox	Open to grad and undergrad students	Hands-on organic synthesis; uses SIL GC-MS/LC-MS for characterization	www	www	www
Chemistry	Hitchcock Center for Chemical Ecology	Center	Christopher Jeffrey (Director), Lora Richards, Matthew Tucker	20+ faculty across chemistry, biology, ecology, agriculture; studies plant defensive chemistry, insect diversity, bio-prospecting in Great Basin; $3M NSF NRT grant	59	328	Phytochemical diversity drives plant-insect community diversity Structure revision of scytonemin imine and its relationship to scytonemin Isolation of new neolignans and	Shared analytical instruments (LC-MS, GC-MS, NMR), synthesis labs, field collection equipment, metabolomics pipelines	NRT fellowships (25+ supported); open to grad	Premier interdisciplinary center bridging chemistry + biology; NRT	www	www	www
Chemistry	Materials Characterization Nevada (MCNV)	Shared Facility	Dev Chidambaram, Krista Carlson (contacts)	Advanced materials characterization facility housing world's second lab-based EXAFS system. XRD, XRF, thermal analysis, surface characterization for materials science and engineering	95	456	EXAFS characterization of novel battery electrode materials XRD phase analysis of advanced ceramic composites Thermal stability of nanostructured	Lab-based EXAFS system, XRD, XRF, DSC/TGA, surface area analyzers, nanoindenter, profilometer	Fee-for-service through NCAR; open to UNR, external, and industry users	World's second lab EXAFS; complements SIL chemistry capabilities. Toys	www	www	www
Chemistry	Microbeam & SEM Laboratory (Mackay School)	Core Facility	Joel DesOrmeau (Manager), Krista Carlson	State-of-the-art microanalysis facility with NSF-funded field-emission EPMA (first in US), FE-SEM, and LA-ICP-MS for geological, materials, and environmental sample analysis at micrometer resolution.	65	356	GHR1 Zircon: a new natural reference material for microbeam U-Pb geochronology Rapid exhumation of Papua New Guinea UHP terrane via eclogite	JEOL JXA iHP200F FE-EPMA (first in US), JEOL JSM 7100FT FE-SEM, LA-ICP-MS, 5 WDS spectrometers, hyperspectral CL, EDS	Fee-for-service; open to UNR, other universities, and industry;	First FE-EPMA installed in US; NSF-funded; critical analytical infrastructure. Toys	www	www	www
Chemistry	Mineral Processing Lab	Research Lab	Pengbo Chu, Barbara Wold (Mining & Metallurgical Engineering)	Ore processing, flotation chemistry, hydrometallurgy, and mineral characterization research supporting Nevada's mining industry. Part of Mackay School.	32	445	Flotation optimization for complex sulfide ores Hydrometallurgical recovery of rare earth elements Mineral liberation analysis using	Flotation cells, ball mills, jaw crushers, leaching reactors, mineral liberation analyzer, assay furnaces	Through Mining & Metallurgical Engineering; industry partnerships	Toys list.	www	www	www

Chemistry	Shared Instruments Laboratory	Core Facility	SIL / Chemistry Department	Liquid-state and solid-state NMR for structural characterization of organic, inorganic, and materials samples	0	0		3x Agilent Varian liquid-state NMR (including 400-MR with auto-switchable probe: 1H, 19F, 13C, 31P), Tecmag Discovery 400 MHz solid-state NMR with	Through SIL; training required; iLab scheduling	NMR is essential for organic/ inorganic chemistry characterization;	www	www	www
Chemistry	Ralph Roberts Center for Economic Geology (CREG)	Research Center	John Muntean (Director, Arthur Brant Endowed Chair)	30-year partnership between Mackay School, Nevada Bureau of Mines & Geology, mining industry, and USGS. Focuses on Carlin-type gold deposit genesis and mineral exploration methods.	4	2	Diversity of Carlin-Style Gold Deposits: applications to exploration The Carlin gold system: exploration applications in Nevada Geochemistry of metasedimentary rock-	Analytical lab access (SEM, EPMA, LA-ICP-MS at Mackay), field exploration equipment, mining industry partnerships, USGS collaboration	Graduate student thesis funding; mining industry internships;	Premier economic geology center; 30-year mining industry partnership. Toys	www	www	www
Chemistry	Shared Instruments Laboratory	Core Facility	Stephen Spain (Director)	Free-access core facility with mass spec, vibrational spectroscopy, thermal analysis, optical spectroscopy, chromatography; Room 016 Chemistry Building				LC-MS (Waters + Agilent TOF), MALDI-8020, Raman microscope, 2x FTIR (Nicolet 380 + Bruker Tensor 27), GC-MS, DSC, TGA, fluorescence,	Free for UNR students/faculty after training; 24/7 card access;	You already have 24/7 access; comprehensive instrument suite across	www	www	www
Chemistry	Tal-Gan Research Group	Lab	Yftah Tal-Gan	Peptide-based tools for bacterial quorum sensing; therapeutic implications for bacterial pathogenesis and virulence	35	1900	Highly potent inhibitors of quorum sensing in Staphylococcus aureus Targeting peptide-based quorum sensing systems for Gram-positive treatment	Peptide synthesizer, HPLC, LC-MS, MALDI-TOF, microplate readers, cell culture, molecular microbiology setup	Open to undergrad, grad, and postdocs; NIH/ NSF funded	Chemical biology; peptide synthesis; uses SIL LC-MS and MALDI; open to undergrads	www	www	www
Chemistry	Tucker Research Group	Lab	Matthew Tucker	Ultrafast 2D IR spectroscopy of protein folding, binding, and function; also studies magnetism and energy dissipation in desert lichen molecules	6	80	Nonequilibrium dynamics of helix reorganization observed by transient 2D IR Site-specific coupling of hydration water and protein flexibility via 2D-IR	Femtosecond lasers, FTIR spectrometers, 2D IR setup, optical tables, cryostats	Open to grad and undergrad students	FTIR/spectroscopy directly relevant to SIL instruments; biophysics bridge. Toys list.	www	www	www
Chemistry	X-ray Crystallography Facility	Core Facility	SIL / Chemistry Department	Single-crystal X-ray diffraction for molecular structure determination	0	0		Bruker Smart APEX diffractometer	Through SIL; sample submission; specialist operation		www	www	www
Chemistry	Yang Research Group	Lab	Ying Yang (NSF CAREER)	Dynamic polymers with reversible bonds for recyclable plastics, self-healing materials, 3D printing, wearable sensors, soft robotics	87	2082	A stretchable mechanically robust polymer with shape-memory-assisted healing Intrinsic self-healing polymers with high E-modulus based on dynamic urea	Polymer synthesis equipment, 3D printers, mechanical testing, spectroscopic characterization	Open to grad and undergrad; 2025 Foundation Early Career	New faculty building lab; dynamic materials are hands-on; likely open to	www	www	www
Chemistry	de Bettencourt-Dias Group	Lab	Ana de Bettencourt-Dias	Luminescent lanthanide complexes; f-element chemistry for energy efficiency, diagnostics, therapy	28	2400	Luminescent lanthanide complexes with dual activity for photodynamic therapy Lanthanide luminescence for biomedical analyses and imaging Ligand design in lanthanide complexes	Fluorescence spectrometers, UV-Vis, X-ray crystallography, glovebox, NMR, CD spectrometer	Open to grad and undergrad students		www		www
Biology	Aquatic Ecosystems Analysis Lab	Lab	Sudeep Chandra, Zeb Hogan	Limnology, aquatic conservation, invasive species management, freshwater fish ecology; Castle Lake station (est. 1959)	114	2643	Lake Tahoe invasive species management and control Aquatic ecosystems conservation and food web ecology Invasive species impacts on aquatic	Water quality probes, CTD profilers, boats, field sampling, isotope analysis	Open to undergrad/grad; Castle Lake has trained 700+		www	www	www
Biology	Cellular and Molecular Imaging Core	Core Facility	Alexander van der Linden (Director), Andrew Yanez (Training	Part of Center for Integrative Neuroscience COBRE. Advanced microscopy, DNA/RNA/protein analysis, and cellular imaging on Fleischmann 3rd floor.	8	40	Confocal microscopy applications in neuroscience Two-photon imaging of neural tissue architecture Tissue clearing and 3D imaging	Leica SP8 confocal with LIGHTNING, Scientifica two-photon, Leica THUNDER 3D, Typhoon Phosphoimager, PCR robot, real-time PCR, cell	UNR researchers; iLab reservation system; training available	Toys list.	www	www	www
Biology	Center for Integrative Neuroscience (COBRE)	Research Center	70+ affiliated faculty across 13 departments	NIH COBRE-funded center (P30 Phase 3, $5.3M) since 2012. Operates molecular imaging, neuroimaging, and VR/AI cores. Supports interdisciplinary PhD in Integrative Neuroscience.	220	8800	Multi-modal neuroimaging studies of brain network organization Neural circuit mechanisms of sensory processing Computational neuroscience	Confocal microscope (Leica SP8), two-photon microscope, THUNDER 3D imaging, VR/AR systems, fMRI, EEG, $9M invested in equipment	70+ faculty in 13 departments; pilot project funding; PhD	Major NIH-funded neuroscience center; $9M equipment investment; Phase	www	www	www
Biology	Center for Molecular Medicine (UNR Med)	Research Center	Multiple department heads (Microbiology, Pharmacology,	107,000 sq ft research facility housing wet labs, rodent vivarium, BSL3 select agent lab. Programs in cancer biology, infectious disease, cardiovascular disease, and neurosciences.	120	4800	Cell biology signaling mechanisms in cardiovascular disease Neuro-immune disease pathogenesis and biomarkers Infectious disease mechanisms in BSL3	BSL3 select agent lab, rodent vivarium, wet labs, medical office suites, cell culture facilities, microscopy	UNR Med faculty and students; partnership with Renown Health	Major UNR Med research hub; BSL3 capability; 107,000 sq ft	www	www	www
Biology	Global Water Center	Center	Sudeep Chandra (Director)	Multidisciplinary center for water sustainability; research on every continent; translates science into actionable solutions	72	1802	Lake Tahoe invasive species management and control Water sustainability and aquatic ecosystems conservation Global freshwater solutions for	Water quality instruments, drone technology, monitoring equipment, predictive modeling	Open to faculty/ students across disciplines; NGO partnerships		www	www	www

Biology	Great Basin Bug Lab	Lab	Matt Forister	Plant-insect ecology, butterfly ecology, diet breadth evolution, phenology, conservation; Science 2025 paper on US butterfly declines	7	54	Increasing neonicotinoid use and butterfly fauna decline Dramatic decline in western butterfly populations linked to pesticides Climate change is a stressor of insect	Field collection equipment, specimen collections, molecular tools, genomic sequencing, ecological software	Open to PhD, Master's, and undergrad students; field + lab research	Very high impact PI (Science 2025); accessible fieldwork; biodiversity +	www	www	www
Biology	Institute for Neuroscience	Institute	Jennifer Hoy (Director), Dennis Mathew, multiple faculty; 70+ affiliated	Multidisciplinary neuroscience center; $25M NIH COBRE funding across 3 phases; 40 active grad students; doctoral program	21	93	Defined cell types in superior colliculus prey capture Layer-specific refinement of visual cortex after eye opening Identification of a brainstem circuit	HD EEG, 2x 3T MRI (at Renown), NIRS, Leica SP8 confocal, Scientifica 2-photon microscope, Leica THUNDER 3D, VR/AR equipment	Grad students in Integrative Neuroscience Program; core facilities	Major neuroscience institute; 2-photon and confocal microscopy;	www	www	www
Biology	Mathew Lab	Lab	Dennis Mathew	Neuroscience of olfaction in Drosophila; neural circuit function; neuromodulation; NIH ENDURE training program	12	52	Internal state affects local neuron function in olfactory behavior Behavior responses to chemical and optogenetic Drosophila stimuli Differential contributions of olfactory	Drosophila culture, electrophysiology rigs, fluorescence microscopy, behavioral arenas, optogenetics	Open to grad and undergrad; runs NIH training program; co-	Drosophila accessible model; NIH training program for undergrads; neuro	www	www	www
Biology	Nevada INBRE (IDeA Network of Biomedical Research Excellence)	Research Network	NIH-funded program directors, multi-institution faculty	NIH program building biomedical research infrastructure across Nevada. Supports early-stage investigators, research labs, data science core, and undergraduate pipeline programs at 5+ institutions.	180	3600	Bioinformatics and computational biology methodologies for Nevada researchers Biomedical research capacity building in EPSCoR states	Bioinformatics servers, computational platforms, data science infrastructure, research funding	Faculty and students at participating Nevada institutions;	Statewide biomedical research infrastructure; NIH-funded. Core	www	www	www
Biology	Ouyang Lab	Lab	Jenny Ouyang	Urban ecology, artificial light pollution effects on birds, circadian biology, physiological adaptation; NSF CAREER $1.2M	37	244	Restless roosts: light pollution affects behavior and sleep physiology Effects of dim artificial light on circadian clock genes in songbird Birds of a feather flock together: social	Light sensors, hormone assay equipment, field monitoring stations, spectroradiometers	Open to grad and undergrad students	NSF CAREER; accessible field biology; bridges biology + physics via light/circadian	www	www	www
Biology	Parchman Lab	Lab	Thomas Parchman	Population genomics: adaptive divergence, hybridization, speciation in conifers, tropical Piper, birds, and fish	57	429	Genome divergence and diversification in geographic mosaic of coevolution Coevolution, local adaptation, and ecological speciation in birds Resource stability and geographic	Molecular biology lab, genomic sequencing, computational resources for genome-scale data	Open to grad and undergrad students		www	www	www
Biology	Voyles Lab	Lab	Jamie Voyles (McMinn Research Professor)	Emerging infectious diseases in amphibians; chytridiomycosis; conservation biology; field research in Panama since 2003	31	260	Pathogenesis of chytridiomycosis causing catastrophic amphibian declines Amphibian fungal panzootic causes catastrophic ongoing biodiversity loss	Pathogen research facilities, physiological analysis, field stations (Panama: Smithsonian SCBI)	Open to grad and undergrad; international field opportunities		www	www	www
Core Facility	Cell Analysis Core (Flow Cytometry)	Core Facility	School of Medicine	Flow cytometry and cell sorting; spatial molecular imaging; cell cycle analysis, proliferation assays	0	0		WOLF microfluidics sorter, Cytoflex 3-laser cytometer, CosMx spatial molecular imager, FlowJo	Fee-based; iLab scheduling; School of Medicine	Toys list.	www		www
Core Facility	Core Analytical Laboratory (CAL)	Core Facility	College of Agriculture + USDA ARS	Analytical chemistry for agricultural, environmental, and plant science; elemental analysis, chromatography	0	0		ICP-AES, ICP-MS, GC (MS/FID/ECD), elemental analyzer, flow injection, UV-Vis, microwave digestion, particle size	Students can train to operate independently; staff can analyze for	Toys list.	www		www
Core Facility	Davidson Cleanroom / Nanofab Lab	Core Facility	Russ Renzas (Director)	Class 1,000 semiconductor/photonic fabrication; lithography, deposition, etch, metrology, wire bonding; 2,937 sq ft				Lithography systems, deposition tools, etching equipment, metrology instruments, wire bonder	UNR researchers; external: industrial rates; Pennington	Nanofabrication for quantum/photonic devices; Physics olympiad alignment. Toys	www		www
Core Facility	Electron Microscopy Facility (EMMF)	Core Facility	Yufeng Zheng (Director)	Scanning and transmission electron microscopy for metals, ceramics, polymers, composites, biological samples	4	1352		Talos F200S S/TEM with EDS, Scios 2 FIB/SEM, FEI Nano600 NanoSEM, sample prep instruments	Contact EMMF; fee-based; iLab scheduling	Electron microscopy gives nanoscale imaging across all sciences;	www		www
Core Facility	Neuroimaging Core	Core Facility	Center for Integrative Neuroscience	MRI/fMRI and neurophysiology; human and small animal imaging	0	0		2x 3T MRI (Philips Ingenia + Siemens Skyra at Renown), 2x HD EEG, NIRS (TechEn)	Available to affiliated faculty/ students; Renown South Meadows	Toys list.	www		www
Core Facility	Nevada Bioinformatics Center	Core Facility	Hans Vasquez-Gross (Director)	Bioinformatics services: NGS analysis, single-cell RNA-seq, transcriptomics, genome resequencing, proteomics data, workshops	0	0	Genomic surveillance of Nevada SARS-CoV-2 variants Plant genomics and water-use efficiency in crassulacean metabolism Cellular paraspeckle component in viral	HPC clusters, bioinformatics software, NGS analysis pipelines, computational workstations	Pay-per-service; workshops open to all; iLab scheduling	Toys list.	www		www

Core Facility	Nevada Genomics Center	Core Facility	Andrew Gorzalski (Director)	DNA sequencing core: Illumina NGS, 10X single-cell, Sanger, fragment analysis; serves 132 Nevada labs + 250 labs nationally/internationally	26	341		Illumina NextSeq 2000, MiniSeq, Sanger sequencing, fragment analysis, BSL-2 lab	Pay-per-service; open to all researchers (academic + industry); iLab	Genomics is central to Biology olympiad; accessible pay-per-service model.	www		www
Core Facility	Nevada Proteomics Center	Core Facility	David Quilici (Director)	Mass spectral proteomic analysis; protein identification, quantitation, and PTM analysis from blood, saliva, tissue	8	47	Quantitative proteomics unveils alterations in neuropathic nerves Mass spectrometry analysis for clinical diagnostics Proteomics of physiological pathways	Bruker timsTOF Pro2, ThermoFisher Eclipse MS, Evosep One LC, gel/IP/solution workflows	Pay-per-service; open to all researchers (academic + industry); iLab	Proteomics bridges chemistry + biology; advanced mass spec complements	www	www	www
Core Facility	Nevada Stable Isotope Lab (NSIL)	Core Facility	Simon Poulson	C, O, H, N, S stable isotope analysis for geological, environmental, and biological samples	25	144	Environmental low temperature geochemistry via stable isotopes Stable isotope analysis of carbon, oxygen, hydrogen, nitrogen, sulfur Isotope geochemistry research	Stable isotope ratio mass spectrometers, sample preparation systems	Fee-based through NCAR; ncar@unr.edu	Toys list.	www		www
External	Biosciences Entrepreneurial Lab (BEL)	Incubator	NCAR	Turnkey wet lab incubator (2,200 sq ft) for biotech startups; fully furnished with standard lab equipment	0	0		Complete wet lab infrastructure, standard lab equipment, shared analytical instruments	Through NCAR incubation program; monthly membership or	Toys list.	www		www
External	DeLaMare Makerspace	Maker Space	DeLaMare Science & Engineering Library	3D printing, laser cutting, PCB milling, soldering, prototyping; free for campus community	0	0		3D printers/scanners, laser cutter, vinyl cutter, PCB mill, soldering stations, sewing stations, hand tools	Free for all UNR community; training required; 180	Free access; 3D printing and prototyping for any science project; very accessible.	www		www
External	Desert Research Institute (DRI)	Research Institute	500+ research faculty/staff; Reno + Las Vegas campuses	Nonprofit sister institution to UNR (NSHE); 40+ specialized labs; atmospheric, earth/ecosystem, hydrologic sciences; >$50M annual research	800	35000	Mineral dust aerosol size distribution and Earth system impacts Dust emissions characterization and atmospheric visibility impact Wildfire smoke air quality and aerosol	Atmospheric science instruments, hydrologic measurement, environmental monitoring, LASSO optical/ aerosol equipment	Research collaboration with UNR; some facilities accessible	Major research institution adjacent to UNR; atmospheric science labs	www		www
External	Innevation Center	Innovation Hub	UNR-affiliated	Startup incubator with co-working, makerspace, conference rooms, mentorship; access to NCAR facilities	0	0		Makerspace tools, co-working spaces, conference rooms	Free for UNR students/ faculty/staff; community: paid	Toys list.	www		www
External	NCAR (National Center for Atmospheric Research)	National Research Center	NSF-sponsored (headquarters: Boulder, CO)	Climate science, meteorology, atmospheric chemistry, space weather; no permanent Reno facility but collaborative partnerships with UNR on drought/climate	0	0	NCEP/NCAR 40-year reanalysis project for global climate record Indicators of global climate change and temperature emissions AMOC Atlantic meridional overturning	NCAR-Wyoming Supercomputing Center, Mauna Loa Observatory, field campaign instruments	Collaborative partnerships; no direct Reno presence; data and models		www		www
External	Nevada Center for Applied Research (NCAR)	Applied Research Center	UNR (600+ companies engaged since 2013)	Coordinates access to all 40+ UNR shared research facilities; supports startups; $201M venture capital raised by affiliates	0	0		Access to all UNR core facilities, Biosciences Entrepreneurial Lab (wet lab incubator), Mechanical Engineering Manufacturing Lab	External companies/ researchers: coordinate through	Gateway to all UNR equipment for paid access; NCAR is the main contact for	www		www
External	Nevada Seismological Laboratory	Research Lab	Christie Rowe (Director)	186-station seismic network across Nevada/E. Sierra; 105 fire watch cameras; Earthquake Early Warning; 50+ years of data			Rapid seismic monitoring deployment for earthquake sequence Nevada earthquake detection and monitoring network Earthquake early warning system for	Seismometers (186-station network), fire watch cameras (105), EEW systems, seismic data archives	Research collaboration; data publicly available; Mackay School		www	www	www

science	lab	technology	toys	www
Computing	BRAIN Core	HPC, AI/ML, VR/AR, Statistics	GPU clusters, cloud AI platforms, ML/DL frameworks (TensorFlow, PyTorch), VR/AR headsets, statistical analysis servers	www
Computing	Power System Research Lab	Power Systems, Renewables, Simulation	SCADA systems, PV arrays, battery storage testbeds, power electronics test benches, real-time digital simulators	www
Computing	Evolution Computing Systems Lab	Evolutionary Computing, ML, Optimization	Computing clusters, ML frameworks, game engines, optimization/simulation software	www
Computing	High Performance Computing - Pronghorn	HPC, GPU Computing, Storage	3,456 CPU cores, 24.8 TiB memory, 44 Tesla P100 GPUs, 3 Tesla V100 GPUs, high-speed interconnect	www
Physics	Center for Civil Engineering Earthquake Research	Seismic Testing, Structural, Geotechnical	5 shake tables, Laminar Soil Box (350 tons), strong floor, reaction wall, 300-kip actuators, DAQ systems	www
Physics	Derevianko Group	Precision Metrology, Laser, Spectroscopy	Precision laser systems, optical lattice clock apparatus, high-precision spectroscopy equipment	www
Physics	Great Basin Center for Geothermal Energy	Geothermal, Geophysics, Field Survey	Geothermal well logging equipment, geophysical instruments, thermal modeling software, GIS systems	www
Physics	High Energy Density Physics	High Energy Density, X-ray, Laser, HPC	National Ignition Facility access, LCLS X-ray laser, European XFEL, DFT/MD computing clusters	www
Physics	Geodetic Lab	GPS/GNSS, Geodesy, Remote Sensing	414-station GPS network, GNSS processing systems, InSAR data, HPC, geodetic data archives	www
Physics	Tscherbul Lab	Quantum Computing, HPC, Simulation	HPC clusters, coupled-channel solvers, quantum master equation codes, Monte Carlo software	www
Physics	Weinstein Lab	Cryogenics, Laser, Vacuum, Spectroscopy	Cryostats, laser systems, optical tables, vacuum chambers, magnetic shielding, spectrometers, matrix isolation	www
Physics	Zebra Pulsed Power Lab	Pulsed Power, High-Power Laser, X-ray Diagnostics, Machining	Zebra (2 TW pulsed power), Leopard laser (50 TW, 300 fs), X-ray diagnostics, streak/ICCD cameras, 800-core cluster, CNC lathe/mill	www
Physics	Nanofabrication Lab	Lithography, Thin Film, Etching, Metrology	Lithography systems, deposition tools, etching equipment, metrology instruments, wire bonder	www
Physics	Electron Microscopy and Microanalysis Facility	Electron Microscopy, FIB, Sample Prep	Talos F200S S/TEM w/ EDS, Scios 2 FIB/SEM, FEI Nano600 NanoSEM, sample prep instruments	www
Physics	DeLaMare Makerspace	3D Printing, Laser Cutting, Electronics, Fabrication	3D printers/scanners, laser cutter, vinyl cutter, PCB mill, soldering stations, sewing stations	www
Physics	Innevation Center	Makerspace, Co-working, Prototyping	Makerspace tools, co-working spaces, conference rooms, prototyping equipment	www
Chemistry	Borotto Lab	Mass Spectrometry, Ion Mobility, Chromatography	timsTOF Pro2, ion mobility MS, tandem MS, IRMPD, CO laser-equipped MS, LC separations	www
Chemistry	Geological Sciences and Engineering Labs	Microthermometry, Raman, Microscopy	Linkam heating/freezing stages (-200 to 1400°C), Raman spectrometer, Olympus BX51 microscope w/ NIR, LA-ICP-MS	www
Chemistry	Geary Lab	Organic Synthesis, Chromatography, Spectroscopy	GC-MS, LC-MS, NMR, column chromatography, Schlenk lines, glovebox	www
Chemistry	Materials Characterization Nevada	X-ray, Thermal, Surface Analysis	Lab-based EXAFS, XRD, XRF, DSC/TGA, surface area analyzers, nanoindenter, profilometer	www

Chemistry	Microbeam Laboratory	Electron Microscopy, Microprobe, Spectroscopy	JEOL JXA iHP200F FE-EPMA, JEOL JSM 7100FT FE-SEM, LA-ICP-MS, 5 WDS spectrometers, hyperspectral CL, EDS	www
Chemistry	Chu Lab	Mineral Processing, Comminution, Leaching	Flotation cells, ball mills, jaw crushers, leaching reactors, mineral liberation analyzer, assay furnaces	www
Chemistry	Shared Instruments Lab	NMR Spectroscopy	3x Agilent Varian liquid-state NMR (400-MR w/ auto-switchable probe), Tecmag 400 MHz solid-state NMR w/ MAS	www
Chemistry	Tal-Gan Group	Peptide Synthesis, Chromatography, Mass Spec	Peptide synthesizer, HPLC, LC-MS, MALDI-TOF, microplate readers, cell culture, microbiology setup	www
Chemistry	Tucker Group	Ultrafast Laser, IR Spectroscopy, Cryogenics	Femtosecond lasers, FTIR spectrometers, 2D IR setup, optical tables, cryostats	www
Chemistry	Yang Group	Polymer Synthesis, 3D Printing, Mechanical Testing	Polymer synthesis equipment, 3D printers, mechanical testers, spectroscopic characterization	www
Chemistry	de Bettencourt-Dias	Luminescence, Spectroscopy, Crystallography	Fluorescence spectrometers, UV-Vis, X-ray crystallography, glovebox, NMR, CD spectrometer	www
Chemistry	Stable Isotope Laboratory	Stable Isotope, Mass Spec, Sample Prep	Stable isotope ratio mass spectrometers, automated sample preparation systems	www
Biology	Cellular and Molecular Imaging Core	Confocal, Two-Photon, Fluorescence, PCR	Leica SP8 confocal w/ LIGHTNING, Scientifica two-photon, Leica THUNDER 3D, Typhoon Phosphoimager, PCR robot, real-time PCR	www
Biology	COBRE Neuroscience	Imaging, VR/AR, Neurophysiology	Leica SP8 confocal, two-photon microscope, THUNDER 3D, VR/AR systems, fMRI, EEG	www
Biology	School of Medicine Research Centers and Core Facilities	BSL-3, Vivarium, Cell Culture, Microscopy	BSL3 select agent lab, rodent vivarium, wet labs, cell culture facilities, microscopy suites	www
Biology	Bug Lab	Field Ecology, Genomics, Specimen Collections	Field collection equipment, specimen collections, molecular tools, genomic sequencing, ecological software	www
Biology	Neuroscience Institute	MRI, EEG, Confocal, Two-Photon, VR/AR	HD EEG, 2x 3T MRI (Renown), NIRS, Leica SP8 confocal, Scientifica 2-photon, THUNDER 3D, VR/AR equipment	www
Biology	Mathew Lab	Drosophila, Electrophysiology, Optogenetics	Drosophila culture, electrophysiology rigs, fluorescence microscopy, behavioral arenas, optogenetics	www
Biology	Nevada INBRE	Bioinformatics, HPC, Data Science	Bioinformatics servers, computational platforms, data science infrastructure. Scientific Core Service Awards. RAIN Technology Access Grants.	www
Biology	Ouyang Lab	Ornithology, Endocrinology, Field Monitoring	Light sensors, hormone assay equipment, field monitoring stations, spectroradiometers	www
Biology	Parchman Lab	Genomics, Molecular Biology, HPC	Molecular biology lab, genomic sequencing platforms, computational resources for genome-scale data	www
Biology	Voyles Lab	Pathogen Research, Physiology, Field Stations	Pathogen research facilities, physiological analysis equipment, field stations (Smithsonian SCBI)	www
Biology	3D Printing Lab	3D Printing	3D Bioprinter Mechanical Property Tester Rheometer	www
Biology	Cell Analysis Core	Flow Cytometry, Cell Sorting, Spatial Imaging	WOLF microfluidics sorter, Cytoflex 3-laser cytometer, CosMx spatial molecular imager, FlowJo	www

Biology	Core Analytical Lab	Elemental Analysis, Chromatography, Mass Spec	ICP-AES, ICP-MS, GC (MS/FID/ECD), elemental analyzer, flow injection, UV-Vis, microwave digestion	www
Biology	Neuroimaging Core	MRI, EEG, NIRS	2x 3T MRI (Philips Ingenia + Siemens Skyra), 2x HD EEG, NIRS (TechEn)	www
Biology	Bioinformatics Center	Bioinformatics, NGS Analysis, HPC	HPC clusters, bioinformatics software, NGS analysis pipelines, computational workstations	www
Biology	Genomics Center	DNA Sequencing, Fragment Analysis	Illumina NextSeq 2000, MiniSeq, Sanger sequencing, fragment analysis, BSL-2 lab	www
Biology	Proteomics Center	Proteomics, Mass Spec, Liquid Chromatography	Bruker timsTOF Pro2, ThermoFisher Eclipse MS, Evosep One LC, gel/IP/solution workflows	www
Biology	Biosciences Entrepreneurial Lab	Wet Lab, Shared Instruments, Incubator	Complete wet lab infrastructure, standard lab equipment, shared analytical instruments	www
Biology	Nevada Center for Applied Research	Core Facilities, Wet Lab Incubator, Manufacturing	Access to all UNR core facilities, BEL wet lab incubator, ME Manufacturing Lab	www
Astronomy	Great Basin Observatory	Telescope, CCD Imaging, Remote Observation	0.7m robotic telescope, CCD cameras, photometry pipeline, remote operation software	www